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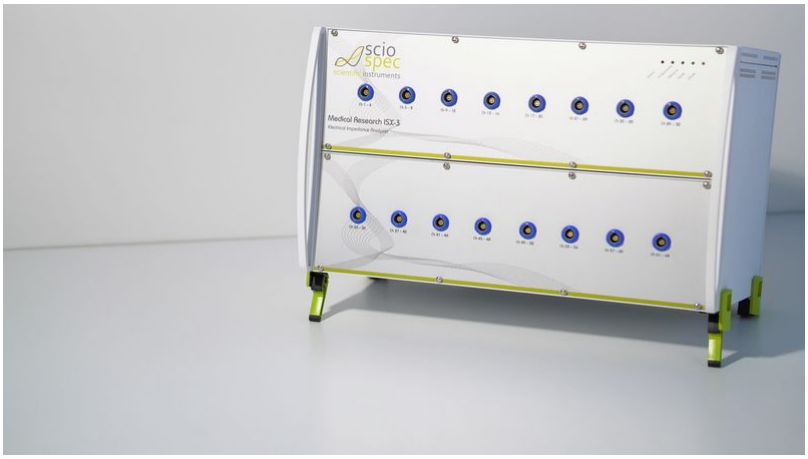
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1 Which products are covered by this manual

This manual is valid for:

- impedance analyzer Medical Research ISX-3
- impedance analyzer Medical Research ISX-3 with MUX32
- impedance analyzer Medical Research ISX-3 with MUX64





2 General

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Please pay attention to the comment.

Please pay attention to the note/warning.

Please pay attention to the warning.

2.1 Warranty

This product is warranted against defects in material and workmanship for a period of one year. During the warranty period any parts or services that show defects with a cause that was present before the transfer of risk shall – at the option of Sciospec – be replaced, reworked or re-performed free of charge (rectification).

For warranty service or repair, this product must be returned to a service facility designated by Sciospec. The buyer shall pay all shipping charges, duties, and taxes for products returned to Sciospec from another country.

Sciospec warrants that its software and firmware designated by Sciospec for use with an instrument will execute its programming instruction when properly installed on that instrument. Sciospec does not warrant that the operation of the instrument, or software, or firmware will be uninterrupted or error free.

Further details on the warranty and transfer of risk are described in the general terms and conditions of Sciospec. No other warranty is expressed or implied. Sciospec specifically disclaims the implied warranties of merchantability and fitness for a particular purpose. Further and/or divergent regulations require contractual agreements in written form.

The foregoing warranty shall not apply to defects resulting from improper or inadequate maintenance by the buyer, operation or service through unauthorized personnel, the buyer-supplied software or interfacing, unauthorized modification or misuse, operation outside the environmental specifications for the product, or improper site preparation or maintenance.

2.2 Intended (normal/contractual) use

The product is intended as an investigational device for indoor use in qualified technical environments and laboratories with all necessary safety measures for electric installations and cabling in place. In order to fulfill the requirements towards electrical safety and electromagnetic compatibility the system is to be installed in a safe

environment with electrical connections (especially power supply) according to general safety regulations. In addition to general safety regulations all safety measures and precautions noted in this document are to be followed.

For outdoor use separate precautions and safety measures are required. Specifically it might be required to mount the system into a protective housing according to the ambient/environmental conditions.

2.3 Non-intended use

Any form of use deviating from the intended use qualifies as non-intended use/misuse/maloperation. Other applications, modes of operations, modifications or types of installations can impair safety and functionality and are thus for safety reasons not permitted for the user or third parties.

2.4 Authorized personnel

In general different levels of qualification of operating and maintenance personnel have to be distinguished.

> Danger through insufficient qualification of operating/service personnel.

The instrument may only be operated, functionally tested and serviced by sufficiently qualified investigators. Work reserved for professional technical personnel may only be done by staff authorized through Sciospec.

2.4.1 Users

Users are all personnel older than 18 years that have been instructed in the operation of the instrument by Sciospec or an authorized representative. They have to have read and understood this manual completely.

User may be tasked with the following:

- General operation of the system through externally accessible interfaces
- Operation of external controls on the instrument
- functional tests through externally accessible interfaces
- to some extent elimination of malfunctions or initiation of measures to eliminate malfunctions through externally accessible interfaces

2.4.2 Professional technical personnel

Professional technical personnel are all persons who in accordance to their professional education, professional experience and contemporary professional occupation

- posses respective knowledge of electrical impedance analyzers
- have been trained, authorized and instructed by Sciospec for maintenance, installation and service tasks on/for the instrument

Further the professional technical personnel have to be qualified for those tasks through their professional education, experience and technical knowledge on electrical impedance analyzers.

In addition to the authorization of users professional technical personnel may be tasked with

- Elimination of malfunctions exceeding the measures permissible for a user
- installations

- maintenance and service work as instructed by Sciospec

2.4.3 Unauthorized personnel

Through operation and service procedures executed by unauthorized personnel substantial danger up to danger to life may arise.

Defects and damage caused by measures performed by unauthorized personnel are not covered by warranty and Sciospec will not be made liable for these.

2.5 Danger and risks

Even with proper professional installation and proper condition of the instrument not all dangers can be covered. Following some of the remaining risks are described.

2.5.1 Electrical currents

Danger to life through electrical currents



- operation and maintenance only to be done by trained personnel
- no access to the instrument for other than authorized personnel
- never bypass fuses or protective parts
- maintenance and installation only in volt-free state
- keep moisture and excessive heat away from the instrument

2.5.2 Risk for danger to personnel

Increased risk for danger to the user exists

- when minors, elderly or disabled persons are allowed to operate the instrument
- when proper instruction, training and monitoring of the user is not possible
- when no authorized personnel exist
- when the number of users to the instrument are very high
- when modifications to the instrument or its installation are done

Modifications, extensions/additions to the instrument are not permissible without prior authorization through Sciospec. Any unauthorized technical or procedural modification voids the permission for operation and the warranty.

2.6 Safety information and precautions

When you notice any of the unusual conditions listed below, immediately terminate operation and disconnect the power cable. Contact Sciospec Scientific Instruments for repair of the instrument. If you continue to operate without repairing the instrument, there is a potential fire or shock hazard for the operator.

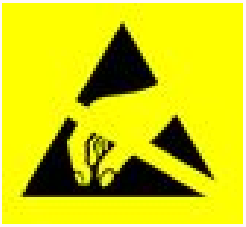
- Instrument operates abnormally.
- Instrument emits abnormal noise, smell, smoke or a spark-like light during the operation.
- Instrument generates high temperature or electrical shock during operation.
- Power cable, plug, or receptacle on instrument is damaged.
- Foreign substance or liquid has fallen into the instrument.

The following measures help assuring safe instrument operation and are to be followed in order to comply with intended use specifications.

⚠ Technical expertise is required. Risk of damage to the instrument exists through unintended use and wrong cabling/installation.

- Do not operate the instrument in the presence of inflammable gasses or fumes. Operation of any electrical instrument in such an environment clearly constitutes a safety hazard.

Electro static discharge warning



This product, like all electronic products, uses semiconductors that can be damaged by electrostatic discharge (ESD). Use care when handling the devices to prevent damage. Damage due to inappropriate handling is not covered by the warranty.
ESD damage is most likely to occur as the test fixtures are being connected or disconnected. Protect them from ESD damage by wearing a grounding strap that provides a high resistance path to ground. Alternatively, ground yourself to discharge any static charge built-up by touching the outer shell of any grounded instrument chassis before touching the test port connectors.
Do not exceed the operating input power, voltage, and current level and signal type appropriate for the instrument being used, refer to technical specifications sections in this manual.

2.7 Replacement parts and equipment

Obtain replacement parts only through Sciospec, authorized customer support services or authorized distributors.

Danger of injury through use of wrong or faulty replacement parts!

The use of wrong, faulty or unauthorized/unintended replacement parts can lead to danger of injury to the operating/service personnel, malfunction or complete failure of the instrument.

Use of unauthorized replacement parts voids the instruments warranty.

3 Allgemeines (Deutsch/German)

Die Inhalte und Werke, die in diesem Handbuch veröffentlicht sind, unterliegen dem deutschen Urheberrecht. Jede Vervielfältigung, Bearbeitung, Verbreitung oder sonstige Verwertung außerhalb der Grenzen des Urheberrechts bedarf der vorherigen schriftlichen Zustimmung des jeweiligen Autors bzw. der jeweiligen Autoren.


Bitte beachten Sie den Kommentar.


Bitte beachten Sie den Hinweis/die Warnung.


Bitte beachten Sie die Warnung.

3.1 Garantie

Für dieses Produkt wird eine Garantie gegen Material- und Verarbeitungsfehler für einen Zeitraum von einem Jahr gewährt. Während der Garantiezeit werden alle Teile oder Leistungen, die aufgrund einer Ursache mangelhaft sind, die bereits vor dem Gefahrenübergang vorlag, nach Wahl von Sciospec kostenlos ersetzt, nachgebessert oder erneut erbracht (Nacherfüllung).

Für Garantieleistungen oder Reparaturen muss dieses Produkt an eine von Sciospec benannte Servicestelle zurückgesendet werden. Der Käufer trägt alle Versandkosten, Zölle und Steuern für Produkte, die aus einem anderen Land an Sciospec zurückgesendet werden.

Sciospec gewährleistet, dass die von Sciospec zur Verwendung mit einem Gerät bestimmte Software und Firmware ihre Programmanweisungen ausführt, wenn sie ordnungsgemäß auf diesem Gerät installiert wurde. Sciospec übernimmt jedoch keine Gewähr dafür, dass der Betrieb des Geräts oder der Software bzw. Firmware unterbrechungsfrei oder fehlerfrei erfolgt.

Weitere Einzelheiten zur Garantie und zum Gefahrenübergang sind in den Allgemeinen Geschäftsbedingungen von Sciospec beschrieben. Es bestehen keine weiteren ausdrücklichen oder stillschweigenden Garantien. Sciospec schließt insbesondere die stillschweigenden Garantien der Marktgängigkeit sowie der Eignung für einen bestimmten Zweck aus. Weitergehende und/oder abweichende Regelungen bedürfen einer schriftlichen vertraglichen Vereinbarung.

Die vorstehende Garantie gilt nicht für Mängel, die auf unsachgemäße oder unzureichende Wartung durch den Käufer, Betrieb oder Service durch nicht autorisiertes Personal, vom Käufer bereitgestellte Software oder Schnittstellen, unbefugte Änderungen oder missbräuchliche Verwendung, Betrieb außerhalb der für das Produkt

angegebenen Umgebungsbedingungen oder eine ungeeignete Installation bzw. Wartung am Einsatzort zurückzuführen sind.

3.2 Bestimmungsgemäße (normale/vertragliche) Verwendung

Das Produkt ist als Untersuchungsgerät für den Einsatz in Innenräumen in qualifizierten technischen Umgebungen und Laboren vorgesehen, in denen alle erforderlichen Sicherheitsmaßnahmen für elektrische Installationen und Verkabelungen vorhanden sind. Um die Anforderungen an elektrische Sicherheit und elektromagnetische Verträglichkeit zu erfüllen, muss das System in einer sicheren Umgebung mit elektrischen Anschlüssen (insbesondere der Stromversorgung) gemäß den allgemeinen Sicherheitsvorschriften installiert werden. Zusätzlich zu den allgemeinen Sicherheitsvorschriften sind alle in diesem Dokument aufgeführten Sicherheitsmaßnahmen und Vorsichtsmaßnahmen zu befolgen.

Für den Einsatz im Freien sind zusätzliche Vorsichts- und Sicherheitsmaßnahmen erforderlich. Insbesondere kann es notwendig sein, das System entsprechend den Umgebungs- bzw. Umweltbedingungen in ein Schutzgehäuse einzubauen.

3.3 Nicht-bestimmungsgemäße Verwendung

Jede Form der Verwendung, die von der bestimmungsgemäßen Verwendung abweicht, gilt als nicht-bestimmungsgemäße Verwendung/Fehlgebrauch/Fehlbedienung. Andere Anwendungen, Betriebsarten, Änderungen oder Installationsarten können die Sicherheit und Funktionalität beeinträchtigen und sind daher aus Sicherheitsgründen für den Benutzer oder Dritte nicht zulässig.

3.4 Autorisiertes Personal

Im Allgemeinen sind unterschiedliche Qualifikationsstufen des Bedien- und Wartungspersonals zu unterscheiden.

> Gefahr durch unzureichende Qualifikation des Bedien-/Servicepersonals.

Das Gerät darf nur von ausreichend qualifizierten Anwendern betrieben, funktionell geprüft und gewartet werden. Arbeiten, die qualifiziertem Fachpersonal vorbehalten sind, dürfen nur von durch Sciospec autorisiertem Personal durchgeführt werden.

3.4.1 Benutzer

Benutzer sind alle Personen über 18 Jahre, die von Sciospec oder einem autorisierten Vertreter in die Bedienung des Geräts eingewiesen wurden. Sie müssen dieses Handbuch vollständig gelesen und verstanden haben.

Benutzer können mit folgenden Aufgaben betraut werden:

- Allgemeine Bedienung des Systems über von außen zugängliche Schnittstellen
- Bedienung externer Bedienelemente am Gerät
- Durchführung von Funktionsprüfungen über von außen zugängliche Schnittstellen
- In gewissem Umfang die Behebung von Störungen oder die Einleitung von Maßnahmen zur Störungsbehebung über von außen zugängliche Schnittstellen

3.4.2 Fachpersonal

Fachpersonal sind alle Personen, die aufgrund ihrer fachlichen Ausbildung, beruflichen Erfahrung und aktuellen beruflichen Tätigkeit

- über entsprechende Kenntnisse im Umgang mit elektrischen Impedanzanalysatoren verfügen
- von Sciospec für Wartungs-, Installations- und Servicearbeiten am bzw. für das Gerät geschult, autorisiert und eingewiesen wurden.

Darüber hinaus muss das Fachpersonal durch seine Ausbildung, Erfahrung und technischen Kenntnisse im Bereich elektrischer Impedanzanalysatoren für diese Aufgaben qualifiziert sein.

Zusätzlich zu den Befugnissen der Benutzer kann Fachpersonal mit folgenden Aufgaben betraut werden:

- Behebung von Störungen, die über die für Benutzer zulässigen Maßnahmen hinausgehen
- Installationen
- Wartungs- und Servicearbeiten gemäß den Anweisungen von Sciospec.

3.4.3 Nicht-autorisiertes Personal

Durch Bedien- und Servicearbeiten, die von nicht-autorisiertem Personal durchgeführt werden, können erhebliche Gefahren bis hin zu Lebensgefahr entstehen.

Mängel und Schäden, die durch Maßnahmen von nicht autorisiertem Personal verursacht werden, sind von der Garantie ausgeschlossen. Sciospec übernimmt hierfür keine Haftung.

3.5 Gefahren und Risiken

Auch bei fachgerechter Installation und ordnungsgemäßem Zustand des Geräts können nicht alle Gefahren ausgeschlossen werden. Nachfolgend werden einige der verbleibenden Risiken beschrieben.

3.5.1 Elektrische Ströme

Lebensgefahr durch elektrische Ströme



- Betrieb und Wartung dürfen nur von geschultem Personal durchgeführt werden.
- Zugang zum Gerät nur für autorisiertes Personal.
- Sicherungen oder Schutzvorrichtungen niemals überbrücken.
- Wartung und Installation nur im spannungsfreien Zustand durchführen.
- Feuchtigkeit und übermäßige Hitze vom Gerät fernhalten.

3.5.2 Gefährdungsrisiken von Personen

Ein erhöhtes Gefährdungsrisiko für den Benutzer besteht,

- wenn Minderjährige, ältere oder beeinträchtigte Personen das Gerät bedienen dürfen,
- wenn eine ordnungsgemäße Einweisung, Schulung und Überwachung der Benutzer nicht möglich ist,
- wenn kein autorisiertes Personal vorhanden ist,
- wenn die Anzahl der Benutzer des Geräts sehr hoch ist,
- wenn Änderungen am Gerät oder an dessen Installation vorgenommen werden.

Änderungen, Erweiterungen oder Ergänzungen am Gerät sind ohne vorherige Genehmigung durch Sciospec nicht zulässig. Jede nicht autorisierte technische oder prozessuale Änderung führt zum Erlöschen der Betriebserlaubnis und der Garantie.

3.6 Sicherheitsinformationen und Vorsichtsmaßnahmen

Wenn Sie eine der unten aufgeführten ungewöhnlichen Situationen feststellen, beenden Sie den Betrieb sofort und trennen Sie das Netzkabel. Wenden Sie sich zur Reparatur des Geräts an Sciospec Scientific Instruments. Wenn Sie das Gerät weiterhin betreiben, ohne es reparieren zu lassen, besteht für den Bediener potenziell Brand- oder Stromschlaggefahr.

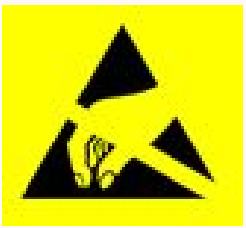
- Das Gerät arbeitet ungewöhnlich oder fehlerhaft.
- Während des Betriebs treten ungewöhnliche Geräusche, Gerüche, Rauch oder funkenähnliche Lichterscheinungen auf.
- Während des Betriebs entstehen ungewöhnlich hohe Temperaturen oder es kommt zu elektrischen Schlägen.
- Netzkabel, Stecker oder die Steckbuchse am Gerät sind beschädigt.
- Fremdkörper oder Flüssigkeiten sind in das Gerät gelangt.

Die folgenden Maßnahmen tragen zu einem sicheren Betrieb des Geräts bei und sind einzuhalten, um die Anforderungen der bestimmungsgemäßen Verwendung zu erfüllen.

⚠ Technische Fachkenntnisse sind erforderlich. Bei unsachgemäßer Verwendung sowie bei fehlerhafter Verkabelung oder Installation besteht die Gefahr von Schäden am Gerät.

- Betreiben Sie das Gerät nicht in der Nähe von entzündlichen Gasen oder Dämpfen. Der Betrieb eines elektrischen Geräts in einer solchen Umgebung stellt eine erhebliche Sicherheitsgefahr dar.

Warnung vor elektrostatischer Entladung



Dieses Produkt verwendet – wie alle elektronischen Geräte – Halbleiterbauelemente, die durch elektrostatische Entladungen (ESD) beschädigt werden können. Gehen Sie beim Umgang mit dem Gerät daher vorsichtig vor, um Schäden zu vermeiden. Schäden, die durch unsachgemäße Handhabung entstehen, sind nicht durch die Garantie abgedeckt.

ESD-Schäden treten besonders häufig beim Anschließen oder Trennen von Testvorrichtungen auf. Schützen Sie diese vor ESD-Schäden, indem Sie ein Erdungsarmband tragen, das einen hochohmigen Ableitpfad zur Erde bereitstellt. Alternativ können Sie sich erden, indem Sie vor dem Berühren der Testport-Anschlüsse das äußere Gehäuse eines geerdeten Gerätegehäuses berühren, um eventuell aufgebaute statische Ladungen abzuleiten.
Überschreiten Sie nicht die zulässige Eingangsleistung, Spannung, Stromstärke oder den für das verwendete Gerät vorgesehenen Signaltyp. Beachten Sie hierzu die technischen Spezifikationen in diesem Handbuch.

3.7 Ersatzteile und Zubehör

Ersatzteile dürfen nur über Sciospec, autorisierte Kundendienststellen oder autorisierte Vertriebspartner bezogen werden.

Verletzungsgefahr durch die Verwendung falscher oder fehlerhafter Ersatzteile!

Die Verwendung falscher, fehlerhafter oder nicht autorisierter bzw. nicht vorgesehener Ersatzteile kann zu Verletzungsgefahr für das Bedien-/Servicepersonal, zu Fehlfunktionen oder zum vollständigen Ausfall des Geräts führen.

Die Verwendung nicht autorisierter Ersatzteile führt zum Erlöschen der Garantie des Geräts.

4 Safety Concept

4.1 General Safety Instructions

⚠ The ISX-3 impedance analyzer in the "medical research" version is intended to facilitate the use of impedance spectroscopy in medical research applications. To that end it includes measures for enhanced electrical safety including medical grade isolation enabling a safe operation of the instrument in medical research settings.

- Safe operation depends on the use case setting including instrument settings, cabling, environment, other equipment etc. Carefully plan the use of the instrument for safe operation!
- Refer to the functional description and technical specification for details on the operational parameters.
- Operation outside the specified ranges may cause damage and/or unsafe operating conditions
- Choosing adequate measurement ranges with sufficient headroom avoids saturation of the measurement port input amplifiers and helps maintaining a safe operating point.
- While not always required, in general the use of AC coupling of the measurement ports is recommended for safe operation. If used without the AC coupling frontend, DC leakage currents might be a concern.
- Assure that the settings of the instrument - especially the excitation parameters - lead to acceptable currents through the measurement object. Ideally this is done on a previously established model (e.g. equivalent impedance network), but if this not feasible the resulting currents should at least be checked for safe operating range during inital setup of the measurements.
- If used in combination with other electrical equipment, this needs to be considered during the assessment of safe operation.

⚠ The Medical Research ISX-3 impedance analyzer is not a medical product and as such its intended purpose does not include diagnosis and/or treatment of patients.

4.2 Functional and Structural Overview

The functional concept of the Medical Research ISX-3 impedance analyzer is to inject an electrical excitation stimulus (apply voltage or inject current) and measure the response of the measurement object in both voltage drop across the object and current flow through the object. From these measurements the complex impedance is then derived. The stimulus is applied as sinusoidal signal with varying frequency. Optionally the amplitude of the excitation signal can also be varied.

The medical research version of the ISX-3 includes several added safety measures specified in accordance with DIN EN 60601-1:2013-12 "Medical electrical equipment – Part 1: General requirements for basic safety and essential performance". The two most of

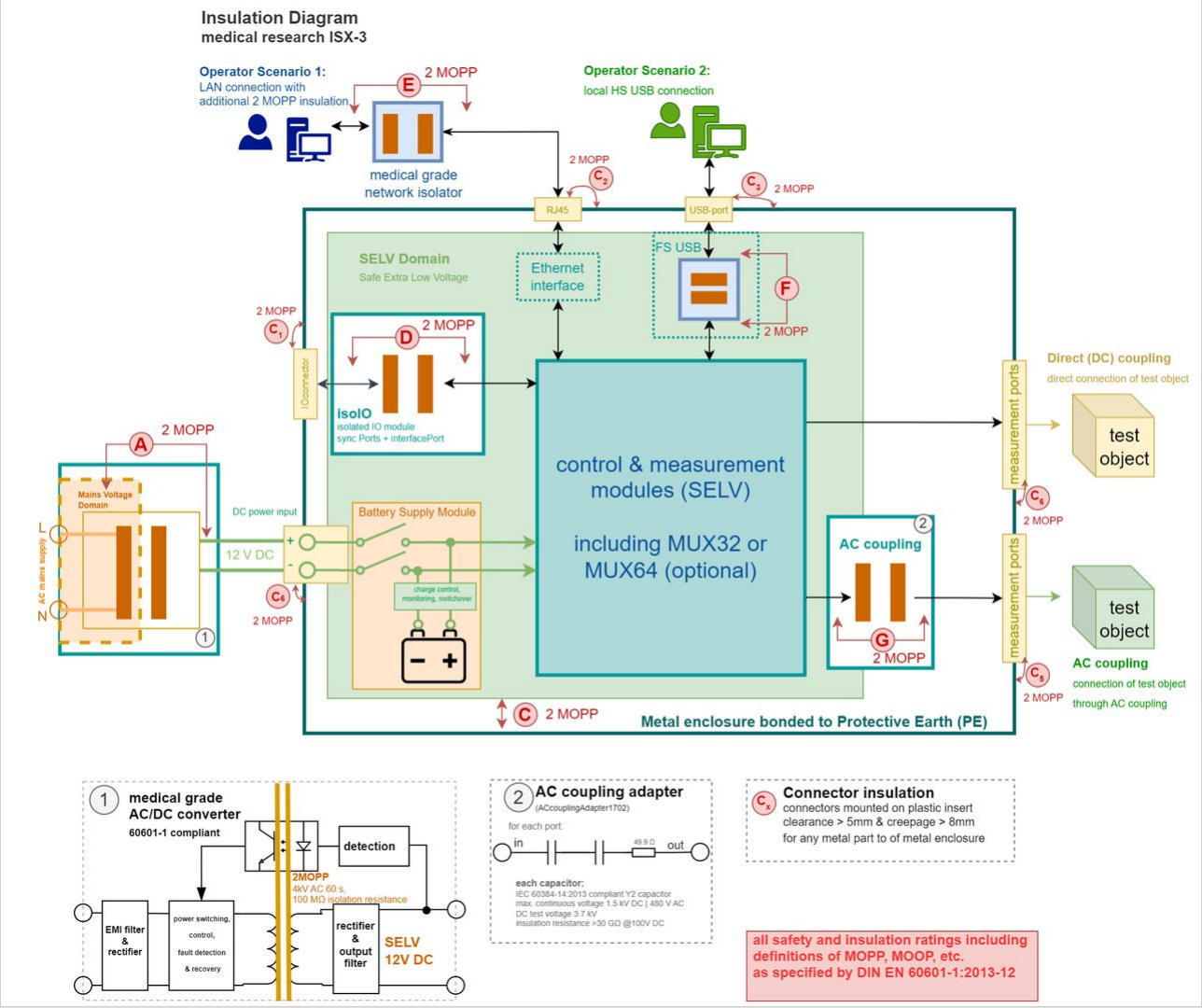
important principles thereby are

- medical grade isolation towards mains and operator side interfaces
- DC blocking through AC coupling on the measurement ports (equivalent to "patient" interface connections)

The Medical Research ISX-3 consists of the following main building blocks:

Component	Function	Safety Notes
metal enclosure	prevention of contact between electrical circuitry and external equipment or humans, shielding of measurement frontend (noise reduction) and mechanical safety	connected to protective earth, all feedthroughs and connectors electrically isolated from enclosure
power supply	external isolated AC/DC converter supplies device with 12 V DC, additionally stand alone battery operation is possible including software controllable electromechanical switch-off for DC input lines	external medical grade 12v AC/DC converter with 2 MOPP isolation ratings must be used to ensure proper safety insulation, for additional safety during live test subject experiments battery operation can be used
measurement and control modules	core functionality of the impedance analyzer including measurement control, signal processing, excitation generation and analog measurement functions	SELV operation
USB (FS) interface	fullspeed USB port for instrument control and data readout from PC	2 MOPP rated USB isolator build into the instrument
Ethernet interface	standard RJ45 Ethernet interface for remote control of the instrument	The interface is not internally isolated. An external medical grade network isolator can be inserted in the communication path to ensure safe isolation ratings.
IsoIO module	low level communication interfaces including hardware synchronization and flexible programmable IO standards including serial (UART, SPI, I2C) for use of the instrument with other external equipment	2 MOPP rated isolation on all ports
AC coupling frontend	AC coupling of measurements ports. Blocks DC currents and adds a high pass filter characteristic limiting low frequency currents. Isolates the instrument galvanically from the measurement object or patient interface. For measurements between 100 Hz and 100 MHz.	IEC 60384-14:2013 compliant Y2 capacitors with high isolation ratings. With the internal circuitry operating in SELV domain this complies with 2 MOPP ratings.

4.3 Isolation Diagram



5 Technical Specification

5.1 General Overview

- single channel (AC coupled) impedance measurement module
- multiplexed second channel (DC coupled)
- impedance measurement capability
  - 100mHz...10MHz (optional: 40 MHz or 100 MHz)
  - mOhm...TOhm
  - connection for 4, 3 or 2 electrode configurations (software configurable)
  - one high resolution signal generator with selectable current or voltage excitation mode
  - automatic ranging
- high resolution DC bias/offset generation (± 1 V)
- high isolation, ultra-low parasitic reed relay switches for channel connect/disconnect with superior performance compared to semiconductor switches
  - >10 TΩ || 0.4 pF isolation at power off or software selected cell disconnect
  - extremely low parasitic capacitance < 0.7 pF offers close to no cell loading
  - max. series resistance 200 mΩ in connected state
  - zero leakage current in on state
  - zero charge injection, no distortion, no noise contributions beyond thermal noise
- integrated battery pack for measurements up to 3 hours
  - 3.6V / 10.05Ah
  - IEC/EN 62133: 2013 approved
- medical grade power supply
  - 12VDC/ 3.33A
  - Medical Safety approved (2xMOPP) according to EN60601-1/EN60601-1-11
- additional peripherals
  - IsoIOport (medical grade isolated, digital IOs with highly flexible digital IO functions + Ultra-fast hardware synchronization through sync in/out ports), connector type: MDR40
- interfaces:
  - isolated USB 2.0 (FS, 12 Mbits/s)
    - Class 3A contact ESD performance per ANSI/ESD STM5.1-2007
    - IEC 60601-1: 250 V rms (reinforced); IEC 60950-1: 400 V rms (reinforced)
  - 10/100 Base-T Ethernet

All specifications are stated for operation at temperatures of 0°C to 40°C.
Warm-up time must be greater than or equal to 30 minutes after power on to comply with all specifications.

Maximum Ratings

Stresses above the listed absolute maximum or maximum ratings may cause permanent damage to the device. This is a stress rating only; functional operation of the device at these or any other conditions above those indicated in the operational section of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

5.2 Measurement specifications

All ports (C, R, WS, W) of channels 1 and 2 are available through BNC coaxial connectors.

measurement port protection



Be aware that the measurement terminals (Counter, Reference, Working Sense and Work Electrode Terminals, including the measurement terminals of the *ExtensionPort*) are very sensitive to electrostatic discharge, over-current and over-voltage. Protection of the terminals of the instrument and strict adherence to the specified maximum ratings has to be ensured by the user. For further information on how to connect specific device under test to the instrument contact Sciospec or an authorized representative directly.

5.2.1 Set point & excitation signal generator (C port)

- software selectable current or voltage source for excitation signal generation
- low distortion single sine signal and superimposed high resolution DC offset

- compliance voltage ±4.9 V (@ 10kΩ load)
- compliance current
  - 200% of selected current range
  - abs. max. ±50mA (continuous)
  - short circuit current max. 65 mA
- Voltage Excitation Mode:
  - Abs. max. applied potential range ± 5 V
  - DC potential set-point resolution 0.0001% of applied voltage range (~19.5bit @2ms settling time)
  - DC potential set-point accuracy ≤ 0.1% || ±100 µV
  - maximum calibrated DC offset error: ± 1 mV
- Current excitation mode
  - Abs max. applied current range ± 50 mA
  - DC current set-point resolution 0.0001% of applied voltage range (~19.5bit @2ms settling time)
  - DC current set-point accuracy ≤ 0.1% || ±1 µA
  - maximum uncalibrated DC offset error: ±40 µA

- AC Excitation Signal Range
  - 1 mV … 1 V (peak amplitude), resolution 0.1 mV in voltage excitation mode
  - 10 µA … 10 mA (peak amplitude) in current excitation mode
  - EIS Excitation Amplitude Error (uncalibrated): ±1% typ. (±8% max.)
  - Excitation Signal Resolution 0.1% of range
  - Frequency Range 100mHz to 10MHz
  - Frequency Resolution
    - Frequency < 10kHz: < 6 mHz
    - 10kHz ≤ frequency ≤ 100kHz < 60 mHz
    - Frequency > 100kHz < 120 mHz
  - Frequency Precision
    - ±100ppm (@25°C)
    - additional ±10ppm over temperature range
    - additional ±5ppm during first year
- Number of Frequency Points per Sweep: 1 to 2048

5.2.2 potential measurement (R & WS ports)

- potential measurement ranges
  - ± 1 V
  - ± 0.09 V
- AC voltage measurement range equals selected potential measurement range
- measured potential resolution < 0.008% of potential range
- measured potential accuracy:
  - max uncalibrated gain error 1%
  - max uncalibrated offset error 2% of potential measurement range
- electrometer amplifier input
  - input impedance ≈ 1 TΩ, 2.3 pF
  - ±1.5 pA input bias current typ. (50 pA max.)

5.2.3 Current measurement (W port)

- 4 current measurement ranges
  - ± 10 mA (100 Ohm range)
  - ± 100 µA (10 kOhm range)
  - ± 1 µA (1 MOhm range)
  - ± 10 nA (100 MOhm range)
- AC current measurement range equals selected current range
- measured current resolution < 0.008% of current range
- max uncalibrated gain error
  - 0.1% for ±10 mA range
  - 0.1% for ±100 µA range
  - 1% for ±1 µA range
  - 5% for ± 10 nA range
- max uncalibrated offset error 2% of current measurement range ± input bias current
- current amplifier input bias current ±10 pA typ. (max. ±1nA)

5.2.4 timing

- max. acquisition rate < 10000 points/s (> 100 µs per frequency)
- DC set point interval > 150 µs (“ramp time resolution”)

5.2.5 impedance spectroscopy

- Impedance Measurement Range: mΩ…GΩ
- Impedance Accuracy (for details refer to respective section)
  - |Z|/Z: 0.1%
  - |Phi|: 0.1°

5.3 Master interfaces

5.3.1 isolated Full Speed USB Interface

standard conformity	USB 2.0, FS (12 Mbits/s)
connector	Mini USB Type B
Protocol	High Speed USB 2.0
ESD Protection	Class 3A contact ESD performance per ANSI/ESD STM5.1-2007
isolation	UL recognition: 5000 V rms for 1 minute per UL 1577 CSA Component Acceptance Notice #5A IEC 60601-1: 250 V rms (reinforced) IEC 60950-1: 400 V rms (reinforced) VDE Certificate of Conformity DIN V VDE V 0884-10 (VDE V 0884-10):2006-12 V_{IORM} = 846 V peak

5.3.2 Ethernet

standard conformity	10/100 Base-T, RJ45
protocol	TCP/IP
isolation	external Isolator provided as accessory UL Recognized Component 5000 V 50/60 Hz for 1 Min. IEC 60601-1 :2005 + A1:2012 Ed. 3.1 (gutted section 17 - EMV) IEC 60601-1-2 :2014 Ed. 4 EN 60601-1 :2006 + A1:2013 Ed. 3.1 (gutted section 17 - EMV) EN 60601-1-2 :2015 Ed. 4

5.4 Relation between the precision setting and the measurement time and measurement accuracy

The precision settings enables the user to adjust the trade off between measurement stability and measurement speed. For more information see chapter Functional Description.

The figure "*Accuracy over Precision Parameter*" shows the influence of the precision settings on the accuracy and time for the measurement of an impedance value at the specified frequency.

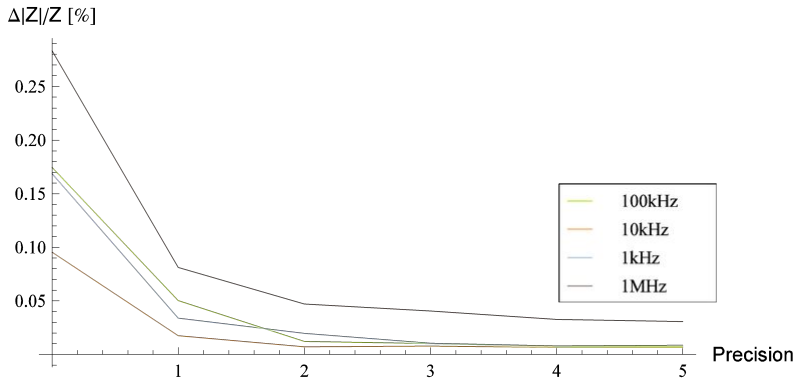


Fig: Accuracy over Precision Parameter

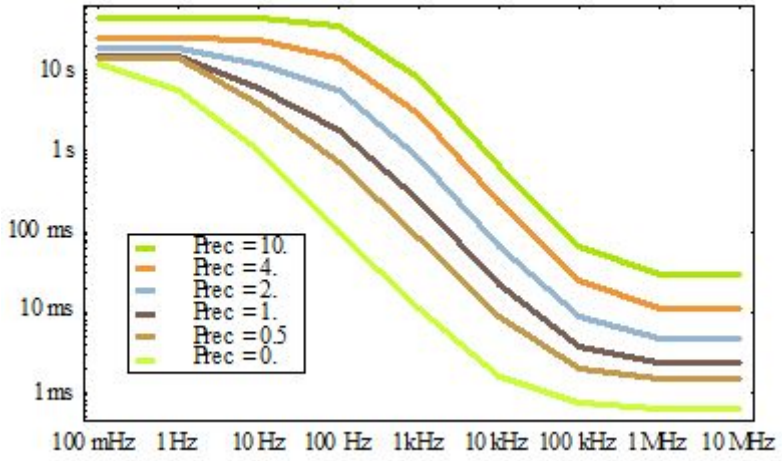


Fig: Measurement Time over Precision Parameter

5.5 Accuracy contour plot

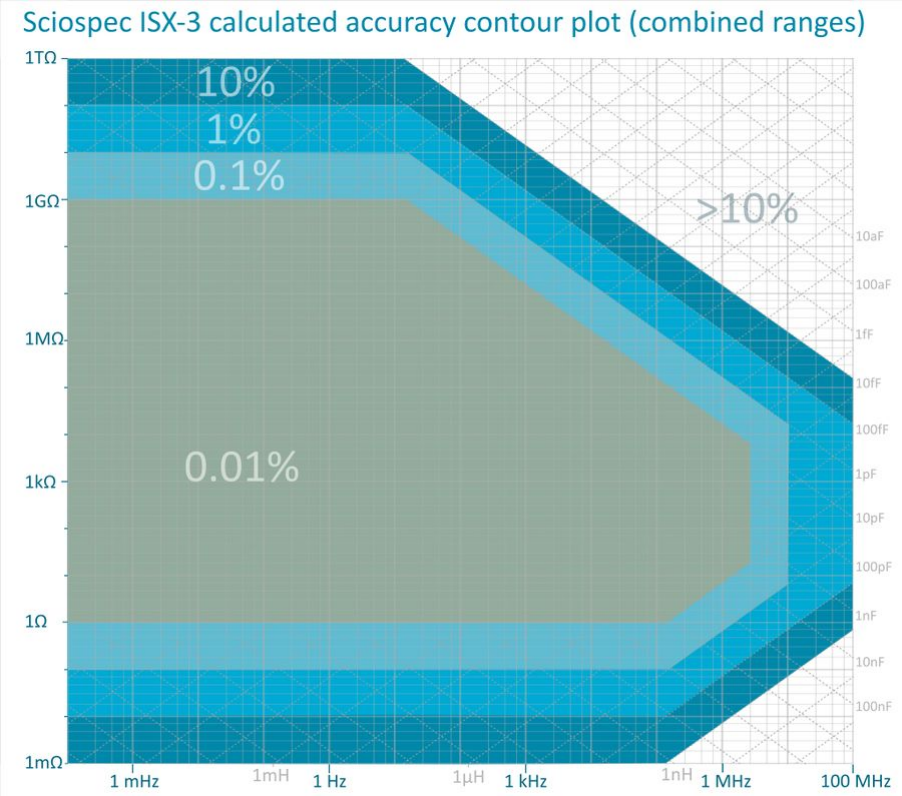
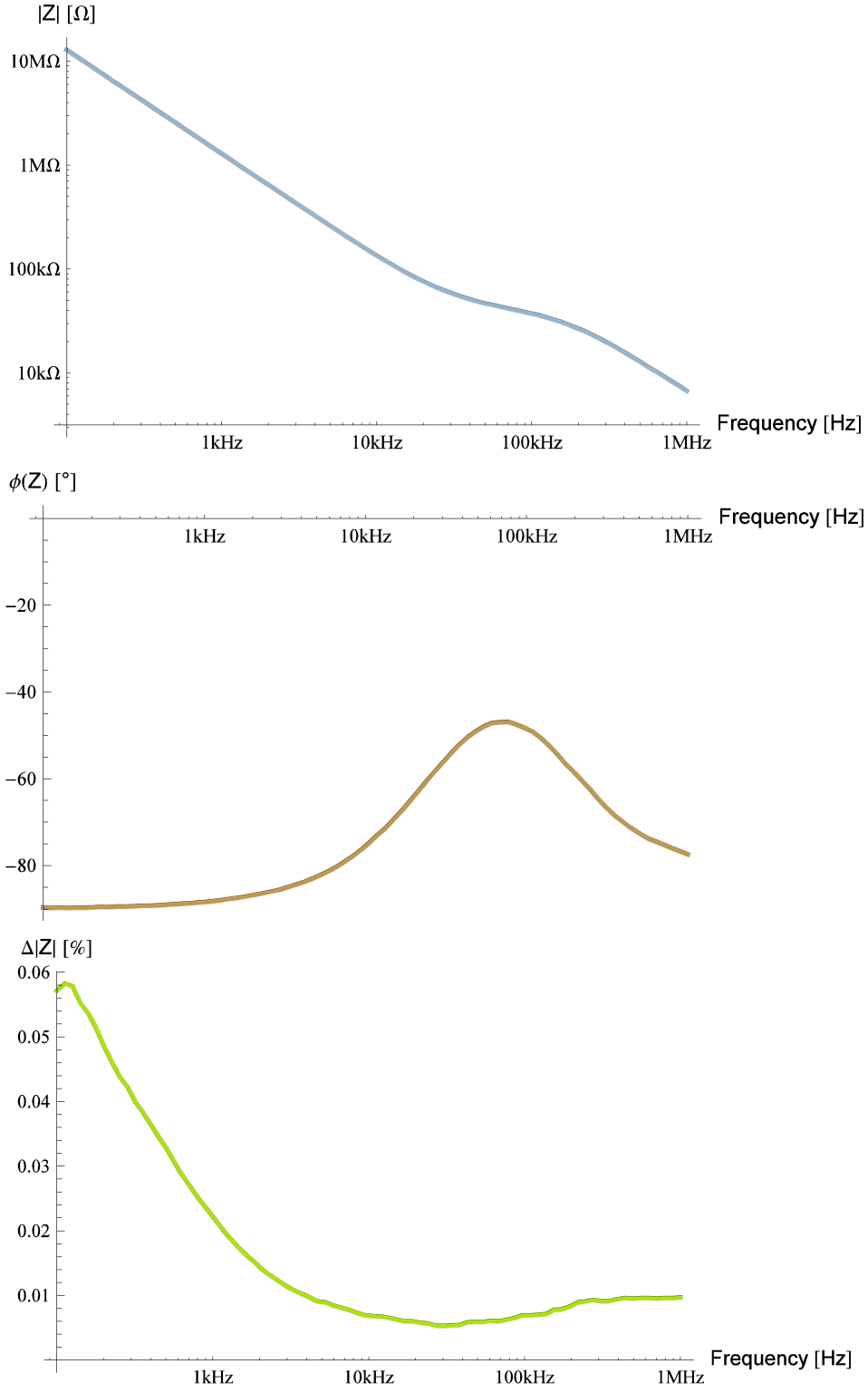


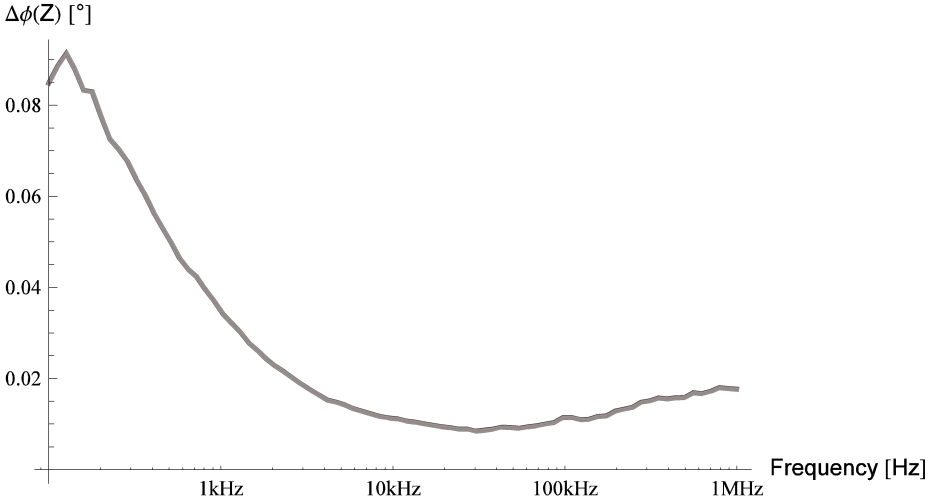
Fig: accuracy contour plot

Note: The accuracy contour plot was determined under lab conditions and should be used for reference purposes. Please note that the true limits of an impedance measurement are influenced by all components in the system, like cables, cell, and the instrument.

5.6 Example Measurement

Frequency sweep: 100Hz - 1MHz, 80 logarithmic Frequency steps, precision 1, amplitude 100mV, measurement range MΩ
Measurement instrument: Sciospec ISX-3 with connected Sciospec MEArack
DUT: multi electrode array, 40µm electrodes, 200µm apart, platinum PBS buffer solution


|Z| [Ω]
10MΩ
1MΩ
100kΩ
10kΩ
1kHz
10kHz
100kHz
1MHz
Frequency [Hz]
ϕ(Z) [°]
Frequency [Hz]
1kHz
10kHz
100kHz
1MHz
−20
−40
−60
−80
Δ|Z| [%]
0.06
0.05
0.04
0.03
0.02
0.01
1kHz
10kHz
100kHz
1MHz
Frequency [Hz]



5.7 AC coupled Channel (First Channel)

The AC coupling interface - Internal adds a fully galvanic isolated interface to the measurement channel of the ISX-3. The circuit is shown in *Fig: Circuit - AcCouplingOption.*,

ⓘ The AC coupling interface should only be used for measurements between 100 Hz to 100 MHz!

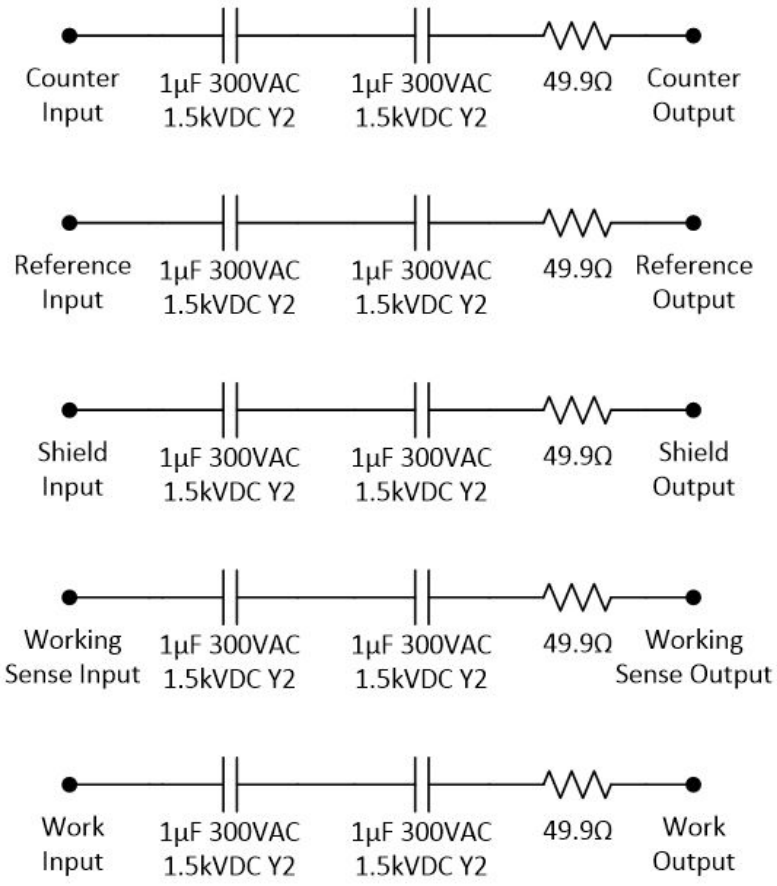


Fig: Circuit – AcCouplingOption (additional each input terminal is connected with a 1MOhm resistor to the cable shield of the input cable.)

5.8 General Specifications Medical Research ISX-3

power requirements	12VDC (typ.), 42W (max), Connector DC Jack (Type: Switchcraft PN: 712A)
Timing accuracy	1% accuracy over the full temperature range
dimensions	248.7mm x 96.7mmx 189.75mm (width x height x depth) see
weight	2.5kg (typical)
operating conditions	0°C to 40°C, <80% relative humidity non condensing, 0...3000m altitude
non-operating conditions	-25°C to 80°C, <80% relative humidity non condensing The temperature gradient should not exceed 1K/min to reach operating conditions.

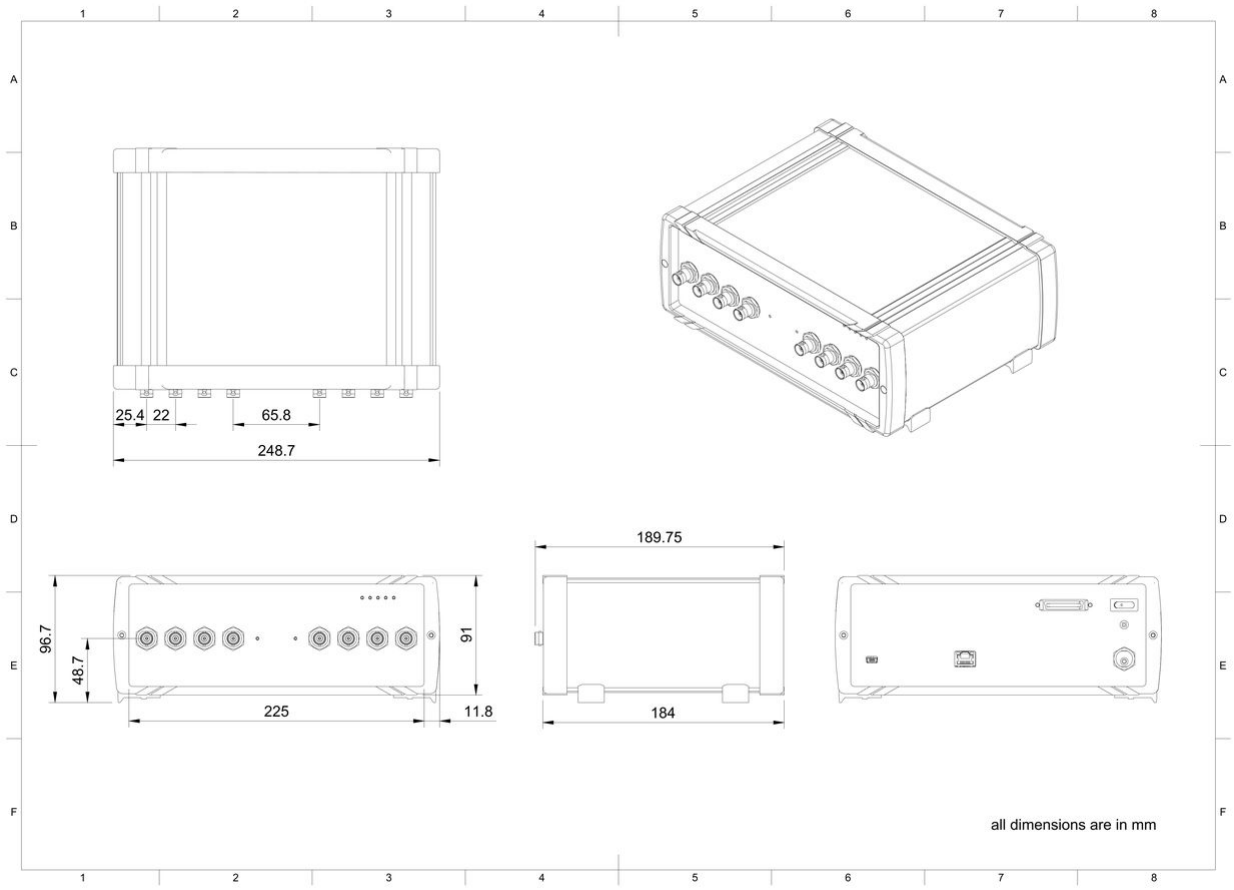


Figure 1: dimensioned drawings of the Medical Research ISX-3

5.9 General Specifications Medical Research ISX-3 with MUX32 or MUX64

power requirements	12VDC (typ.), 42W (max), Connector DC Jack (Type: Switchcraft PN: 712A)
Timing accuracy	1% accuracy over the full temperature range
dimensions	449 mm x 155.41 mm x 286.26 mm (width x height x depth) see figure below
weight	8kg (typical)
operating conditions	0°C to 40°C, <80% relative humidity non condensing, 0...3000m altitude
non-operating conditions	-25°C to 80°C, <80% relative humidity non condensing The temperature gradient should not exceed 1K/min to reach operating conditions.

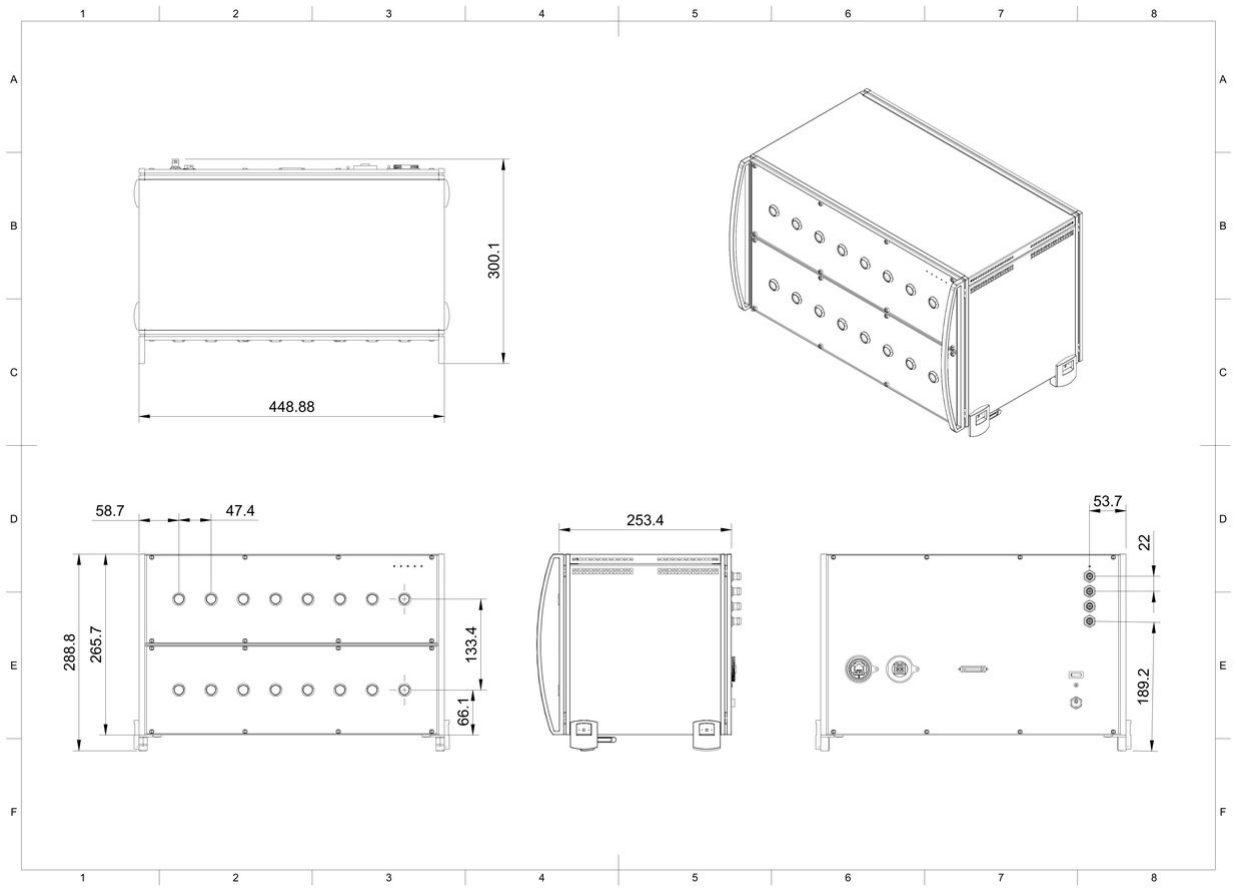


Figure 2: dimensioned drawings of the Medical Research ISX-3 with MUX32 or MUX64

6 Functional Description

6.1 Basics of Impedance Measurements

An impedance measurement is a fundamental method to determine the electrical properties of a device under test (DUT). With this method, the AC electrical impedance Z is gathered which is composed of a frequency-dependent resistance part Z_{real} as well as a frequency-dependent reactance part Z_{imag} according to

$Z(f) = Z_{\text{real}}(f) + jZ_{\text{imag}}(f)$.

The real part $Z_{\text{real}}(f)$ describes the Ohmic lossess of the DUT, whereas the imaginary part $Z_{\text{imag}}(f)$ encompasses the stored energy in form of capacity and inductivity.

Our LCR bridges, the LCR-1 as well as the LCR-3, measure the impendance at a single frequency to derive equivalent model parameters such as the capacity (C), the inductivity (L), the resistance (R) as well as loss parameters (e.g., dissipation factor D or quality factor Q). The excitation is realized via a small sinusoidal signal at constant frequency and amplitude. The impedance of the DUT is then determined from voltage and current measurements and their phase correlation.

To yield a more comprising characterization of the DUT over a particular frequency range, our impedance analyzers offer the possibility to record an impedance spectrum. With these devices, the frequency of the excitation signal is progressively incremented and, once more, the impedance is evaluated. Consequently, sophisticated frequncy-dependent processes, such as polarisation effects, charge transport, and relaxation behavior can be thoroughly analyzed.

In dependence on the application, the measurement results can be visualized, e.g., using

- a Bode plot (amplitude and phase of the impedance over frequency),
- a Nyquist plot (imaginary part of the impedance over real part of the impedance),
- a time trace plot (amplitude and phase of the impedance over time).

Typical fields of application are

- characterization of electronic components (e.g., capacitors, inducatances, resistors),
- investigation of electrochemical systems (e.g., sensors, fuel cells),
- analysation of materials (e.g., dielectric properties, electrical conductivity),
- bio-technological measurements (e.g, cell cultures, tissue samples).

In dependence on the respective use case, there are three typical topologies for impedance measurements:

- 2-point configuration,
- 3-point configuration,
- 4-point configuration.

6.2 Accuracy and precision

6.2.1 General concept

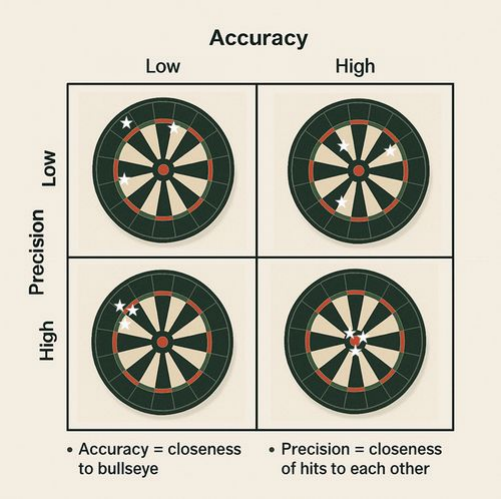
Two core concepts that need consideration during measurements (e.g., of currents, voltages, impedances), and which are often interchanged, are accuracy and precision.

- **Accuracy** describes how close a measured value is to the true or reference value. A highly accurate measurement means that systematic errors (e.g., calibration offsets, instrument drift) are minimized.
- **Precision**, in contrast, refers to the repeatability of the measurement. If the same quantity is measured multiple times under the same conditions, a precise instrument will yield very similar results—even if these results are not necessarily close to the true value. Precision is influenced by factors such as instrument noise, stability of test leads, and environmental conditions.

To elucidate both concepts with an example from the topic of impedance measurements, suppose the true impedance of a calibration resistor is 50.0 Ω.

- An instrument that is precise but inaccurate—that is, repeatedly giving the wrong value due to systematic errors—might consistently report 52.3 Ω.
- An accurate but imprecise instrument, which on average measures close to 50.0 Ω but with large variation, might give values scattered between 48 Ω and 52 Ω.
- Only an instrument that is both accurate and precise would repeatedly produce results tightly clustered around 50.0 Ω.

The attached plot furthermore shows a graphical visualization of both concepts.



6.2.2 Accuracy contour plot

A powerful tool for graphically highlighting the strengths of a measurement device in terms of accuracy is the so-called accuracy contour plot, which can be found in the chapter "Technical descriptions". In the context of impedance measurements, an accuracy contour plot visualizes measurement uncertainty across a range of frequencies and impedance values.

In such a plot, the vertical axis represents impedance values in logarithmic scaling, while the horizontal axis indicates the frequency range, also in logarithmic scaling. Contour lines then enclose combinations of frequency and impedance values that share the same relative error, expressed as a percentage. By inspecting these contours, one can quickly identify the regions in which the instrument achieves high precision and where its performance degrades.

Sciospec's accuracy contour plots additionally include tilted lines that allow the user to assess the precision of capacitance and inductance measurements. The capacitance range can be read from the right vertical axis,

whereas the inductance range can be derived from the lower horizontal axis, which also displays the frequency range.

It should be noted that although the plot is termed an accuracy contour plot, it in fact represents the precision ranges of the measurement device. Nevertheless, since the designation *accuracy contour plot* has become established in common parlance, we will adhere to this terminology. High accuracy can, in turn, be achieved by applying a proper compensation procedure prior to measurement.

6.3 Port Configurations

There are several ways to physically connect the DUT to the analyzer, commonly referred to as port configurations. Each configuration determines how current is delivered and where voltage is sensed, affecting both the simplicity and accuracy of the measurement. Choosing the right one is a key step in reliable and accurate impedance measurements. To help you do this, the three most commonly used configurations are detailed below.

Quick summary

There are **three standard configurations**:

- **2-Point:** Easiest setup, but introduces errors of the wires and contacts as well as potential double layer polarization effects in the current path.
- **3-Point:** Adds a reference voltage sense for improved accuracy, commonly used in electrochemical applications studying chemical dynamics at the work electrode interface.
- **4-Point:** Separates current and voltage measurement completely for the highest accuracy; recommended for critical measurements or when wiring parasitics would affect results.

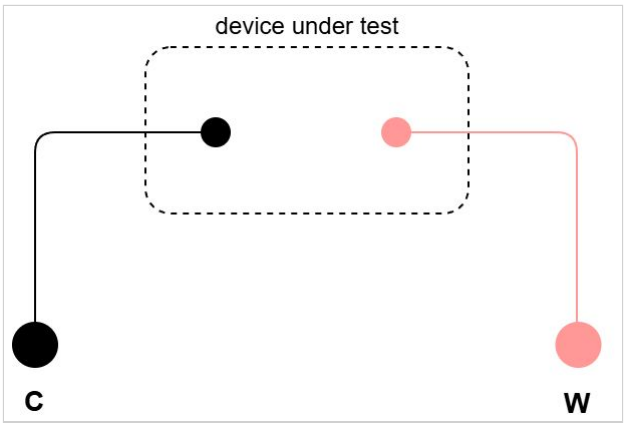
6.3.1 Port Types

Most impedance measurement instruments use a combination of these four fundamental port types:

- Counter Electrode (C): This port provides the excitation signal, driving a controlled current through the device under test.
- Work Electrode (W): The primary return path for the excitation current, connected to one end of the DUT.
- Reference Electrode (R): This high impedance voltage sensing input measures the electrical potential at a specific point side of the DUT facing the counter port.
- Sense Electrode or Working Sense (WS): Like the reference electrode, this provides a high-impedance potential measurement point, this time at work port facing side of the DUT.

6.3.2 2-Point Configuration

In a 2-point configuration only Counter (C) and Work (W) ports are used. These two ports serve both as current-carrying and voltage-sensing electrodes.



The simple wiring with only two connections is a clear advantage of this configuration, especially in high channel count applications.

However, there is also a potential drawback. The current flowing through the wires results in a voltage drop that effectively adds as series resistance/impedance to the DUT. While short wires usually do not present a problem, longer wiring, additional series components like clamps and connectors or switches in series with the wires will add to this effect. At higher frequencies often the inductive components of these elements amount to noticeable parasitic impedance in series with the DUT.

Polarization and double layer capacitance

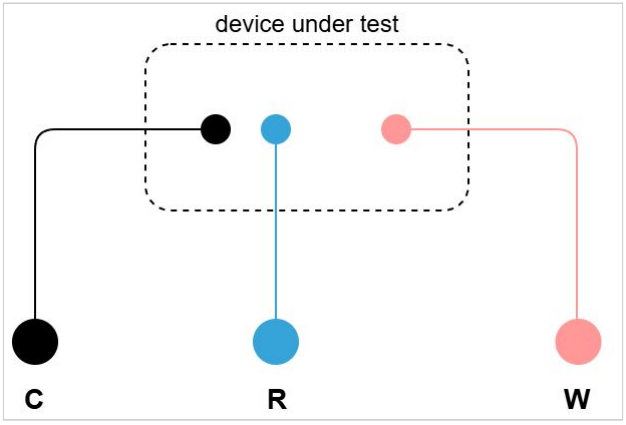
The formation of double layer capacitance is another very prominent effect that has to be considered for selecting a port configuration. When current flows across material interfaces with different charge carrying behaviour, this leads to the redistribution of the charge carriers close to the interface. This is commonly referred to as polarization. The observed behavior is similar to that of a capacitance (also often modelled as constant phase elements) in series with the current path. This phenomenon is most commonly associated with metal-to-fluid interfaces because of their importance in electrochemistry, but the concept extends to many different material interfaces where charge carriers can redistribute. As any series capacitance, this can lead to a very high impedance especially at lower frequencies. If uncompensated the resulting measurement error can be significant, in some cases leading to the DUT impedance being completely dominated by the polarization impedance and effectively hidden in the noise floor of the measurement.

6.3.3 3-Point Configuration

The 3-point configuration adds the Reference electrode connection to the previously described scheme of a 2-point configuration.

With the addition of one more connection, the potential measured across the DUT is now obtained as the difference between the Reference and Work electrodes. Thus, any parasitic effect along the counter port side of the current path is effectively eliminated from the measurement result - including cable effects, double layer capacitance, and any contact impedances. However, the same parasitic effects on the other side of the DUT (along

the current path through Work) still remain as part of the measured impedance. The tradeoff here is the higher complexity of the setup due to an added connection.

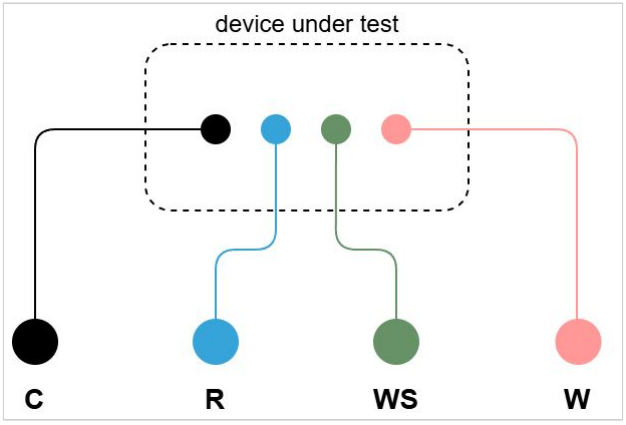


This configuration is most commonly applied in electrochemical setups where the elimination of the double layer formation on the Counter electrode is crucial for proper measurement results. Here, the surface chemistry including any double layer on the working electrode is what is being investigated and thus cannot be also eliminated.

6.3.4 4-Point Configuration

The 4-point configuration uses all four port types described above. Now, the voltage across the DUT is measured as difference between the Reference and Working Sense ports. This eliminates any series parasitics along the current path, both on the Counter port side and the Work port side of the DUT.

This comes at the expense of requiring 4 connections including wires, contacts/electrodes and any other intermediate routing elements (e.g., switches). Each additional wire and input port can also introduce parasitic capacitance, especially noticeable at higher frequencies. Nonetheless, in many scenarios the 4-point-configuration yields the best measurement accuracy. This is largely due to the exclusion of series parasitics. In many cases these dominate the overall impedance in the current-carrying path and in simple two-port configurations often can largely degradate the sensitivity towards the actual DUT´s impedance. Still it is recommended to minimize lead and contact resistances to prevent issues with respect to the compliance voltage (in case of current excitation) or the sensitivity (in case of voltage excitation).



6.3.5 Practical Considerations

As a rule of thumb, if the electrode interface is NOT the investigation target (electrochemical interface studies or specific sensor types), in many cases the 4-point configuration is a good choice. In particularly challenging measurement settings, the correct method of determining the best configuration is to analyze the equivalent circuit structure of the overall measurement setup. From this, the effect of parasitic elements on the obtained impedance measurement can be estimated and the port configuration chosen accordingly.

> ⓘ Even if your device under test requires a 2-point connection it might still be advisable to route your measurement signals as a 4-point configuration as close to the measurement object as possible before shorting the ports to obtain a 2-point connection. This can, for example, be achieved through Kelvin clip probes.

On devices with separate C, R, W, and WS ports, 2- and 3-port configurations are achieved by shorting the WS and W ports and/or the C and R ports. This can be achieved through external adapters, using specialized clamps and probes or direct short circuit wiring between the instrument ports. On most Sciospec devices including multiplexers and adapters, this can simply be achieved through software-controlled settings. The signal routing is then accomplished internally removing the neccessity for additional external wiring.

	Number of connections	Measured voltage path	Eliminates parasitic wire impedance, connectors, double layer capacitance, etc.	Typical applications	Key tradeoffs
2-point	2	C ↔ W	no	high channel count, simple setups	lowest accuracy, simplest wiring
3-point	3	R ↔ W	Counter port side only	electrochemical studies	moderate accuracy, added setup complexity
4-point	4	R ↔ WS	Counter and Work port side	highest accuracy in most common settings, precision measurement	most complex setup, extra capacitance

6.4 Multiplexed versus true parallel measurements

There are two common concepts engineers encounter when signals from multiple channels or devices under test (DUTs) need to be acquired automatically: **multiplexed measurements** and **true parallel measurements**. While both approaches are valid, they differ significantly in architecture, performance, and application domain. This chapter highlights their key characteristics.

6.4.1 Multiplexed measurements

A **multiplexed measurement** setup consists of a single measurement unit (e.g., ISX-3 with multiplexer, LCR-3) that is sequentially connected to multiple channels by means of an electronic switch or multiplexer. The process works as follows: The multiplexer connects channel 1 to the measurement unit, a measurement is performed, then it switches to channel 2, and so on, depending on the configured number of channels.

Each transition from one channel to the next requires a finite switching time, typically in the range of microseconds to milliseconds depending on the hardware. For slow or stable signals, this delay is negligible. However, for dynamic signals it can result in measurement errors or missed events.

Furthermore, the switching elements (e.g., relays, FET switches) inevitably introduce additional parasitic impedances such as resistance, capacitance, and inductance. These parasitics can distort impedance measurements, especially at high frequencies, or limit the achievable accuracy.

In applications requiring high accuracy - such as precision resistance or impedance testing - a 4-wire (Kelvin) connection can be used. This technique eliminates the influence of contact resistances and parasitic effects of the switching elements on the actual measurement path, thereby improving accuracy even in multiplexed setups.

Typical applications for multiplexed measurements include:

- Automated testing of passive components (capacitors, resistors, inductors) in production environments
- Quality control routines with large numbers of relatively stable DUTs
- Long-term monitoring of slowly varying parameters (e.g., leakage currents, temperature-dependent resistances)
- Cost-sensitive test systems with many channels but moderate timing requirements

6.4.2 True parallel measurements

In contrast, **true parallel measurement** setups provide each channel with its own dedicated measurement unit. This allows all channels to be acquired simultaneously, without the need for switching or sequential scanning.

An important requirement in such systems is synchronization. To ensure that results from multiple channels are truly comparable, the measurement units must share a common reference clock or be triggered with a precise synchronization signal. Proper synchronization guarantees that all values represent the exact same point in time; an indispensable feature for transient or high-speed applications.

Typical applications for true parallel measurements include:

- Synchronized impedance or LCR measurements on multiple DUTs in research and development
- High-speed transient analysis, where time correlation between channels is essential
- Multi-channel sensor arrays (e.g., material testing, biomedical instrumentation)
- Characterization of components under dynamic operating conditions, such as power electronics or RF circuits
- Reliability and stress testing, where multiple DUTs must be monitored simultaneously in real time

6.4.3 Comparison

The following table summarizes the advantages and disadvantages of both approaches:

	multiplexed measurements	true parallel measurements
Advantages	• Cost-efficient • Compact systems • Suitable for stable/slow signals • Scalable to many DUTs	• Genuine simultaneity • Accurate time correlation • Reliable with transient or dynamic signals • No switching artefacts
Disadvantages	• Time-shifted results • Parasitic effects • Limited speed with many channels	• Higher cost • More complex hardware • Increased size and power consumption

6.4.4 Hybrid solutions

Some measurement systems combine the advantages of both concepts. For example, Sciospec's CSX-64 provides 8 fully independent, parallel impedance or electrochemical measurement channels plus a 64-port any-to-any multiplexing matrix.
The following table outlines the principle approaches of Sciospec's measurement devices:

device	measurement type	number of ports	number of modules
ISX-3 plus (internal/external) multiplexer	multiplexed	up to 128	
LCR-3	multiplexed	up to 128	
ISX-5 without multiplexers	truly parallel		up to 8
ISX-5 with multiplexers	hybrid	up to 64 per module	up to 8
CSX-64	hybrid	64	8
EIT	truly parallel	up to 256	
CardioEpiX	truly parallel	512	

6.5 Overcurrent / Overvoltage Detection

The device will send a system message (see Acknowledge messages in COMInterface documentation) if a overcurrent or overvoltage event was detected during a measurement.

-  Overcurrent Detected: Value of DC current on W-ports exceeds capability of configured current range
-  Overvoltage Detected: Value of DC voltage difference between R and WS port exceeds capability of configured voltage range
-  Overcurrent and Overvoltage Detected

6.6 Compensation

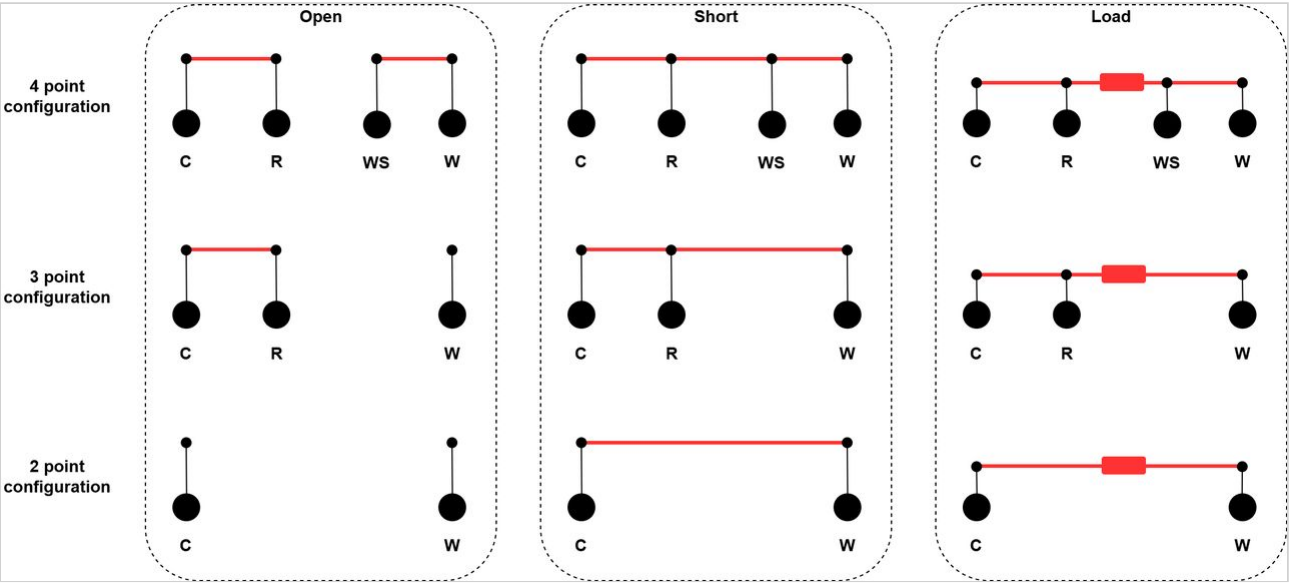
Impedance compensation can be performed to reduce the influence of parasitic effects on the measurement results. These effects include, but are not limited to, cable resistance, contact resistance, and stray capacitances, which may significantly affect measurement accuracy if not compensated.

After configuring a new measurement setup, it is recommended to execute the compensation routine prior to the first measurement. The user shall ensure that the appropriate calibration conditions are correctly established before starting the routine. The compensation is then carried out using an open–short–load procedure. During this process, the system sequentially measures the open, short, and load conditions in order to determine correction parameters for subsequent measurements. The routine is executed for each configured frequency point. Different types of compensation configurations can be realized by selecting appropriate port

configurations (Chapter "Compensation in the Software"). The calculated compensation data are stored and remain valid as long as the measurement setup remains unchanged. Any modification of relevant parameters, such as the frequency settings, port configuration, or cabling requires a new compensation.

If no compensation is performed, or if the compensation is no longer valid, the measurement results may be significantly affected by parasitic effects.

A schematic representation of the connection principles for 4-point, 3-point, and 2-point configurations is shown below.



6.7 Settings description

6.7.1 DC Bias

The "DC Bias" functionality allows to adjust the DC-offset-voltage across the measuring object. More precisely, the offset voltage between the R (reference) and Ws (work sense) ports will be adjusted. The DC Bias can be activated/deactivated and the desired value set from the software or using the Sciospec COMInterface. When the DC Bias is activated, the system regulates the DC Bias to the set value one single time. If the measuring section or the measuring object changes, this procedure has to be repeated to regulate the system to the new conditions. The DC Bias can be set between + 1 V and - 1 V.

Prerequirements for DC Bias adjustment

- The measurement object must be connected to the device before DC Bias adjustment is performed.
- A measurement setup must be set.
- No measurement may be active.
- Current direction: C = + , W = -

DC Bias is only available on the measurement port 1 (BNC). It is not available for the ExtensionPort.

The absolute value of excitation amplitude + DC Bias does not exceed 1V (limit of voltage measurement). |excitation amplitude + DC Bias| ≤ 1 V.

6.7.2 Precision

The precision settings enables the user to adjust the trade off between measurement stability (low standard deviation of measured impedance values) compared to high measurement speed. A low precision settings correspond to fast measurements with lower stability. High precision setting corresponds to greater stability at longer measurement times. The precision setting is directly correlated to the relative bandwidth of the measurement. The setting can be found in the measurement setup config panel in the software or alternatively in the set setup command (0xB6) of the Sciospec ComInterface.

For relation between the precision setting, accuracy, measurement timing and the range setting see Technical Specification chapter.

6.7.3 Measurement delay options and phase synchronization

There are different options to define delays during a measurement. Basically there are two possibilities. Either way the measurement pauses after a full sweep is completed (sweep delay, synchronization time) or after a specific frequency point has been measured (point delay). In the case of a none zero point delay, the next frequency excitation will be used and before the actual measurement starts the defined time will elapse. Both parameters can be set in the software or by a corresponding command via USB or Ethernet (see Command description command 0xB9 and command 0xB6).

The optional phase synchronization feature ensures that change of frequency in the excitation signal, which is exerted on the device under test, with no change in phase and without any glitches. This is for example needed for precise measurements of the impedance of resonant devices under test. The disadvantage of this option is that a longer time for the measurement is needed (see Command description command 0xB6).

Phase synchronization

The phase synchronization feature is only available for excitations frequencies greater than 100 kHz.

6.8 Master Interfaces

6.8.1 Full Speed USB

The device creates a virtual COM port named "serial USB device", which can be used like a serial port.

Any COM port obtains a unique number by the operating system (e.g., "COM6"). Which number is assigned to a certain port is decided by the operating system and cannot be altered.

The port can be used either to connect the Sciospec software or any terminal program to the device. There is no need to configure the general settings for the serial port (e.g., baud rate, stop bit count, etc.). Any setting of your terminal program will work.

To connect the Sciospec software to the virtual COM port, choose "serial" in the connection dialog and then choose the desired port.

Driver installation

For Microsoft Windows 8 or more recent versions, there is no need to install any driver to connect to the device. After establishing a USB connection between the device and a PC, the new serial COM port will be available.

For Microsoft Windows 7 or XP, the driver named "Sciospec FS USB driver" provided in the Sciospec software package has to be installed before connecting the device to the PC. If you encounter a compatibility or verification warning during installation, choose to continue and install the driver anyway. When the installation has finished, you can connect the device to your PC. The new serial COM port will then be available.

Finding the correct COM port

If more than one COM is active on the PC, you have to determine which one belongs to the Sciospec device. An overview of known COM ports is given by the Windows device manager (shortcut for Windows 8 or more recent: 'WIN + X') under "Ports (COM & LPT1)". Plugging in the USB cable while observing the port list shows which number is assigned to it.

Example: C-Code for establishing a connection

```
Handle hSerial = CreateFile(  "\\\\.\\COM6",
                              GENERIC_READ|GENERIC_WRITE,
                              0,
                              NULL,
                              OPEN_EXISTING,
                              FILE_ATTRIBUTE_NORMAL,
                              NULL );
          if (hSerial == INVALID_HANDLE_VALUE)
          {
                  fprintf(stderr, "Error\n");
                  return 0;
          }
            return 1;
```

Ethernet

The Sciospec device uses a bidirectional socket communication in which it implements the socket server and the connected PC the socket client. It is delivered with a ready to use ethernet configuration. It supports DHCP to receive an IP address from a DHCP server. Since DHCP is enabled by default, the device will obtain an IP address if you connect it to your ethernet router and can instantly be operated via this interface. This is the default setting

but can be deactivated using the software or the “set Ethernet configuration” command. When DHCP is deactivated a static IP address must be set by the user. All Sciospec devices use the following port.

port	5000

The Sciospec device also can identify itself in a local area network by answering a UDP broadcast call. It answers with its device id. From this answer you may obtain the IP address by checking the UDP package. The broadcast message to be send is the "Get Device ID" command and needs to be send on the following port

port	8888

Example: Establishing a connection using the COMinterface

1. Connect the device Ethernet network and via USB to your computer.
2. Open terminal program and connect to the USB port of the device. (see "Setting up a connection to the device via USB")
3. Send the "get IP address" command **[BE] [01] [01] [BE]**.
4. The device will return the IP address, which it received from the DHCP server. (See command description for return syntax or disabling DHCP.)
5. Disconnect the terminal programm.
6. Use the obtained IP address and the port to establish a socket connection from your software to the device.
7. From now on the device can be controlled by regular COMinterface commands.

Example: C-Code for establishing a connection

```
    int sock;

/**This block is needed only if using WinSocket (Windows) to initialize the socket*/
    printf("\nInitialising Winsock...\n\n");
    WSADATA wsa;
    if (WSAStartup(MAKEWORD(2,2),&wsa) != 0){
        printf("Failed. Error Code : %d",WSAGetLastError());
        return 1;
    }
/** End of WinSocket initializing*/


    if( (sock = socket(AF_INET, SOCK_STREAM , 0)) == INVALID_SOCKET ){
         printf("Could not create socket");
         return -1;
    }

    /*Initialize address of device
     * IP address needs to be adapted, depending on local network and Ethernet
settings of the device*/
```

```
    struct sockaddr_in server;
    server.sin_addr.s_addr = inet_addr("192.168.100.115");
    server.sin_family = AF_INET;
    server.sin_port = htons( 5000 );

    //Connect to remote server
    if (connect(sock , (struct sockaddr *)&server , sizeof(server)) < 0){
        printf("connect error");
        return -1;
    }
```

Example: Java-Code for UDP broadcast call

```
String host = "255.255.255.255";    //Broadcast address
int port = 8888;                    //the destination port of the broadcast
char data[] = {0xD1, 0, 0xD1};      //bc char is unsigned
String message = new String(data);

try{
    InetAddress adds = InetAddress.getByName(host);
    DatagramSocket ds = new DatagramSocket();
    DatagramPacket dp = new DatagramPacket(message.getBytes(), message.length(),
adds, port);
    ds.send(dp);
    ds.close();
}
catch (UnknownHostException e){
    e.printStackTrace();
} catch (SocketException e) {
    e.printStackTrace();
} catch (IOException e){
    e.printStackTrace();
}

byte[] buf = new byte[1024];    //store the message sent

try {
    //Bind port
    DatagramSocket ds = null;
    DatagramPacket dp = null;
    ds = new DatagramSocket(port);
    dp = new DatagramPacket(buf, buf.length);

    System.out.println("The listening broadcast port is open:"+port);

    ds.receive(dp);
    ds.close();

    System.out.print("Recieved Msg:");
```

```
    for(int j=0;j<dp.getLength();j++) {
         System.out.print(" "+String.format("%02X", buf[j]));
    }

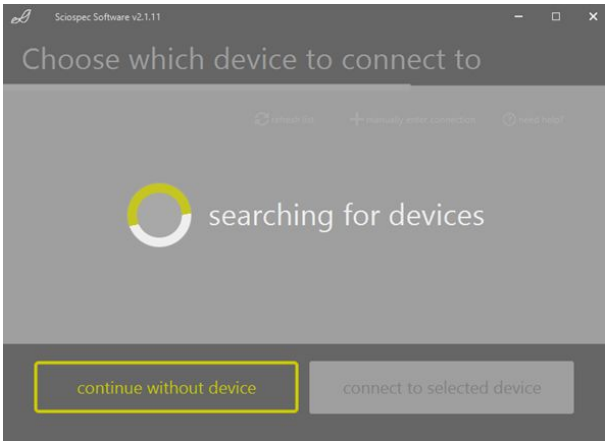
    System.out.println("");
    System.out.print("Received from:"+dp.getAddress());
}
catch (SocketException e) {
    e.printStackTrace();
} catch (IOException e){
    e.printStackTrace();
}
```

7 Measurement Software Description

7.1 Establishing Connection

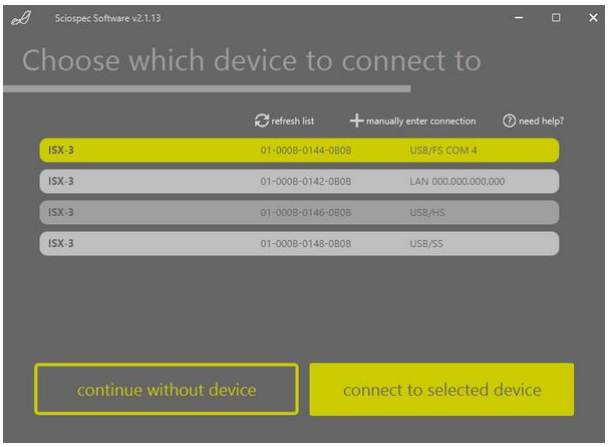
Once the software is started, the connection dialog appears. Please make sure, that the device is powered and connected to the computer by one of the available connection interfaces.

The connection dialog is shown in the image below.

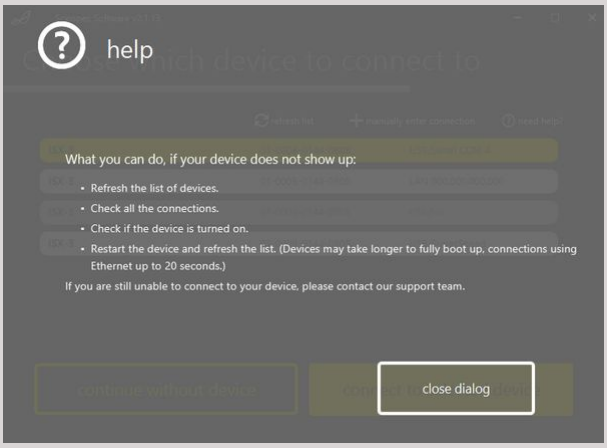


In this scenario, the software will automatically search for available devices on all available interfaces, the dialog box will afterwards display a list of available devices. Each device will be listed with its serial number and connection type (LAN, USB FS (FullSpeed), USB HS (HighSpeed), or USB SS (SuperSpeed)).

To enhance visibility, the selected device will be highlighted in green within the dialog box as soon as it appears. Additionally, the first entry in the list will be pre-selected by default. To proceed, simply click the "connect to selected device" button.



In case the desired device does not appear in the list, we have designed a convenient solution: By clicking on the "need help?" button within the dialog box a dedicated screen presenting you a range of potential solutions to address the issue will appear. This resourceful screen will provide you with instructions and resolve connectivity concerns or device visibility issues you might encounter.



After highlighting a device, you may press the "connect to selected device" button. The main window of the software will open up.

Don't connect

You also can run the software without an actively connected device. This way, you may visualize and analyze previously recorded data. To do so, click the "continue without device" button.

7.2 Home Screen

The software will be opened in the home screen as displayed in the figure below.



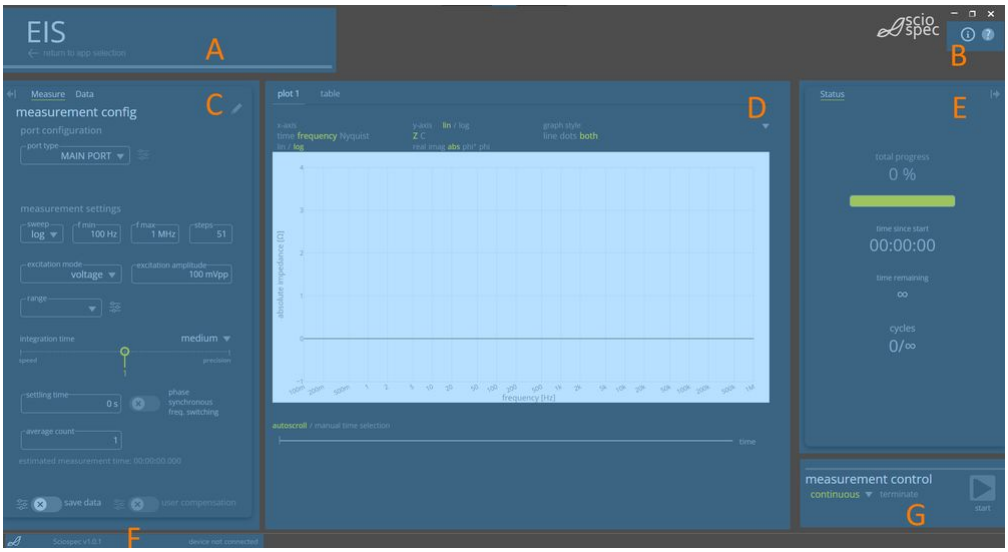
On the home screen, users are provided with a collection of applications (apps) displayed by individual icons. Each app provides a unique feature set for executing a usecase or running a specific measurement. A summary of the functionality is presented in an info box once an app is selected.

> ⓘ Available apps can differ depending on the connected device.

To launch an app, simply click on its icon and then select "launch app".

7.3 App Layout

All apps follow a common layout. The major elements are marked in the picture below.



- A - App name and link to home screen
- B - Help and information
- C - Configuration and data management panel
- D - Visualization pane
- E - Status panel
- F - Device status pane
- G - Measurement control pane

The panels "configuration and data management" and "status" are collapsable and the visualization pane gets increased in size accordingly.

7.3.1 User interface elements

The following user interface elements are used throughout all apps:

	toggle switch

	settings button
	collapse button
	backwards button
	information button
	help button
	edit button
	add button
	delete button

7.3.2 App name and link to home screen

The app name is presented. Below the app name, the user can click the backwards button to return to the home screen and select another app.

7.3.3 Help and information

Information dialog

Clicking the information button opens an information dialog. See the picture below for an example.



The information that is presented is devided into

- device information,
- licence information (software),
- contact information.

Device information

All relevant information are shown to the user. This includes information about the general information such as "Device Name" and "Device ID" as well as status information about the device connection. Additionally, information about the device firmware as well as the software version are presented to the user.

Licence information

A list of all used software packages and their respective licence is given.

Contact information

Information of how to get in touch with Sciospec Scientific Instruments GmbH is listed.

Help button

Clicking the help button opens the user manual of the connected device.

7.3.4 Configuration and data management panel

The panel can consist of multiple tabs.

The "Measure" tab holds all parameters needed to configure a measurement with the selected app including measurement settings and port configuration settings. Provided settings vary among the different apps and are dependent of the connected device. This is because not all devices share an identical feature set as, for example, port configuration options.

The "Data" tab provides data management features to the user. This includes data import and export functionality.

ⓘ This panel is collapsable.

7.3.5 Visualization pane

In the center of the overall layout the visualization pane is placed. Recorded measurement data are presented to the user. The way data is presented depends on the intended use case of the app. Different ways of visualization are accessable by the tab structure.

7.3.6 Status panel

This panel holds information about the general status of a running measurement, such as a progress indicator or the remaining measurement time.

> This panel is collapsable.

7.3.7 Device status pane

At the bottom of the screen, information about the device status are presented. It is shown, whether the device is connected or disconnected. In addition to that, the software version is shown. This pane is shown in all apps.

7.3.8 Measurement control pane

The measurement control pane holds all the elements needed to start and stop a measurement. This pane is available in all apps. A detailed description of the provided functionalities is given in the section "General information" of the software description.

7.3.9 Edit and extended settings

Different parts of the software can be edited such as labels of measurements. The edit functionality is indicated and can be accessed by the edit button.

Some aspects have extended setting options. Those can be accessed by selecting the settings button, which is placed next to the parameter with available extended options.

7.4 General information

7.4.1 Running a measurement



To start a measurement, select the start button placed in the bottom right corner of the screen.

Depending on the connected device, different measurement control settings are available to the user:

- continuous,
- single,
- interval.

Continuous mode

The configured setup will be measuremed continuously by starting over and over again until stopped manually by clicking the start button again. Alternatively, the user can specify a fixed number of measurements or a duration. After reaching either the specified number or the duration, the measurement will be stopped automatically.

Single mode

In single mode the configuration will be measured once.

Interval mode

The user can specify a time interval between consecutive measurements. Once the interval time is exceeded, the configured measurement is started again. Similar to the options available for continuous mode, either a fixed number of measurements or a duration can be specified. After reaching either the specified number or the duration, the measurement will be stopped automatically.

7.4.2 Visualization

Data is visualized in a standardized plotting canvas. Axes settings, such as labels and options to customize, strongly depend on the app in use.

On the top right is a menu with different selectable options presented to the user. The attaching table displays a rough overview.

Menu item	Function
	autoscale
	reset axes
	rectangle zoom
	pan
	export screenshot
	export data

Zooming action

By default "rectangle zoom" is activated. Therefore, drawing a rectangle on the plotting canvas with the left mouse button clicked will reset the axes of the plotter to the selected area. Drawing a horizontal or vertical line will only zoom in the *x*-axis or *y*-axis, respectively, while the other axis is kept the same.

Clicking the autoscale button will result in automatically setting the axes minimum and maximum values to be the minimum and maximum values of the displayed data. The reset axes button resets the scales to ±10% of the maximum/minum value of the displayed data. Switching between the autoscale view and the ±10% view can also be done by a double click onto the plotting canvas.

Pan action

To move the plotting canvas without resetting the scaling of the axes, select the pan tool. By clicking onto the canvas with the left mouse button, the canvas can be moved towards the wanted direction. This action is also accessible when the rectangle zoom action is selected by using the right mouse button.

Export

The currently shown plotting canvas can be easily exported to an image file by clicking the export screenshot button. Subsequently, a directory and filename can be specified. In addition to that, the displayed data can be exported to an .xlsx-file by clicking the export data button. Once more, a directory and filename can be specified.

Legend

A legend with labels of displayed datasets is shown in the top right corner of the plotting canvas. To exclude a dataset from being shown, click on the respective item of the legend. To show the dataset again, click on the dataset again.

7.4.3 Data management

The „Data“ tab of the configuration and data management panel provides access to data management features. A list of data sets is presented. Datasets can be exported by selecting the export button. Additionally, datasets can be imported by selecting the import button.

Data file format

Sciospec's native data format uses the HDF5 framework. The data file extension is .scio5. A data file holds all information of a conducted experiment, which can consist of multiple different measurement configurations or experimental steps. Each experimental step consists of

- a setup group as well as
- a data group.

The setup group holds all relevant information of the performed measurement. This includes metadata such as

- device information,
- software version,
- execution date,

as well as all the settings needed to parametrize the conducted measurement again. These settings depend on the chosen app.

The data group holds all recorded measurement data. The type and structure of data depends on the conducted measurement or chosen app.

For further information about the HDF5 format please refer to https://www.hdfgroup.org/solutions/hdf5/.

"Autosave" feature

Autosave allows to automatically save the measured data to a previously selected folder. To use the autosave feature, select the save data toggle switch and specify the saving directory. For changing the directory, select the settings button next to the toggle switch. Once a measurement is started, a subfolder with the name of the measurement configuration will be created and all data will be saved there.

The major advantage of this feature is that the required program memory will not increase significantly over the runtime of the experiment and all measured data is stored to the hard drive. A fault condition will not compromise the already measured data. In case the “Autosave” feature is not active, the measured data will only be held in the program memory.

ⓘ It is highly recommended to use the “Autosave” feature when performing long experiments.

7.4.4 Unit Prefix

All parameters can be entered with a unit prefix. Valid unit prefixes are listed in the table below.

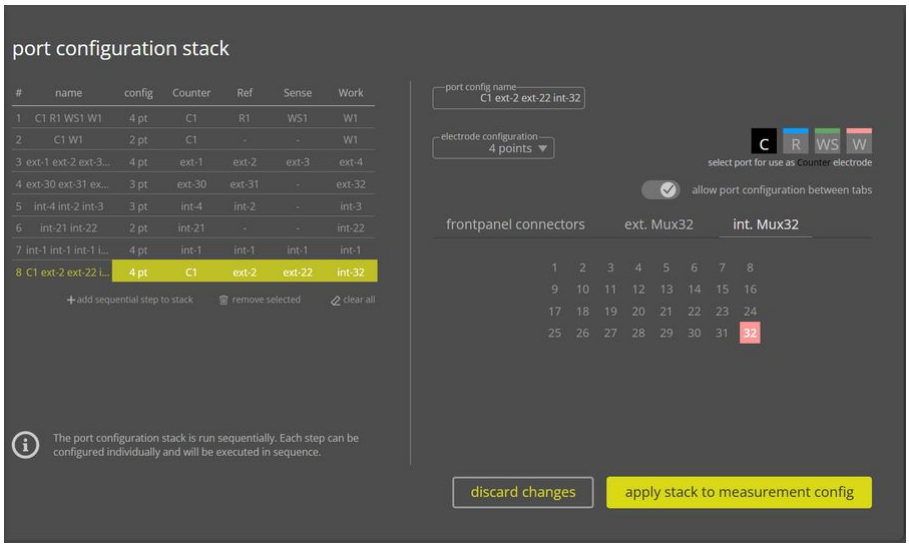
Prefix	Symbol	Power
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
milli	m	10^{-3}
micro	µ / u	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

7.5 Port Configuration in the Software

Many devices developed by Sciospec support automated measurements across multiple channels. While selecting “MAIN PORT” in the “Port Type” dropdown field configures a default 4-point measurement using the four BNC

connectors on the front panel of the connected device, more advanced channel selection features are available in “Advanced” mode.

Selecting this option enables the settings button adjacent to the dropdown field. Activating this button opens the port configuration tab within the software (see attached image). This tab is divided into two main sections: the port configuration stack and the port selection.



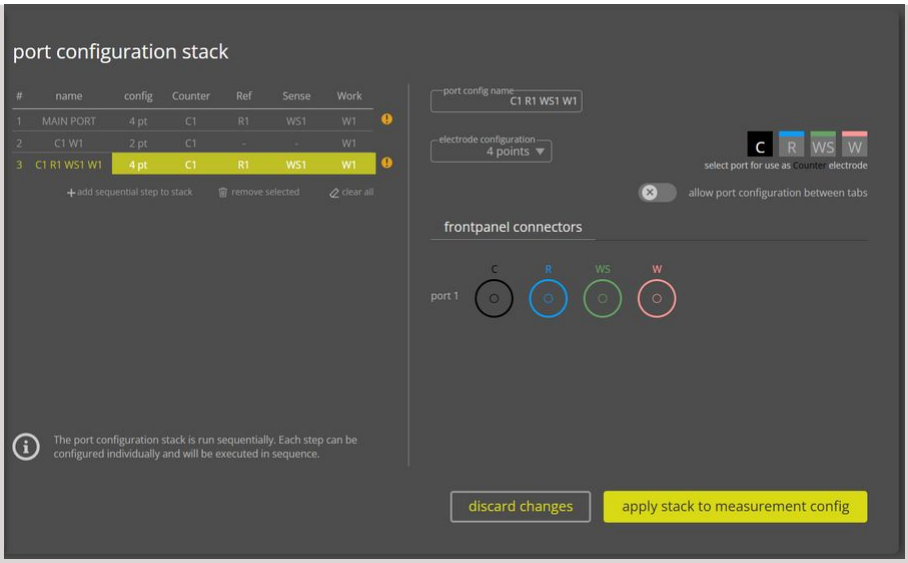
7.5.1 Port configuration stack

The port configuration stack is displayed as a table. Each row represents a step (i.e., a valid port configuration) that is executed sequentially during a multiplexed measurement process. The columns indicate the respective step number, the selected electrode configuration, and the assignment of electrodes to Counter, Reference, Sense, and Working. Detailed information on these functionalities is provided in “Port Configurations” in Chapter “Function Description”. Additional steps can be added by selecting “add sequential step to stack”, while a selected step can be removed using “remove selected”. The entire port configuration stack can be cleared by selecting “clear all”. Selecting a specific step enables its individual configuration.

The upper image shows an example of a port configuration stack consisting of eight steps. Various configurations are illustrated to demonstrate the extended channel configuration capabilities of a device equipped with a main port, a 32-channel external multiplexer, and a 32-channel internal multiplexer.

The first six steps present typical measurement configurations, including 2-point, 3-point, and 4-point setups. Step 7 shows a configuration in which all four electrodes are assigned to a single port, resulting in a short circuit. Step 8 demonstrates a configuration in which a 4-point measurement is distributed across different connector types (main port, external multiplexer, internal multiplexer). This configuration is enabled by activating the “Allow port configuration between tabs” toggle switch.

ⓘ If a port configuration is selected more than once, the software indicates this issue with an orange exclamation mark. Hovering over the exclamation mark displays a notification informing the user of a duplicate channel selection.



7.5.2 Port selection view

The port selection view provides all controls required to configure a single step within the port configuration stack:

- The name of the selected step is displayed in the “port configuration name” input field. The user may assign a custom name by editing this field.
- The electrode configuration type can be selected as 2-point, 3-point, or 4-point in the "electrode configuration" dropdown field. Depending on the selected configuration, the electrode functionalities Counter (C) and Working (W) and/or Reference (R) and Sense (WS) are displayed as color-coded boxes.
- The available connectors are listed below. The presentation of these connectors depends on the connected device. For example, the LCR-1 displays only the front panel connectors, whereas the ISX-3, when equipped with internal and external multiplexers, additionally provides tabs for the internal multiplexer (including the corresponding number of channels) and for an external multiplexer with its respective channel count. The assignment of electrode functionalities is performed by selecting the corresponding connector. The assignment process automatically begins with the Counter electrode. Once a valid selection has been made, the software proceeds automatically to the next electrode functionality. This progression is indicated by highlighting the electrode functionality box that is to be configured next. Alternatively, the user can manually select a specific electrode functionality by clicking on the corresponding colored box.

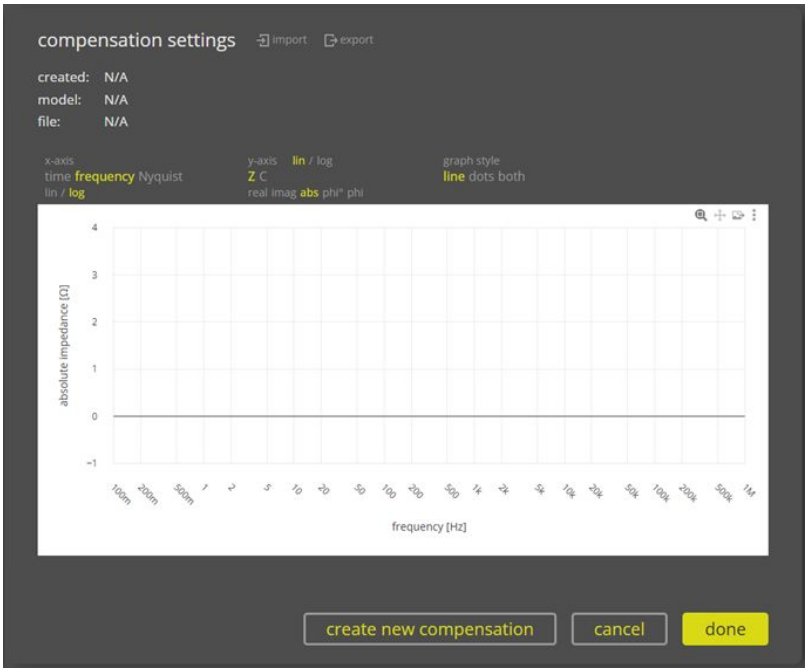
The configured settings can be confirmed by selecting “apply stack to measurement config”. To discard the new configuration and revert to the previously applied settings, select “discard changes”.

ⓘ Port configurations can also be defined across different tabs. For example, the Counter electrode can be assigned to a front panel connector, while the Reference electrode as well as the Working Sense electrodes are assigned to a port of the external multiplexer, and the Work electrode finally to a port of the internal multiplexer in a 4-point configuration (see step 8 in the first image of this chapter for an example). To enable this functionality, activate the “allow port configuration between tabs” toggle switch.

7.6 Compensation in the Software

Impedance compensation in the software is performed using an open–short–load procedure to reduce the influence of parasitic effects on the measurement results. These effects include, but are not limited to, cable resistance, contact resistance, and stray capacitances, which may significantly affect measurement accuracy if not compensated.

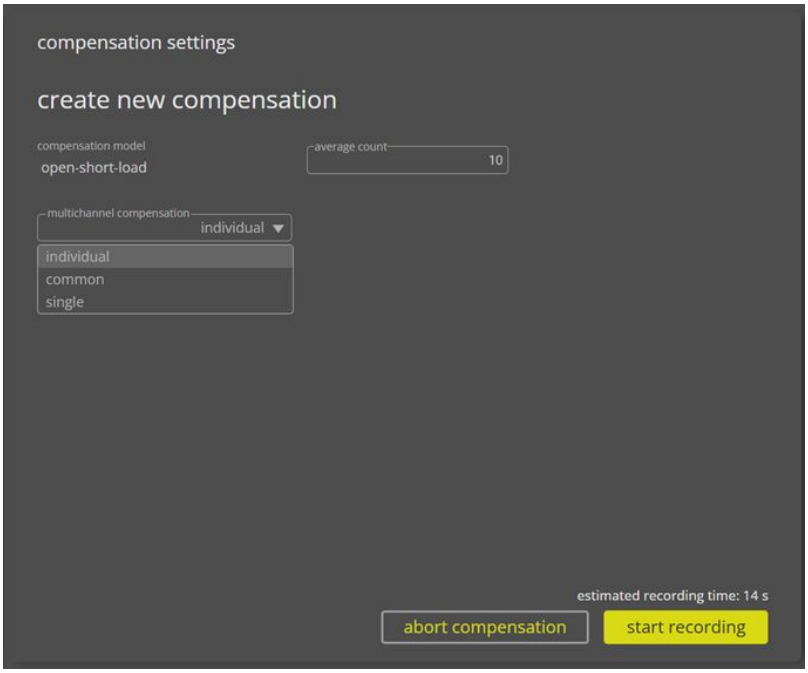
The compensation functionality within the software is implemented using different compensation types. Selecting the settings button located in the lower-right corner of the configuration panel (next to the "save data" toggle switch") opens the compensation settings, as illustrated in the image below.



Previously recorded compensation data are visualized within this menu. In addition, the user can export these data or import previously recorded compensation datasets.

The controls located above the plot and in the upper-right corner of the plotting area allow interaction with the displayed data for improved visualization. Furthermore, selecting entries in the legend enables individual compensation spectra to be shown or hidden.

If new compensation data need to be recorded (e.g., due to changes in measurement settings), the “create new compensation” button shall be selected. This action opens a new tab, as illustrated in the following image.



The available compensation types can be selected via the “multichannel compensation” dropdown field. The individual types are described in detail below. Upon successful completion of the compensation routine, the user can define whether the compensation shall be automatically applied to the measurement data by enabling or disabling the “use compensation” toggle switch. The user can return to the most recently recorded compensation spectra by clicking the respective settings button in the configuration panel.

ⓘ Compensation cannot be performed if the “current range” in the “measurement settings” of the configuration panel is set to “auto”.

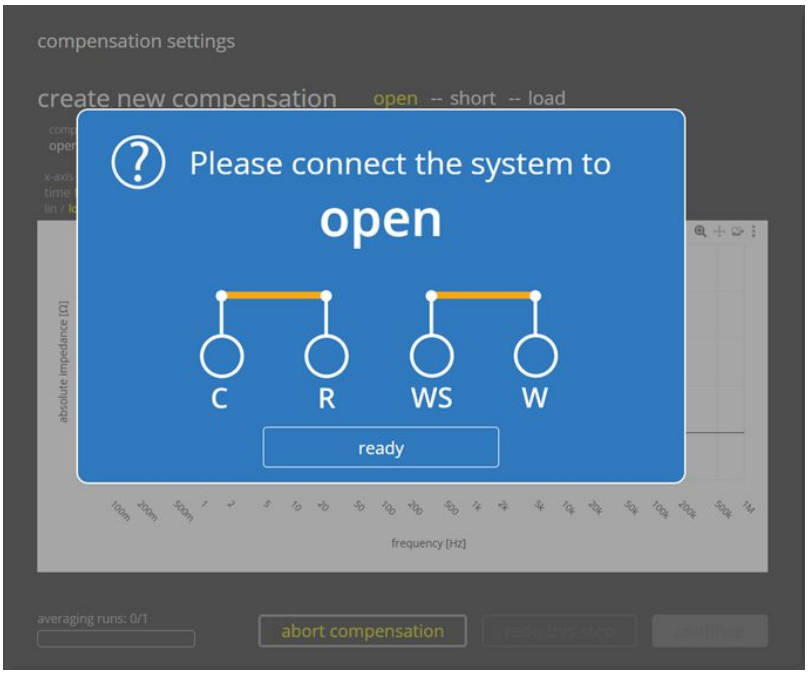
ⓘ For accurate compensation data, the “average count” should be set as high as reasonably possible while maintaining an acceptable “estimated recording time”. In particular, the open spectrum may exhibit significant fluctuations; therefore, averaging is recommended to obtain stable and reliable results.

ⓘ Since the LCR app performs measurements at a single frequency, the compensation data are displayed in tabular form rather than as a plot.

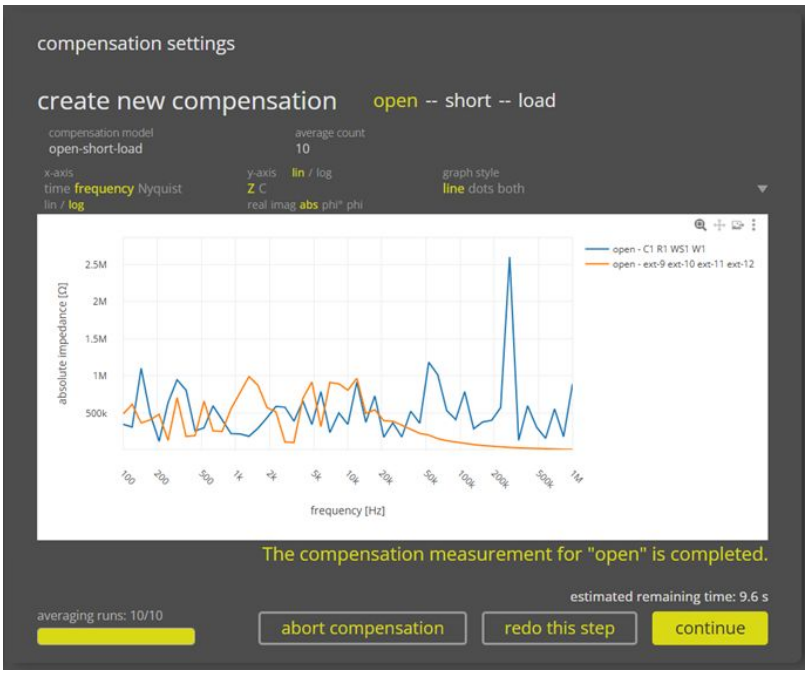
7.6.1 Multichannel compensation: individual

The multichannel compensation type “individual” can be used when each configured channel is to be compensated separately. In this case, every configured channel records its own open spectrum, short spectrum, and load spectrum individually. The software guides the user through the compensation process step by step, as illustrated in the following procedure:

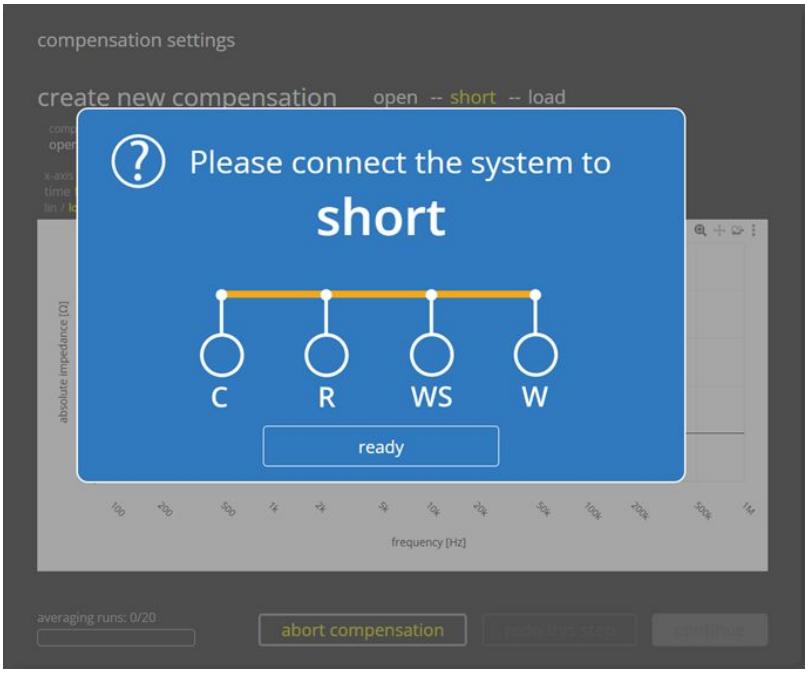
1. After selecting individual compensation and clicking the “start recording” button, a pop-up window appears prompting the user to prepare the setup for an open measurement.



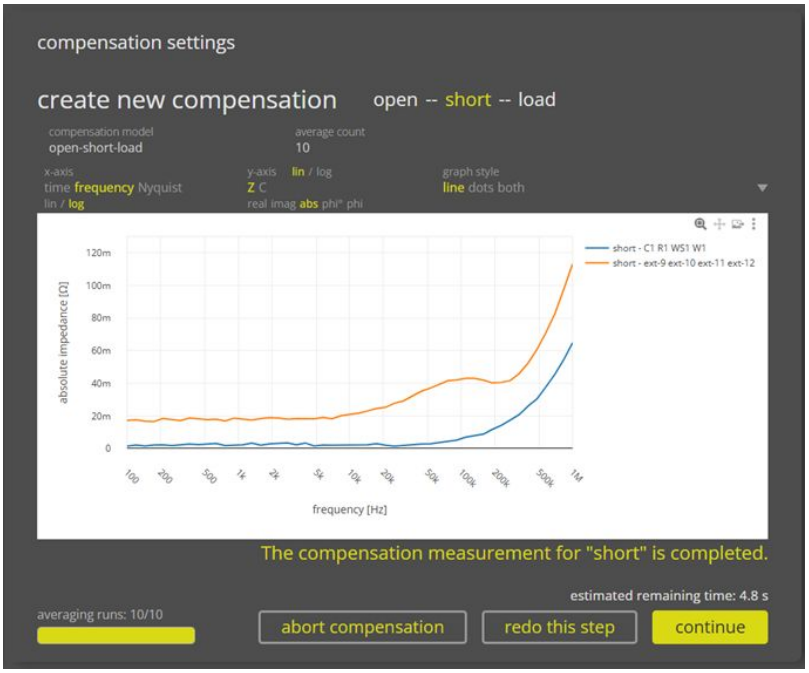
2. In this step, the user prepares all configured channels for an open measurement. Once the setup is complete, clicking “ready” records the open spectrum data, which is then displayed as a plot for each configured channel. The example image shows the open spectra for the main port (blue curve) as well as for a channel combination on an external 32-channel multiplexer (orange curve).



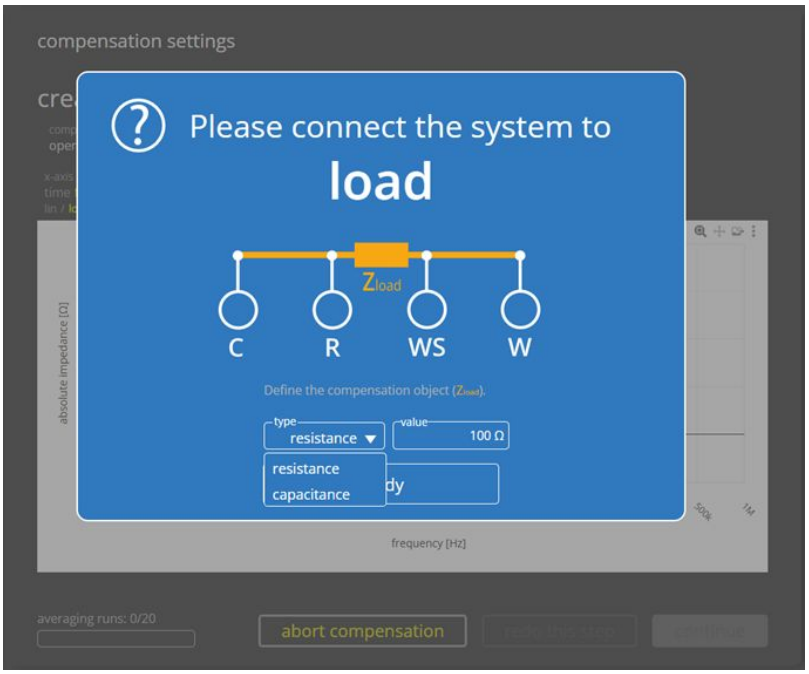
3. If the user is not satisfied with the result, the step can be repeated by clicking “redo this step”. Otherwise, the user proceeds by clicking “continue”.
   Repeating the step reopens the pop-up window for the open measurement as described in Step 1.
   Proceeding opens a new pop-up window prompting the user to prepare the setup for the short measurement (see following image).



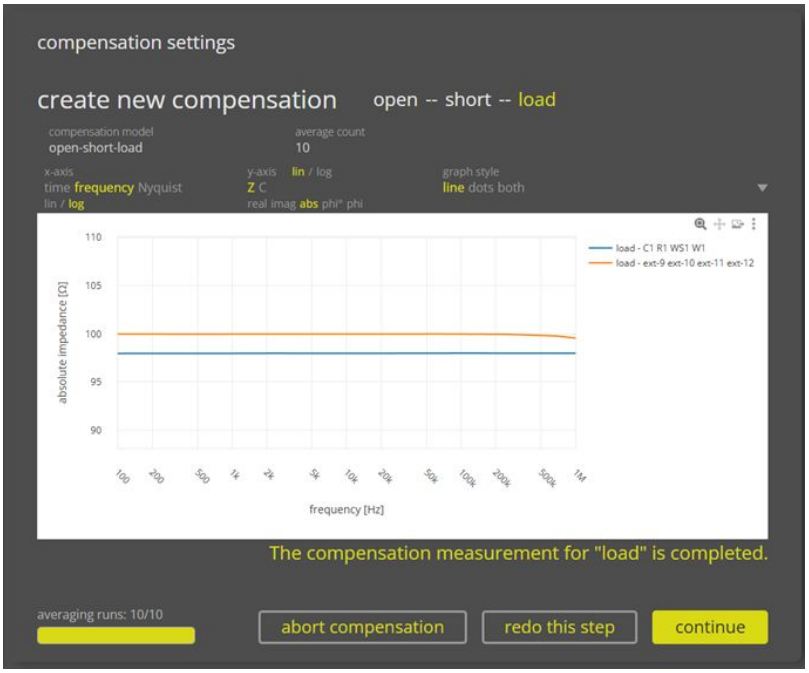
4. As in the open measurement step, the recorded data is displayed as a plot (see example image).



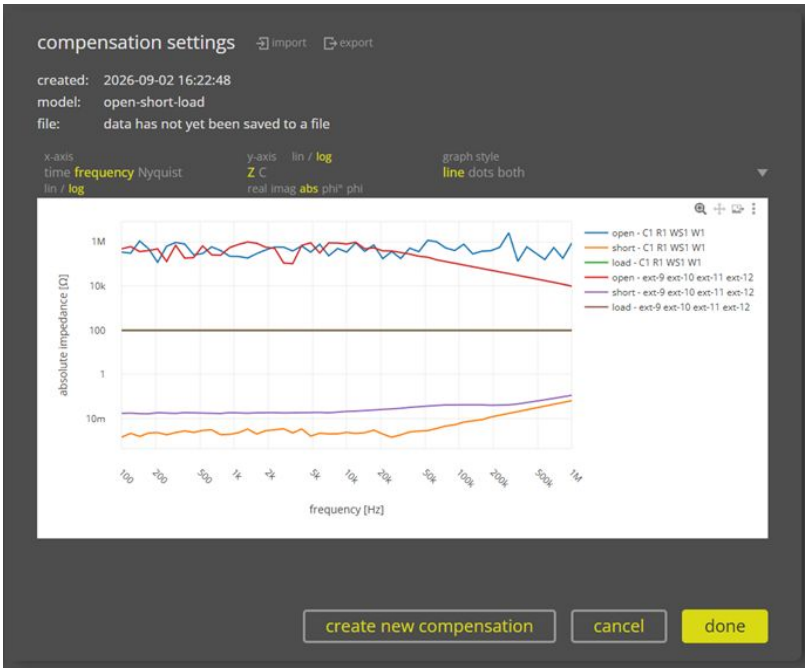
5. The user can either repeat this step or proceed to the load measurement.
   Selecting “continue” opens a new pop-up window prompting the user to prepare the setup for the load measurement. At this stage, the user must additionally specify whether the load is a resistance or a capacitance and enter the corresponding value (e.g., 100 Ω), as shown in the following image.



6. After completing the setup, the user clicks “ready” to record the load spectrum. The recorded data is then displayed as a plot (see following image).



7. After confirming this step, the compensation process is completed. All recorded spectra are displayed together. Using a logarithmic scale for the y-axis can improve the distinction between the individual spectra (see example image).



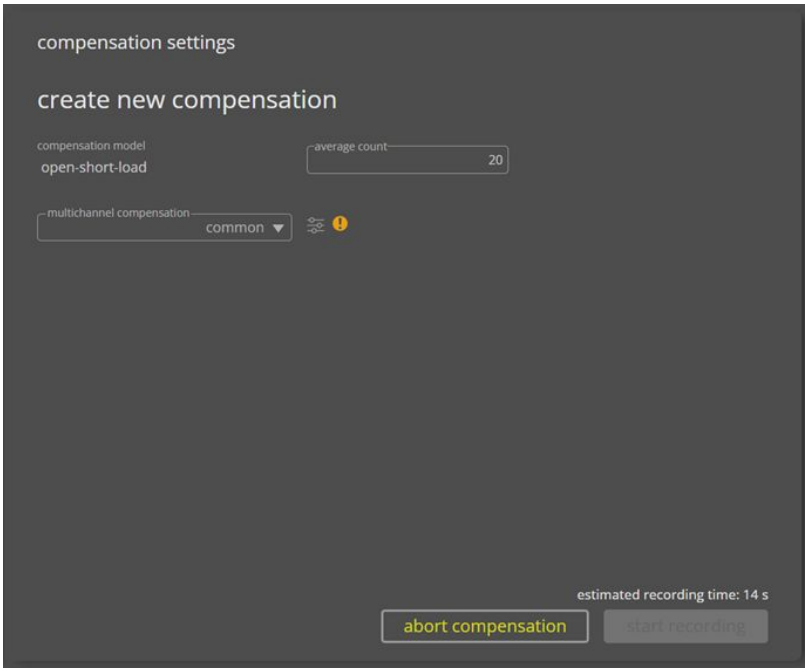
After completing the compensation procedure, the user can export the data for further processing, interact with the plot to improve visibility, or save the data directly from the plot view. If the compensation results are satisfactory, the process can be finalized by clicking the “done” button.

7.6.2 Multichannel compensation: common

In the multichannel compensation type “common”, the user assigns one channel for the open measurement, one channel for the short measurement, and one channel for the load measurement. Unlike the “individual” compensation procedure, all three spectra are recorded in a single step after the channel assignment. The recorded data is then applied to all channels in the port configuration stack.

> ⓘ The channels defined in the port configuration stack may differ from those used for the common compensation. This represents a key difference compared to the direct process used in the “individual” compensation procedure.

This type of compensation is implemented in the software as follows: After selecting “common” under “multichannel compensation”, a settings button and an orange exclamation mark appear next to the dropdown field, indicating that the channel configuration must be adjusted for this compensation process (see attached image).

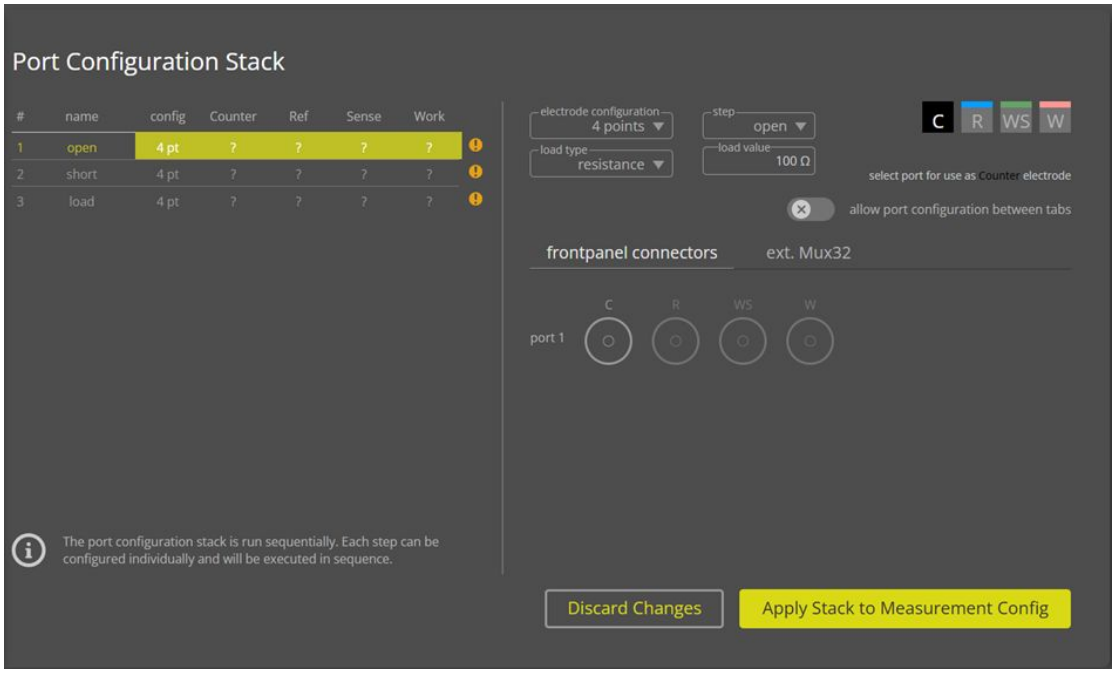


Clicking the “settings” button opens a tab that is similar in layout and functionality to the port configuration tab. By default, the port configuration stack contains three steps: one for the open spectrum, one for the short spectrum, and one for the load spectrum (see Chapter “Port Configuration in the Software” for more details). Initially, none of these steps are configured and are therefore indicated by an orange exclamation mark. Selecting a step allows the user to configure it by choosing from the available channel combinations (e.g., front panel connectors or channels of an external multiplexer). Note that the port selection view for common compensation differs slightly from the measurement channel selection view. In addition to the “electrode configuration” dropdown (for selecting 2-point, 3-point, or 4-point measurements), three further input fields are available:

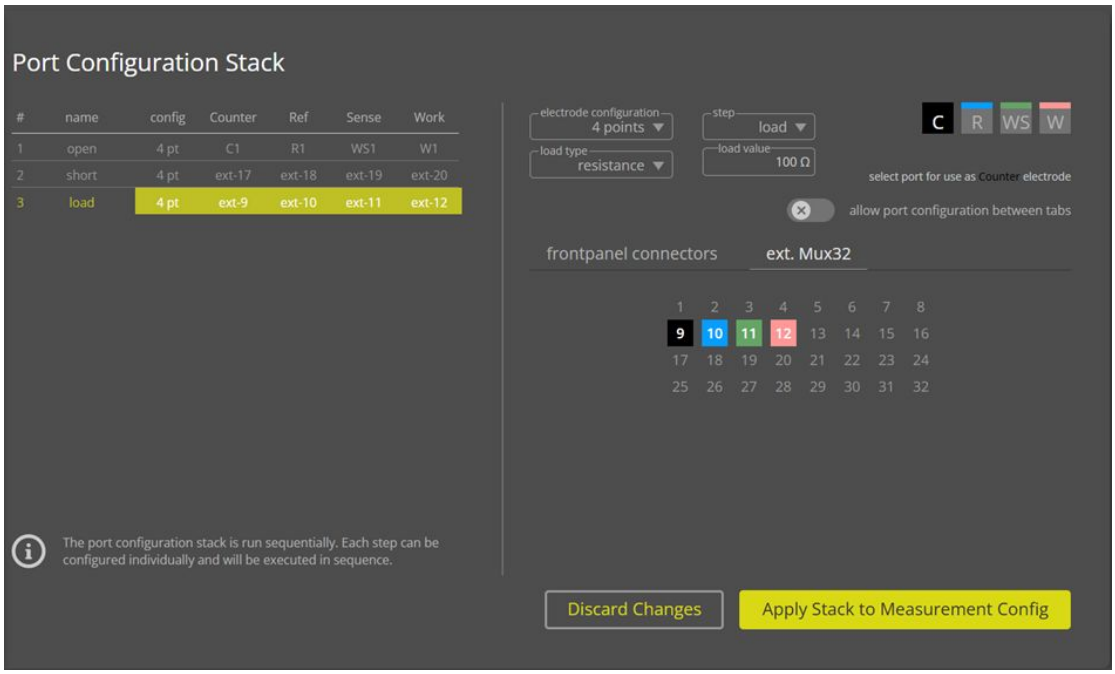
- “step”: Allows switching between the different steps in the port configuration stack (e.g., to modify channel assignments).
- “load type”: Specifies whether the load is a resistance or a capacitance.

- “load value”: Defines the theoretical value of the selected load.

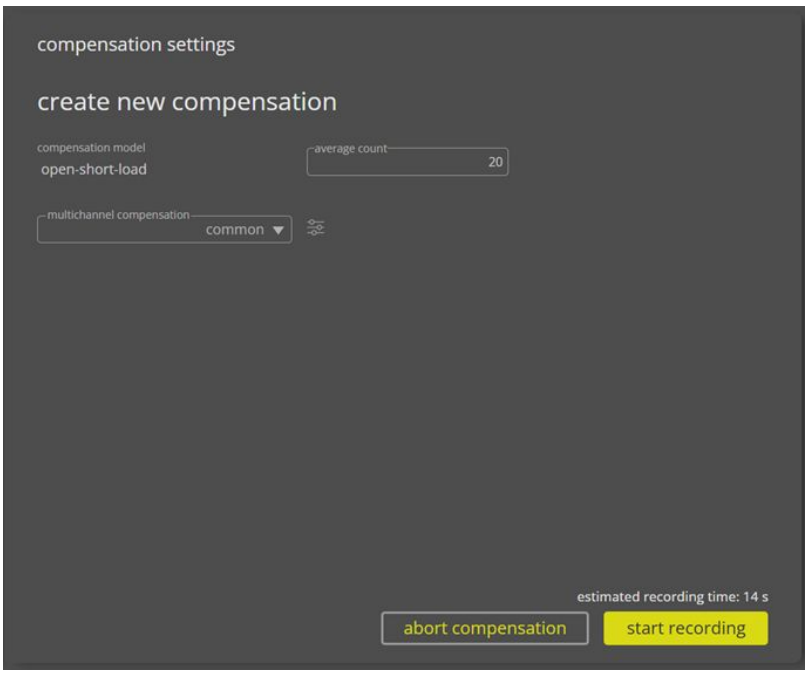
The “load type” and “load value” fields are only relevant for the load measurement step.



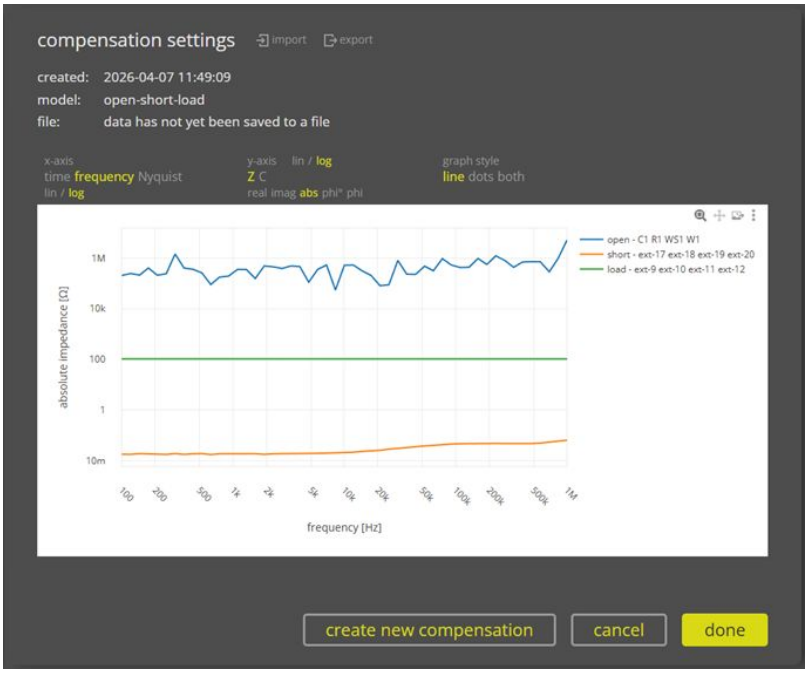
Once the three steps are properly configured, the orange exclamation marks disappear and the channel selection can be confirmed by pressing "Apply Stack to Measurement Config".



This action closes the port configuration tab and returns the user to the compensation settings tab. The “start recording” button is now enabled. Clicking this button records all three compensation spectra automatically.



Once the spectra have been recorded, they are displayed in the same manner as for the “individual” compensation type, with all options available as described above.

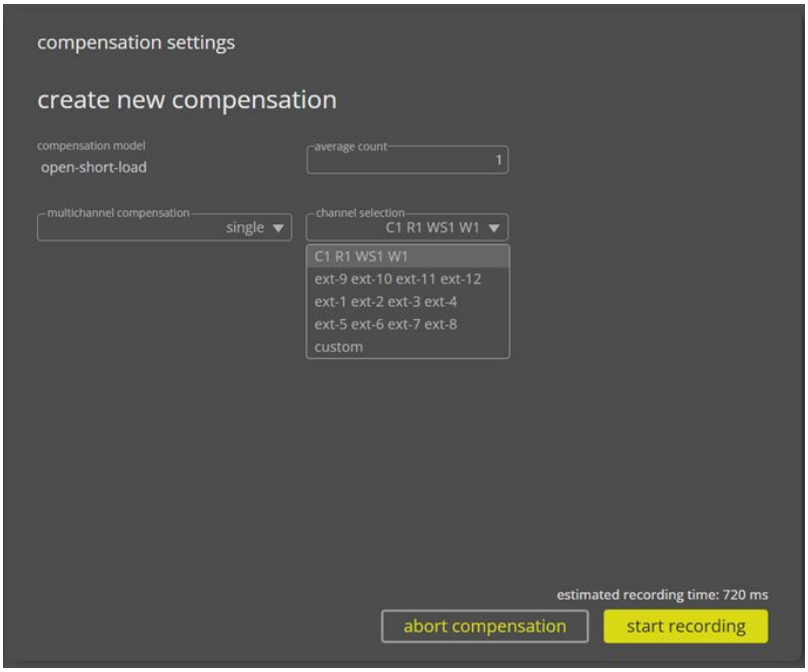


7.6.3 Multichannel compensation: single

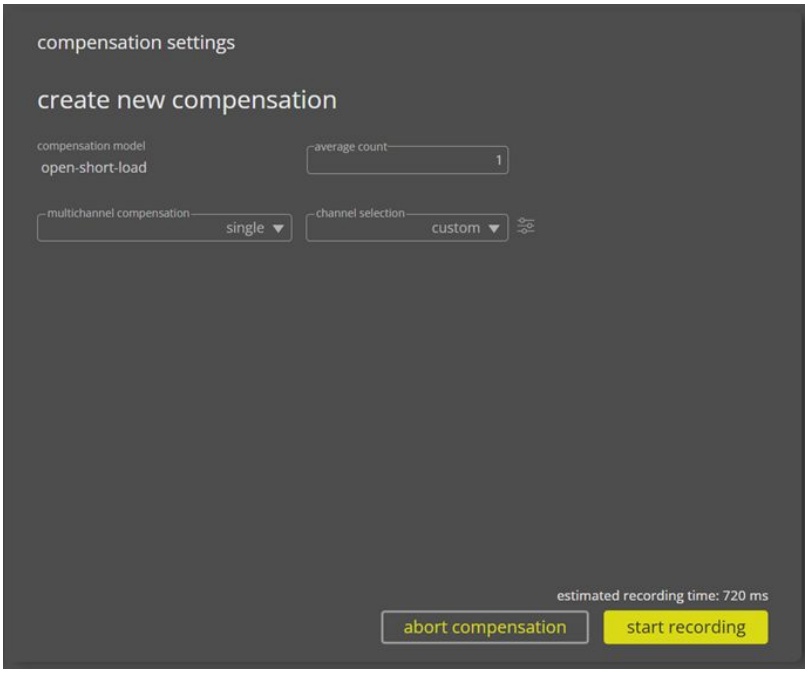
The multichannel compensation type “single” is similar to the “individual” compensation type. However, the open, short, and load spectra are recorded for a single predefined channel, and the resulting data is applied to all channels in the port configuration stack.

> ⓘ The channels defined in the port configuration stack may differ from the one used for the single compensation. This represents a key difference compared to the direct process used in the “individual” compensation procedure.

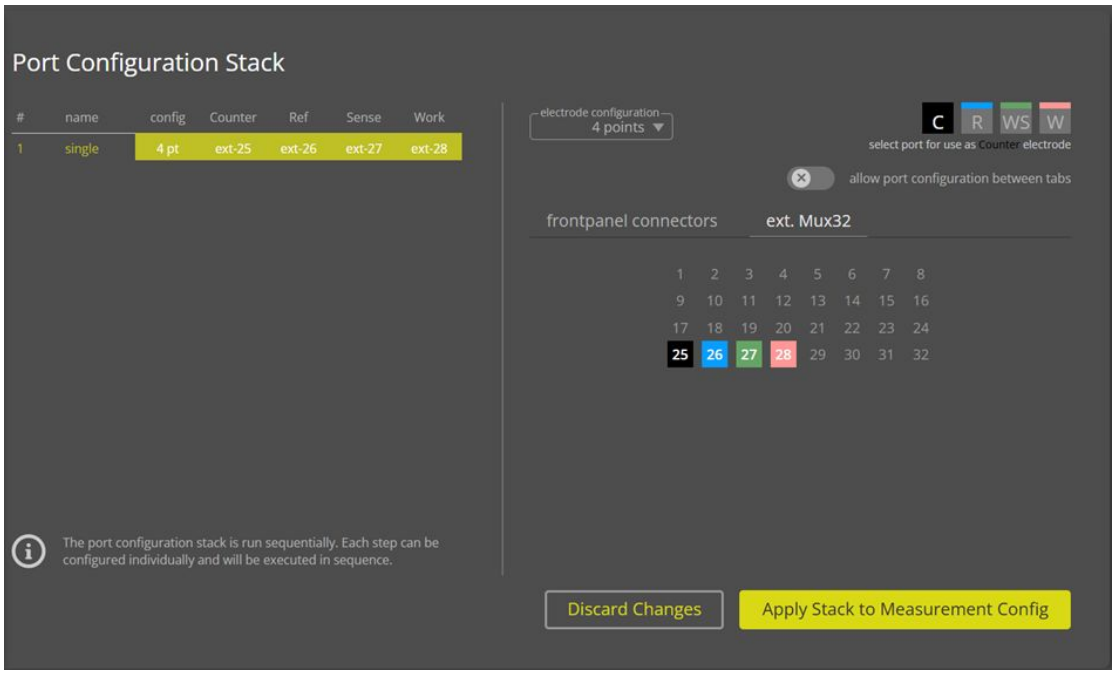
The software implements this type of compensation as follows: After selecting “single” in the “multichannel compensation” dropdown field, an additional dropdown field appears, providing suggested channel selections based on the configuration defined in the port configuration stack. The example shown in the following image illustrates that the measurement channels of the main port, as well as three additional port configurations using an external multiplexer, have been configured.



Alternatively, The user may select the “custom” option to define a different set of channels independently of the port configuration. In this case, a further settings buttons occurs next to the "channel selection" dropdown field, as illustrated in the attached image.



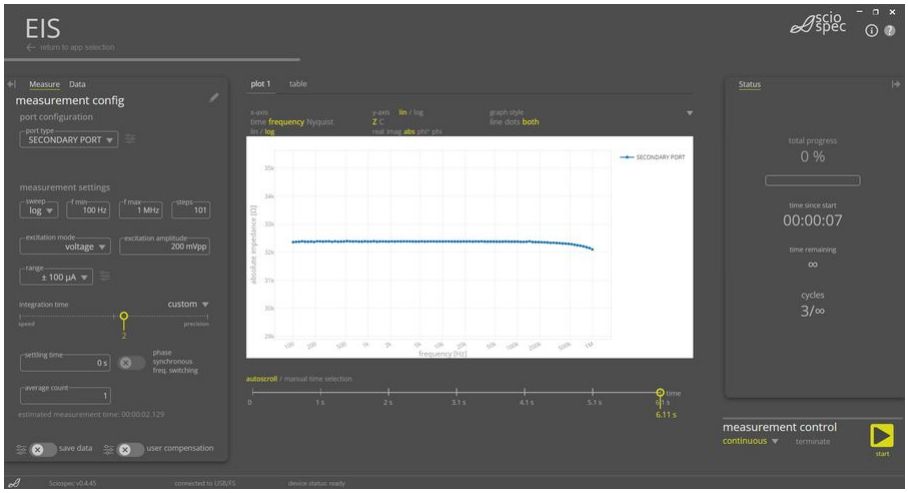
Clicking this additional “settings” button opens a tab similar to the port configuration tab, allowing the user to define a custom channel combination independent of the channel selection used for recording the measurement data.



Once a channel is selected, the subsequent compensation procedure is analogous to that described for the “individual” compensation type.

7.7 EIS App

The EIS App enables the acquisition of impedance spectra and the monitoring of impedance changes over time. A screenshot of the overall appearance is given below.



7.7.1 Configuring a measurement

The configuration panel on the left is divided into two major parts:

- port configuration (channels settings),
- measurement settings.



Port configuration

Port configuration options depend on the connected device and connected accessories. A description of the port configuration options is provided in a seperate section of this document.

Measurement settings

Frequency settings

The measurement settings include frequency settings with different sweep types

- logarithmic sweep,
- linear sweep,
- single frequency measurements.

and parameters such as

- fmin (start frequency of a sweep),
- fmax (stop frequency of a sweep),
- steps (number of steps in between fmin and fmax).

> ⓘ Valid input values for fmin, fmax, and steps depend on the connected device. Please refer to the technical specifications and/or the functional description of this document.

Excitation settings

Depending on the connected device there are different excitation type settings available

- voltage,
- current,
- no control,
- potentiostat,
- galvanostat.

Additionally, the excitation amplitude can be configured as a peak-to-peak value.

> ⓘ Valid input values for the excitation amplitude depend on the connected device and the chosen excitation type. Please refer to the technical specifications and/or the functional description of this document.

Range settings

With the provided options the user can configure the measurement range settings. This strongly depends on the connected device. Range settings may include

- current range,
- voltage range.

Setting these parameters directly affects what kind of DUTs can be measured, since the determination of complex impedance is done by performing a voltage and a current measurement. A disadvantageous selection might cause bad a signal-to-noise ratio (SNR) or clipping of the input stages. Hence, the measurement ranges should also be set in accordance with the set excitation amplitude.

Some devices offer the option "auto-ranging". Activating auto-ranging, dynamically sets the measurement ranges to give the best SNR. A detailed description of this feature is given in the functional description section of this document.

> ⓘ Available current or voltage ranges depend on the connected device. Please refer to the technical specifications and/or the functional description of this document.

Measurement accuracy settings

The user has access to different parameters to adjust the overall measurement accuracy. Setting the integration time has a strong influence on the repeatability of complex impedance determination and on the measurement speed. The user can choose from presets or manually tune the value of the integration time.

Integration time	Measurement speed	Setting
low	fastest	0
medium (default)	fast	1
high	slow	2
very high	slowest	3
custom		0-10

> ⓘ A detailed explanation of the concept of accuracy is given in the functional description of this document.

The parameter "settling time" introduces a hold time between the start of excitation and the start of the acquisition of the impedance data. Furthermore, the option "phase synchronous change freq. switching" can be enabled. Both - settling time and phase synchronous frequency switching - can be used to compensate for settling effects of devices under test, e.g., resonant objects.

> ⓘ Valid input values for the settling time depend on the connected device. Please refer to the technical specifications and/or the functional description of this document.

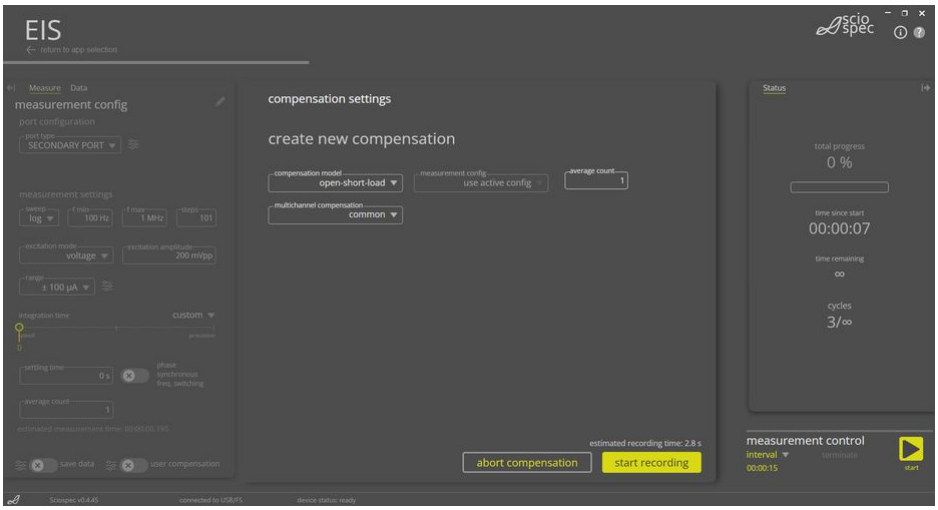
In addition to adapting the integration time and settling effects, the user can define an average count. In accordance with the provided setting, multiple datasets will be averaged. With a setting of 1 no averaging will be applied.

Finally, based on the provided settings an estimation of the expected overall measurement time is presented.

7.7.2 Performing a compensation routine

To compensate for parasitic impedance effects induced by the cabling, it is possible to compensate with an easy open-short-load compensation. Depending on the connected device performing this compensation routine is available to the user. A description of the compensation routine and how to configure the required connections is provided in the functional description of this document. The compensation procedure has to be done for every individual measurement configuration. Nevertheless, the recorded compensation data can be stored to the device and reloaded for later use.

The compensation routine can be started in the software by selecting the user configuration toggle switch and click on compensate. The software then prompts to set the open configuration, followed by short and load configuration. It is possible to use an ideal resistor as a reference load object. To get best results, choose a load object in the impedance range of the device under test.



> ⓘ The compensation is only valid for the specific measurement settings and port configuration. It will be automatically turned off when changing any of those settings.

ⓘ The compensation routine feature is not available for all devices. Check the functional description section of this document to find out, whether this feature is available for the connected device.

7.7.3 Overcurrent / overvoltage detection

As stated in the range settings description, disadvantageous settings can cause clipping of the input ranges, which will cause faulty measurement data. The software will display a warning icon in the device status pane of the software window while the measurement is running if an overcurrent or an overvoltage event is detected. A detailed description of the mechanism is given in the functional description of this document.

Overcurrent and overvoltage events are also saved with the measurement data.

ⓘ The overcurrent/overvoltage detection feature is not available for all devices. Check the functional description section of this document to find out, whether this feature is available for the connected device.

7.7.4 Visualization of measurement data

The measured data is presented in the plotter area of the visualization pane. Different settings to customize the appearance of measurement data are configurable for the user. These settings include:

- axis settings,
- graph style settings.

Below the plotter area a horizontal scrollbar is accessible. By changing the position of the cursor, the user can select a specific dataset recorded in the experiment either based on the time stamp of the recorded dataset or the frequency. This depends on the configured x-axis setting frequency or time (see below).

Axis settings

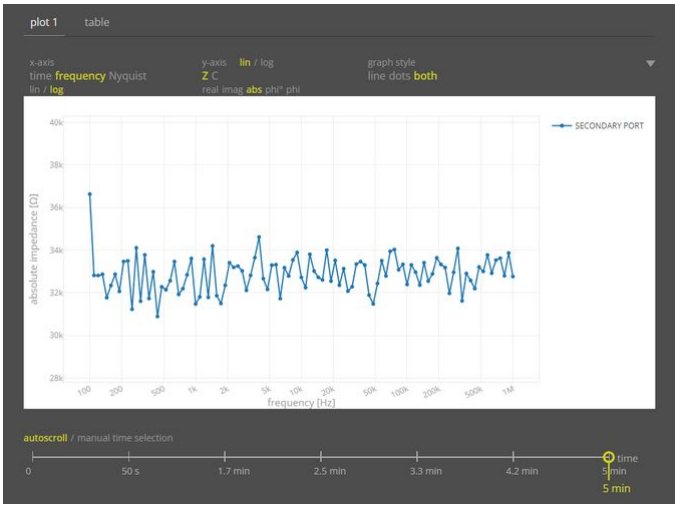
The x- and y-axis can be customized individually by selecting the presented options.

The x-axis can be set to show frequency values allowing to observe the trend of the configured y-axis value in correlation to frequency. Alternatively the x-axis can be configured to show time values. This allows for observation of the configured y-axis value at a selected frequency (horizontal scrollbar) over time. As a third option the plotter can be configured to show a Nyquist plot. For the options frequency and time, the user can choose and configure the spacing to be either linear or logarithmic.

If the axis setting is not configured to be Nyquist plot, the y-axis can be configured to show either impedance or capacitance. Additionally, options for showing the real, the imaginary, and the absolute value as well as the phase in radians or degrees are provided. Just like for the x-axis, the spacing can be set to be either linear or logarithmic.

Graph style

The graph style can be customized to show either lines, dots or both.



7.7.5 Data structure

Each measurement configuration of the .scio5 file includes two groups, the setup group and the data group. In the following the content of each group for data generated with the EIS app is described

Setup group

Metadata

The following entries of the setup group give the metadata of the measurement configuration

- setupInfo
- name
- date
- timestamp
- deviceInfo
- internalMux
- softwareVersion

"SetupInfo" is a compound data set with column labels. It contains information about the saved setup type as well as the version of the setup. "Timestamp" holds the date in milliseconds. "deviceInfo" again is a compound dataset with column labels. It contains information such as device ID, device type, device manufacturer and the firmware version of the device. The entry "internalMux" holds information about used extension ports or internal mux moduls. "SoftwareVersion" contains the version of the backend part of the software, that was used to create the hdf5 file.

Measurement settings

The following entries of the setup group give hold all information about the measurement settings

- channelSettings
- frequencyRows
- impedanceMeasurement
- measureconfigSettings
- settings

- measurementMode

"ChannelSettings" holds information about the selected channels. The entry "frequencyRow" contains information about the frequency settings of the measurement configuration, such as minimal frequency, maximum frequency, frequency steps, scale settings, etc. "measureconfigSettings" holds information about the electrode configurations used for the chosen measurment port. "Settings" contains all parameters not related to the frequency settings. "MeasurementMode" is set to "impedance measurement" for the EIS app-

Data group

For each configured port exist an individual port group.

Every port group holds multiple datasets. These are

- frequency
- measurementData
- overflowDetection
- timestamp
- timestampInMicroSec
- type

"Frequency" holds all the frequencies measured. "MeasuremenData" contains all the recorded real and imaginary parts. It is a three-dimensional dataset, Each row indicates the data at each timestamp, and each column indicates the data at each frequency value. The third dimension destinguishes real and imaginary values. "Timestamp" contains information about the date of the spectra while information about the start of the individual measurement is hold in "timestampInMicroSec". The "overflowDetection" dataset contains information about overcurrent or overvoltage incidents. "Type" states that the data group is an impedance spectrum.

8 Sciospec Communication Interface

The Sciospec Communication Interface (COMinterface) enables the user to access all functions of the device by using any of the available master interfaces. The actual command structure is identical in all connection types.

8.1 Syntax

The general structure of each communication with a Sciospec device:

- The communication is done by frames
- Each communication frame is constructed as follows

- 1 byte command-Tag (Frame-Start)
- 1 byte number of data-bytes (0...255)
- 0...255 data-bytes
- 1 byte Command-Tag (Frame-End)

- The command-tag identifies the command (see Command list)
- Frame-Start and –End must be identical

Example: "System-Ready-Message"

CMD-Tag	Number of bytes	Data	CMD-Tag
0x18	0x01	0x84	0x18

8.2 Acknowledge messages

- Communication-frames with incorrect syntax will cause a “Frame-Not-Acknowledge” message
- If the transmission of a communication-frame is interrupted for more than 10 ms a “Timeout” message is send
- Every invalid command-tag will cause a “Not-Acknowledge” message
- Every valid command is acknowledged with an acknowledge command [ACK]
- For commands with a return value the returning frame comes before the acknowledge message
- When commands are sent during the current measurements, measurement data can be transmitted between the command and the following returning frame and the acknowledge-message (commands are handled asynchronously)
- Before sending a new command, the resulting acknowledge or not acknowledge of the previous command has to be awaited.

The ACK-Frame:

0x18	0x01	[ACK]	0x18

General System Messages	
0x01	Frame-Not-Acknowledge: Incorrect syntax

General System Messages	
0x02	Timeout: Communication-timeout (less data than expected)
0x04	Wake-Up Message: System boot ready
0x11	TCP-Socket: Valid TCP client-socket connection
0x81	Not-Acknowledge: Command has not been executed
0x82	Not-Acknowledge: Command could not be recognized
0x83	Command-Acknowledge: Command has been executed successfully
0x84	System-Ready Message: System is operational and ready to receive data

Device Specific System Messages	
0x90	Overcurrent Detected Value of DC current on W-ports exceeds capability of configured current range
0x91	Overvoltage Detected Value of DC voltage difference between R and WS port exceeds capability of configured voltage range

ⓘ For use of the impedance measurement module within a ISX-5/MSX-8 platform system refer to the respective system manual for guidance on specific communication within those systems. Commands and functionality are the same as described here, but additional framing and setup commands will be required due to the multi slot system structure within ISX-5/MSX-8.

8.3 Abbreviations

abbreviation	full name
[CT]	command tag
[LE]	length
[OB]	option byte
[CD]	command data

8.4 Command list

The leading hex code of each command heading represents the [command code] of the respective function.

0x90 - Save Settings

0x97 - Set Options
0x98 - Get Options
0xA1 - Reset System
0xB0 - Set FE Settings
0xB1 - Get FE Settings
0xB2 - Set ExtensionPort Channel
0xB3 - Get ExtensionPort Channel
0xB5 - Get ExtensionPort Module
0xB6 - Set Setup
0xB7 - Get Setup
0xB8 - Start Measure
0xB9 - Set Sync Time
0xBA - Get Sync Time
0xBD - Set Ethernet Configuration
0xBE - Get Ethernet Configuration
0xCF - TCP connection watchdog
0xD0 - Get ARM firmware ID
0xD1 - Get Device ID
0xD2 - Get FPGA firmware ID
0xB3 - Get ExtensionPort Channel

8.5 Command description

8.5.1 0x90 - Save Settings

General Syntax

[CT] 00 [CT]

Return

ACK

Description

Saves the following parameters permanently into the flash memory of the ISX-3:

- NTC Parameters 1 and 2
- Parameter stack synchronization time

This command can only be used if no measurement is currently running.

8.5.2 0x97 - Set Options

General Syntax

[CT] [LE] **[OB] [CD]** [CT]

Return

ACK

[OB]

Function	code
Activate time stamp in ms	0x01
Activate time stamp in µs	0x02
Activate current range	0x04

ⓘ This setting will only be saved temporarily. After a system restart, the setting will be reset to default.

Active time stamp in ms

Configuration of the Instrument. Activate time stamp in ms of measured data. Time stamp added to measurement data in 4Byte unsigned integer. If measured time exceed 0xFFFFFFFF (about 1193h) time will start over at 0x00000000.

Syntax

- Syntax set: **[CT] 02 01 [CD] [CT]**

[CD]

- 0x01 Enable Time Stamp
- 0x00 Disable Time Stamp

Depending on this setting the return frame of the measured data changes (see command B8)

Remarks:
It is not possible to change this setting while a measurement is running. This setting cannot be saved persistently. It is not possible to activate the time stamp in µs and in ms. Only the last configuration will be valid.

Active time stamp in µs

Configuration of the Instrument. Activate time stamp in µs of measured data. Time stamp added to measurement data in 5Byte unsigned integer. If measured time exceed 0xFFFFFFFFFF (about 305,4h) time will start over at 0x0000000000.

Syntax

- Syntax set: **[CT] 02 02 [CD] [CT]**

[CD]

- 0x01 Enable Time Stamp
- 0x00 Disable Time Stamp

Depending on this setting the return frame of the measured data changes (see command B8)

Remarks:
It is not possible to change this setting while a measurement is running. This setting cannot be saved persistently. It is not possible to activate the time stamp in µs and in ms. Only the last configuration will be valid.

Activate current range

Configuration of the Instrument. Activate current range of measured data. The current range is added to the measurement data as 1Byte unsigned integer. A description of possible values can be found in "0xB0 - Set FE Settings".

Syntax

- Syntax set: **[CT] 02 04 [CD] [CT]**

[CD]

- 0x01 Enable current range
- 0x00 Disable current range

Depending on this setting the return frame of the measured data changes (see command B8)

Remarks:
It is not possible to change this setting while a measurement is running. This setting cannot be saved persistently.

8.5.3 0x98 - Get Options

General Syntax

[CT] [LE] **[OB]** [CT]

Return

[CT] [LE] **[OB] [CD]** [CT]

ACK

[OB]

Function	code
Time stamp	0x01 || 0x02
frequency range	0x03
current range	0x04

Time stamp

Returns the currently configured options of the instruments.

Syntax

- Syntax get: 98 01 **[OB]** 98
  - Return: 98 02 **[OB] [timeStamp]** 98

[OB]

- Time stamp option byte
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:

- 0x01
- 0x02

[timeStamp]

- Currently configured time stamp option
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:
  - 0x00 - time stamp disabled
  - 0x01 - time stamp in ms
  - 0x02 - time stamp in µs

frequency range

Returns available configurable frequency range of the instruments.

Syntax

- Syntax get: 98 01 03 98
  - Return: 98 09 03 **[MinF] [MaxF]** 98

[MinF]

- Minimum configurable frequency
- Data format: floating point single precission
- Length: 4 Byte
- Unit: Hz

[MaxF]

- Maximum configurable frequency
- Data format: floating point single precission
- Length: 4 Byte
- Unit: Hz

current range

Returns the currently configured options of the instruments.

Syntax

- Syntax get: 98 01 04 98
  - Return: 98 02 04 **[current range]** 98

[current range]

- Currently configured current range option
- Data format: unsigned integer
- Length: 1 Byte
  - 0x01 Enable current range output
  - 0x00 Disable current range output

8.5.4 0xA1 - Reset System

complete restart of the system

General Syntax

[CT] 00 [CT]

Return

ACK
Wake-Up Message
System-Ready-Message

8.5.5 0xB0 - Set FE Settings

Frontend configuration

> The device has a stack length of 128. Sending the set FE settings command more than the length of the channel stack, results in a NACK return. Send following command to empty the stack: **B0 03 FF FF FF B0**

General syntax

[CT] [LE] [CD] [CT]

Return

ACK

Depricated syntax

Syntax set: B0 [LE] **[measurement mode] [measurement channel] [current range] [voltage range]** B0

[measurement mode]

- set the electrode configuration
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:
  - 0x01 - 2 point configuration
  - 0x02 - 4 point configuration
  - 0x03 - 3 point configuration

[measurement channel]

- set the measurement channel of the device
- Data format: unsigned integer
- Length: 1 Byte
- interpretation / value
  - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
  - 0x02 - Extension port
  - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[current range]

- set the current range to be used for the measurement
- Data format: unsigned integer

- Length: 1 Byte
- Interpretation / value:
    - 0x00 - auto range
    - 0x01 - ± 10 mA
    - 0x02 - ± 100 µA
    - 0x04 - ± 1 µA
    - 0x06 - ± 10 nA

[voltage range]

- set the voltage range to be used for the measurement
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:
    - 0x01 - ± 1 V
    - 0x02 - ± 90 mV

ⓘ The parameter of [voltage range] is optional and does not need to be sent. If [voltage range] is not sent the default value is used.

Current syntax

- Syntax set: B0 [LE] **[measurement mode] [current range] [voltage range] [channel settings]** B0

[measurement mode]

- set the electrode configuration
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:
    - 0x01 - 2 point configuration
    - 0x02 - 4 point configuration
    - 0x03 - 3 point configuration

[current range]

- set the current range to be used for the measurement
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:
    - 0x00 - auto range
    - 0x01 - ± 10 mA (100)
    - 0x02 - ± 100 µA (10k)
    - 0x04 - ± 1 µA (1M)
    - 0x06 - ± 10 nA (100M)

[voltage range]

- set the voltage range to be used for the measurement
- Data format: unsigned integer
- Length: 1 Byte

- Interpretation / value:
  - 0x01 - ± 1 V
  - 0x02 - ± 90 mV
- Default: 0x01

[channel settings]

- set the active channels for the measurment. The content is dependent on the desired [measurement mode]
- Data format: unsigned integer
- Lenght:
  - [measurement mode] 0x01: 6 Byte
  - [measurement mode] 0x02: 12 Byte
  - [measurement mode] 0x03: 9 Byte
- Interpretation:
  - [measurement mode] 0x01: [C] [Cext] [W] [Wext]
  - [measurement mode] 0x02: [C] [Cext] [R] [Rext] [WS] [WSext] [W] [Wext]
  - [measurement mode] 0x03: [C] [Cext] [R] [Rext] [W] [Wext]

[C]

- activate the main C port
- Data format: unsigned integer
- Length: 1Byte
- Interpretation / value:
  - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
  - 0x02 - Extension port
  - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[Cext]

- activate the C port of an extension port module
- Data format: unsigned integer
- Length: 2Byte
- Interpretation / value: This can be any number that is valid for used ExtensionPort module. If no ExtensionPort is used this value must be set to zero.

[R]

- activate the main R port
- Data format: unsigned integer
- Length: 1Byte
- Interpretation / value:
  - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
  - 0x02 - Extension port
  - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[Rext]

- activate the R port of an extension port module
- Data format: unsigned integer
- Length: 2Byte

- Interpretation / value: This can be any number that is valid for used ExtensionPort module. If no ExtensionPort is used this value must be set to zero.

[WS]

- activate the main WS port
- Data format: unsigned integer
- Length: 1Byte
- Interpretation / value:
  - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
  - 0x02 - Extension port
  - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[WSext]

- activate the WS port of an extension port module
- Data format: unsigned integer
- Length: 2Byte
- Interpretation / value: This can be any number that is valid for used ExtensionPort module. If no ExtensionPort is used this value must be set to zero.

[W]

- activate the main W port
- Data format: unsigned integer
- Length: 1Byte
- Interpretation / value:
  - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
  - 0x02 - Extension port
  - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[Wext]

- activate the W port of an extension port module
- Data format: unsigned integer
- Length: 2Byte
- Interpretation / value: This can be any number that is valid for used ExtensionPort module. If no ExtensionPort is used this value must be set to zero.

8.5.6 0xB1 - Get FE Settings

Returns the currently selected frontend configuration

General syntax

[CT] [LE] [OB] [CT]

Return

[CT] [LE] [CD] [CT]
ACK

[Option byte]

Function	code
channel count	0x00
channel configuration	0xXX

channel count

Returns the number of configured channels

- Syntax get: B1 01 00 B1
  - Return: B1 02 **[channel count]** B1

[channel count]

- number of configured channels
- Data format: unsigned integer
- Length: 2 Byte

channel configuration

Returns the configuration for the specified channel number

- Syntax get: B1 02 **[channel number]** B1
  - Return: B1 [LE] **[measurement mode] [current range] [voltage range] [channel settings]** B1

[channel number]

- number of the channel configuraiton to get information on
- Data format: unsigned integer
- Length: 2 Byte

[measurement mode]

- mode of the electrode configuration
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:
  - 0x01 - 2 point configuration
  - 0x02 - 4 point configuration
  - 0x03 - 3 point configuration

[current range]

- activated current range of [channel number]
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:
  - 0x00 - auto range
  - 0x01 - ± 10 mA (100)
  - 0x02 - ± 100 µA (10k)
  - 0x04 - ± 1 µA (1M)
  - 0x06 - ± 10 nA (100M)

[voltage range]

- activated voltage range of [channel number]
- Data format: unsigned integer
- Length: 1 Byte
- Interpretation / value:
    - 0x01 - ± 1 V
    - 0x02 - ± 90 mV

[channel settings]

- get the channel settings of of [channel number]. The content is dependent of [measurement mode]
- Data format: unsigned integer
- Lenght:
    - [measurement mode] 0x01: 6 Byte
    - [measurement mode] 0x02: 12 Byte
    - [measurement mode] 0x03: 9 Byte
- Interpretation:
    - [measurement mode] 0x01: [C] [Cext] [W] [Wext]
    - [measurement mode] 0x02: [C] [Cext] [R] [Rext] [WS] [WSext] [W] [Wext]
    - [measurement mode] 0x03: [C] [Cext] [R] [Rext] [W] [Wext]

[C]

- state of the main C port
- Data format: unsigned integer
- Length: 1Byte
- Interpretation / value:
    - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
    - 0x02 - Extension port
    - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[Cext]

- activated C port of an extension port module
- Data format: unsigned integer
- Length: 2Byte
- Interpretation / value: This can be any number that is valid for used ExtensionPort module. If not configured this value is zero

[R]

- state of the main R port
- Data format: unsigned integer
- Length: 1Byte
- Interpretation / value:
    - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
    - 0x02 - Extension port
    - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[Rext]

- activated R port of an extension port module
- Data format: unsigned integer

- Length: 2Byte
- Interpretation / value: This can be any number that is valid for used ExtensionPort module. If not configured this value is zero.

[WS]

- state of the main WS port
- Data format: unsigned integer
- Length: 1Byte
- Interpretation / value:
  - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
  - 0x02 - Extension port
  - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[WSext]

- activated WS port of an extension port module
- Data format: unsigned integer
- Length: 2Byte
- Interpretation / value: This can be any number that is valid for used ExtensionPort module. If not configured this value is zero.

[W]

- state of the main W port
- Data format: unsigned integer
- Length: 1Byte
- Interpretation / value:
  - 0x01 - Main port (BNC port or "port 1" on ISX-3mini)
  - 0x02 - Extension port
  - 0x03 - ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)

[Wext]

- activated W port of an extension port module
- Data format: unsigned integer
- Length: 2Byte
- Interpretation / value: This can be any number that is valid for used ExtensionPort module. If not configured this value is zero.

8.5.7 0xB2 - Set ExtensionPort Channel

This command is depricated and should not be used. The functionality of this command moved to "0xB0 - Set FE Settings"

Set the ExtensionPort channel settings.

General syntax

[CT] 04 **[CP] [RP] [WS] [WP]** [CT]

Return

ACK

ExtensionPort- channel settings

[CP]	(C-Port): Counter - Port Selection
[RP]	(R-Port): Reference - Port Selection
[WS]	(WS-Port): Working Sense - Port Selection
[WP]	(W-Port): Work - Port Selection

See documentation of the connected extension module for detailed information.

8.5.8 0xB3 - Get ExtensionPort Channel

This command is depricated and should not be used. The functionality of this command moved to "0xB1 - Get FE Settings"

Read the currently set ExtensionPort configuration.

General syntax

[CT] 00 [CT]

Return

[CT] 04 **[CP] [RP] [WS] [WP]** [CT]
ACK

ExtensionPort- channel settings

CP	(C-Port): Counter - Port Selection
RP	(R-Port): Reference - Port Selection
WS	(WS-Port): Working Sense - Port Selection
WP	(W-Port): Work - Port Selection

See documentation of the connected extension module for detailed information.

8.5.9 0xB5 - Get ExtensionPort Module

Read the type of the currently connected extension module.

General syntax

[CT] 00 [CT]

Return

Optional channelCount is only valid in case of module = Mux32any2any2202

[CT] [LE] **[ExtensionModule] [optional ChannelCount Ext*] [InternalModule] [optional ChannelCount Int*]** [CT]

ACK

[LE]

- 0x02 unsigned integer
- 0x04 unsigned integer if ExtensionModule = Mux32any2any2202
- 0x04, unsigned integer if InternalModule = Mux32any2any2202
- 0x06, unsigned integer if ExtensionModule = Mux32any2any2202 and InternalModule = Mux32any2any2202

[ExtensionModule]

This code represents the connected extension module.

Function	code
no module connected	0x00
MEArack	0x01
MuxModule32	0x02
ECIS Adapter	0x03
ExtensionPortAdapter	0x05
SlideChipAdapter	0x06
Mux32any2any (external)	0x07
DaQEisMux	0x08
Mux32any2any2202 (external)	0x09

[InternalModule]

This code represents the internal module.

Function	code
no module connected	0x00
MuxModule16x4	0x01
MuxModule32x2	0x02

Function	code
Mux32any2any (internal)	0x07
Mux32any2any2202	0x09

[optional ChannelCount*]: 2byte unsigned integer

8.5.10 0xB6 - Set Setup

General syntax

[CT] [LE] **[OB] [CD]** [CT]

Return

ACK

[OP]

	Function	code
Initialization and Configuration Commands	Init	0x01
	Add single frequency point	0x02
	Add frequency list	0x03
	Set amplitude	0x05
Compensation Commands	Start compensation	0x10
	Compensation acknowledge	0x11
	Compensation not acknowledge	0x12
	Compensation redo Open	0x13
	Compensation redo Short	0x14
	Compensation redo Load	0x15
	Compensation-interaction Load value	0x16
	Set compensation data	0x17
	Set compensation Channel	0x18
Loading Setup from Slot	Loading setup from slot	0x20
DC Bias	DC Bias	0x30

	DC Bias not-acknowledge	0x32
	DC Bias set value	0x33

Init

This option resets the currently configured setup and an empty setup is initialized.

Syntax:

- Syntax set: **[CT] 01 01 [CT]**

Add single frequency point

This command is used to add a single frequency point to the currently configured setup.

Syntax

- Syntax set: **[CT] [LE] 02 [CD] [CT]**

[CD]

- [LE] = 0x0D (frequency, precision, amplitude)

frequency			precision			amplitude		
MSB	...	LSB	MSB	...	LSB	MSB	...	LSB

In this case the point delay (= 0 ms) and phase sync is not used.

- [LE] = 0x0D **+** extended options length

frequency			precision			amplitude			point delay				use phase sync				excitation type			
MSB	...	LSB	MSB	...	LSB	MSB	...	LSB	EOP =01	MSB	...	LSB	EOP =02	MSB	...	LSB	EOP =03	MSB	...	LSB
obligatory									optional				optional				optional			

Frequency: frequency in Hz (4Byte float)

Frequency must be > minimum frequency for measuring AC impedance or frequency must be equal to zero for measuring DC resistance.

Precision: precision value (4Byte float)

Amplitude:

- for voltage excitation: amplitude in V (peek value, 4Byte float)
- for current excitation: amplitude in A (peek value, 4Byte float)
- values outside of the following amplitude ranges will be acknowledged and corrected internally to the corresponding minimum or maximum
  - voltage excitation range: 0.1 mV to 1 V

- current excitation amplitude range: 1 µA to 10 mA
- please note that the specified excitation ranges are smaller then ranges listed above. Sciospec will only guarantee the correctness of the specified excitation ranges stated in the Technical Specifications.

Extended Options EOP:

- For each optional additional setting an EOP identifier needs to be send before the actual setting. Therefore any of the following options or multiple can be send.
- Point delay (EOP=01): delay between this frequency and the next frequency in µs (4Byte unsigned integer)
- Use phase sync (EOP=02): phase synchronous switch between this and the next frequency (4Byte unsigned integer)
  - Use: 0x0000001
  - Don't use: 0x00000000
- excitation type (EOP=03): Defines weather the excitation amplitude is in Volt or in Ampere (4Byte unsigned integer)
  - voltage excitation: 0x00000001 (default)
  - current excitation: 0x00000002

Examples

- B6 0D 02 46 FA 00 00 3F 80 00 00 3E 80 00 00 B6
  - Frequency = 32 kHz
  - precision = 1.0
  - amplitude = 0.25 V
  - NO extended options have been used
- B6 12 02 46 FA 00 00 3F 80 00 00 3E 80 00 00 **02** 00 00 00 01 B6
  - Frequency = 32 kHz
  - precision = 1.0
  - amplitude = 0.25 V
  - phase synchronization = used
  - One extended option has been used

Add frequency list

This command is used to add multiple frequencies to the currently configured setup.

Syntax

- Syntax set: **[CT] [LE] 03 [CD] [CT]**

[CD]

- [LE] = 0x16

start-frequency			stop-frequency			count			scale	precision			amplitude		
MSB	...	LSB	MSB	...	LSB	MSB	...	LSB	MSB	MSB	...	LSB	MSB	...	LSB

In this case the point delay (= 0 ms) and phase sync is not used.

- [LE] = 0x16 + extended options length

start-frequency			stop-frequency			count			scale	precision			amplitude			point delay				use phase sync				excitation type			
MSB	...	LSB	MSB	...	LSB	MSB	...	LSB	MSB	MSB	...	LSB	MSB	...	LSB	EOP = 01	MSB	...	LSB	EOP = 02	MSB	...	LSB	EOP = 03	MSB	...	LSB
obligatory																optional				optional				optional			

- Start-Frequency: start-frequency of the frequency block in Hz (4Byte float)
- Stop-Frequency: stop-frequency of the frequency block in Hz (4Byte float)
- Count: number of frequency steps used (4Byte float, rounded to the next smaller integer)
- Scale: linear or logarithmic scale (1Byte integer)
  → linear: 0
  → logarithmic: 1
- Precision: precision value (4Byte float)
- Amplitude:
  - for voltage excitation: amplitude in V (peek value, 4Byte float)
  - for current excitation: amplitude in A (peek value, 4Byte float)
  - values outside of the following amplitude ranges will be acknowledged and corrected internally to the corresponding minimum or maximum
    - voltage excitation range: 0.1 mV to 1 V
    - current excitation amplitude range: 1 µA to 10 mA
    - please note that the specified excitation ranges are smaller then ranges listed above. Sciospec will only guarantee the correctness of the specified excitation ranges stated in the Technical Specifications.

Extended Options EOP:

- For each optional additional setting an EOP identifier needs to be send before the actual setting. Therefor any of the following options or multiple can be send.
- Point delay: delay between this frequency and the next frequency in µs (4Byte unsigned integer)
- Use phase sync: phase synchronous switch between this and the next frequency (4Byte unsigned integer)
  - Use: 0x0000001
  - Don't use: 0x00000000
- excitation type (EOP=03): Defines weather the excitation amplitude is in Volt or in Ampere (4Byte unsigned integer)
  - voltage excitation: 0x00000001 (default)
  - current excitation: 0x00000002

Example

- B6 20 03 44 7A 00 00 4B 18 96 80 41 20 00 00 01 3F 80 00 00 3E 80 00 00 **01** 00 00 03 E8 02 00 00 00 00 B6
- start frequency = 1 kHz
- stop frequency = 10 MHz
- count = 10

- scale = logarithmic
- precision = 1.0
- amplitude = 0.25 V
- point delay = 1000 µs
- phase sync = disable

Set amplitude

This command sets the amplitude.

Syntax

Two options are available:

- Syntax set all amplitudes with one command: **[CT] 06 05 [ExcitationType] [Amplitude] [CT]**
- Syntax set amplitudes row-wise: **[CT] 08 05 [Row] [ExcitationType] [Amplitude] [CT]**

[ExcitationType]

- Length: 1 byte
- 0x01 - voltage
- 0x02 - current

[Amplitude]

- Length: 4 byte
- Data format: float
- for voltage excitation: amplitude in V (peek value, 4Byte float)
- for current excitation: amplitude in A (peek value, 4Byte float)
- values outside of the following amplitude ranges will be acknowledged and corrected internally to the corresponding minimum or maximum
    - voltage excitation range: 0.1 mV to 1 V
    - current excitation amplitude range: 1 µA to 10 mA
    - please note that the specified excitation ranges are smaller then ranges listed above. Sciospec will only guarantee the correctness of the specified excitation ranges stated in the Technical Specifications.

[Row]

- Length: 2 byte
- 0 ≤ row ≤ maximum row count

start compensation

A detailed example of the required communication necessary to compensate a setup can be found in Example of impedance compensation procedure.

This command starts the compensation of the setup.

Syntax

- Syntax set: **[CT] [LE] 10 [sendCompensationData] [CT]**

[sendCompensationData]

- Length: 1 byte
- Data format: unsigned integer

- 0x01 = enable
  0x00 = disable (default)
- this is optional and can be omitted
- example: B6 01 10 B6 or B6 02 10 00 B6 start compensation without sending the compensation data.
- example: B6 02 10 01 B6 start compensation with sending the compensation data.

compensation acknowledge

The compensation acknowledge (CACK) is used to answer a “Compensation-Interaction-Request”

Syntax

- Syntax: **[CT] [LE] 11 [avg count] [CT]**

[avg count]

- Length: 1 byte
- Data format: unsigned integer
- number of averaging (default: 1)
- this is optional and can be omitted
- example: B6 01 11 B6 or B6 02 11 00 B6 compensation acknowledge without averaging.
- example: B6 02 11 0A B6 compensation acknowledge with averaging of 10 spectra.

Compensation not-acknowledge

The compensation acknowledge (CNACK) is used to answer a “Compensation-Interaction-Request”

Syntax

- Syntax: **[CT] 01 12 [CT]**

Compensation redo Open

This command repares the measurement to redo the open measurement for aquire the compensation data. This command is only applicable

- if the system is idle and aquiring open-shot-load compensation data is completed
- or aquiring open-short-load compensation data is in progress and open measurement was just completed.

Syntax

- Syntax set: **[CT] 01 13 [CT]**
- Return
  - ACK & compensation-interaction-request open

Compensation redo Short

This command repares the measurement to redo the short measurement for aquire the compensation data. This command is only applicable

- if the system is idle and aquiring open-shot-load compensation data is completed
- or aquiring open-short-load compensation data is in progress and short measurement was just completed.

Syntax

- Syntax set: **[CT] 01 14 [CT]**

- Return
    - ACK & compensation-interaction-request short

Compensation redo Load

This command repares the measurement to redo the load measurement for aquire the compensation data. This command is only applicable

- if the system is idle and aquiring open-shot-load compensation data is completed
- or aquiring open-short-load compensation data is in progress and open measurement was just completed

Syntax

- Syntax set: **[CT] 01 15 [CT]**
- Return
    - ACK & compensation-interaction-request load

compensation-interaction Load value

The connected load value must have been send in the command data part [CD] of the command.

Syntax

- Syntax set: **[CT] 06 16 [LoadType] [Value] [CT]**

[LoadType]

- Length: 1 byte
- 0x01 - resistance
- 0x02 - capacitor

[Value]

- Length: 4 byte
- Data format: float
- Unit: Ohm for [LoadType] = resistance, Farad for [LoadType] = capacitor

Example

- [CT] 06 16 01 44 7A 00 00 [CT] → resistance = 1000 Ohm

set compensation data

This command configures the compensation data of the configured setup for a specific frequency and a specific channel.

> ⓘ This changes the status of the measurement configuration to compensated (Check get compensation data status: B7 02 17 01 B7. Compensation data is applied to the measurment data. Make sure all frequencies of all channels have valid compensation data (Check get compensation data: B7 03 17 02 [channel] B7).

Syntax

- Syntax reset compensation data: **[CT] 02 17 01 [CT]**

- Syntax set data: **[CT] [LE] 17 02 [Channel] [Row] [Open Re] [Open Im] [Short Re] [Short Im] [Load Re] [Load Im] [CT]**

[Channel]	[Row]	[Open Re]	[Open Im]	[Short Re]	[Short Im]	[Load Re]	[Load Im]
1 byte	2 byte	4 byte float	4 byte float	4 byte float	4 byte float	4 byte float	4 byte float

set compensation channel

This command configures the channel setting used for compensation. For multi channel configuration, the individual compensation mode can be configured (separate compensation data per channel). Alternatively a channel can be configured to aquire the compensation data. This can be set after the compensation intercation message is received.

Syntax

- Syntax - set individual compensation mode (default after start compensation): **[CT] 01 18 [CT]**
- Syntax - set compensation channel: **[CT] 0E 18 [measurement mode] [measurement channel C] [2Byte ext C] [measurement channel R] [ext R] [measurement channel Ws] [ext Ws] [measurement channel W] [ext W] [CT]**

[measurement mode]

Function	code
4 point configuration	0x02
3 point configuration	0x03
2 point configuration	0x01

[measurement channel]

Function	code
BNC Port (ISX-3mini: Port 1)	0x01
ExtensionPort	0x02
ExtensionPort2 (ISX-3mini: Port 2, ISX-3: optional, *InternalMux*)	0x03

[ext channel]

ExtensionPort channel (internal or external). This can be any number (2B unsigned integer) that is valid for used ExtensionPort module. If no ExtensionPort is used this value must be set to zero.

Loading setup from slot

It is possible to load (recalling) all information required for a measurement from the internal storage of the impedance analyzer. The system holds up to 255 setup configurations including any compensation data. For addressing, the configurations are numerated from slot 1 to slot 255.

Syntax

- Syntax loading from slot: **[CT] 02 20 [Slot] [CT]**

[Slot]

- Length: 1 byte

Example

- [CT] 02 20 05 [CT] → load configuration from slot 5

DC Bias

This command activates or deactivates the DC Bias on the configured channel. When the DC Bias is activated DC Bias is set to 0V and then regulated towards the selected value. This process can be aborted by sending the “DC Bias-Not-Acknowledge”.

If the DC Bias is reached a “DC Bias-Acknowledge” is send from the ISX-3 / ISX-3 mini.

If the DC Bias could not reached “DC Bias-Not-Acknowledge” is send from the ISX-3 / ISX-3 mini.

Syntax

- Syntax set: **[CT] 02 30 [CD] [CT]**

[CD]

- Length: 1 byte
- 0x00 - deactivate
- 0x01 - activate

DC Bias not-acknowledge

This command is to abort the DC Bias regulation.

Syntax

- Syntax: **[CT] 01 32 [CT]**

DC Bias set value

This command sets the value for the DC Bias.

Syntax

- Syntax set: **[CT] 05 33 [bias value] [CT]**

[bias value]

- Length: 4 byte
- Data format: float
- Unit: V (volt)
- range: + 1 V to - 1 V

Example

- [CT] 05 33 3F 80 00 00 [CT] → Sets the bias value to 1 V.

ⓘ How to regulated the DC Bias after the measuring section or the measuring object changed:

- send the **DC Bias set value** command (Syntax set: [CT] 05 33 [bias value] [CT]) → not necessary if the set value doesn't change
- send the **DC Bias** command (Syntax: [CT] 02 30 [CD] [CT])

8.5.11 0xB7 - Get Setup

General syntax

[CT] [LE] **[OB] [CD]** [CT]

Return

[CT] [LE] **[OB] [CD]** [CT]

ACK

[OP]

Function	code
Get total number of frequencies	0x01
Get information of frequency point	0x02
Get frequency list	0x04
compensation-interaction-request Open	0x13
compensation-interaction-request Short	0x14
compensation-interaction-request Load	0x15
compensation data	0x17
save setup to slot	0x20
Get DC Bias	0x33

Get total number of frequencies

This command reads the total number of frequencies configured in the setup.

Syntax

- Syntax get: **[CT] 01 01 [CT]**
  - Return: **[CT] 03 01 [CD] [CT]**

[CD]

- number of rows configured
- Length: 2 byte
- Data format: unsigned integer

Get information of frequency point

This command gets information of a configured point of the setup.

Syntax

- Syntax get: **[CT] XX 02 [CD] [EOPs] [CT]**
  - Return: **[CT] XX 02 [Frequency] [Precision] [Signal amplitude] [EOP] [EOP-value] [CT]**

[CD]

- row number
- Length: 2 byte
- Data format: unsigned integer

[Frequency]

- Length: 4 byte
- Data format: float
- Unit: Hz

[Precision]

- Length: 4 byte
- Data format: float

[Signal amplitude]

- Length: 4 byte
- Data format: float
- voltage amplitude in V
- current amplitude in A

[EOPs]

- optional list of extended options
- each extended option can be on of the following

EOP	
0x01	point delay
0x02	phase sync
0x03	excitation type

Get frequency list

This command gets a list of frequencies configured in this setup.

Syntax

- Syntax get: **[CT] 01 04 [CT]**
  - Return: **[CT] [LE] 04 [4 Byte float frequency$_1$] [...] [4 Byte float frequency$_N$] [CT]**

[4 Byte float frequency$_1$] [...] [4 Byte float frequency$_N$]

- *In the case of "get frequency list" cmd one data frame is limited to a total of 252 bytes of data. The returning command will be split into multiple separate frames, if this number is exceeded. For example if the setup contains 64 frequency points the first 62 will be transmitted in a frame containing 249 bytes (=62*4+1) and in a separate frame containing 9 bytes (=2*4+1) bytes of data.*

compensation-interaction-request Open

This is a message send by the ISX-3 to request an open configuration on the measurement channel. This request must be answered with a CACK to continue the compensation or with a CNACK to abort the compensation.

compensation-interaction-request Short

This is a message send by the ISX-3 to request a short configuration on the measurement channel. This request must be answered with a CACK to continue the compensation or with a CNACK to abort the compensation.

compensation-interaction-request Load

This is a message send by the ISX-3 to request a load configuration on the measurement channel. This request must be answered with a CACK to continue the compensation or with a CNACK to abort the compensation.

Before the answer to this request is send the load value and type must have been send to the device (Option 0x16).

compensation data

This command gets the status of the compensation data and to read the compensation data.

Syntax

- Syntax get compensation data status: **[CT] 02 17 01 [CT]**
  - return **[CT] 03 17 01 [compensation data staus] [CT]**
- Syntax get compensation data : **[CT] 03 17 02 [channel] [CT]**
  - return **[CT] 1D 17 02 [channel] [frequency index] [real Open] [imag Open] [real short] [imag short] [real load] [imag load] [CT]**

[channel]

- channel number: 1 .. number of configured channels
- Length: 1 byte
- Data format: unsigned integer

frequency index

- index of frequency row: 0 .. number of configured frequencies -1
- Length: 2 byte
- Data format: unsigned integer

[real Open] [imag Open] [real short] [imag short] [real load] [imag load]

- real or imagninary part of compensation data
- Length: 4 byte
- Data format: 32Bit float

save setup to slot

This command saves the setup-configuration.

Syntax

- Syntax: **[CT] 02 20 [Slot] [CT]**

[Slot]

- Slot number
- Length: 1 byte

Get DC Bias

This command reads the currently configured DC Bias from the device.

Syntax

- Syntax get: **[CT] 01 33 [CT]**
  - Return: **[CT] 05 33 [DC Bias] [CT]**

[DC Bias]

- Length: 4 byte
- Data format: float
- Unit: V

8.5.12 0xB8 - Start Measure

Starts the measurement.

General syntax

[CT] [LE] **[OB] [CD]** [CT]

Return

If time stamp and current range are disabled (see command 0x97 and 0x98)

[CT] 0A **[ID] [Real part] [Imaginary part]** [CT]

Else if time stamp in ms is enabled (see command 0x97 and 0x98)

[CT] 0E **[ID] [Time stamp] [Real part] [Imaginary part]** [CT]

Else if time stamp in µs is enabled (see command 0x97 and 0x98)

[CT] 0F **[ID] [Time stamp] [Real part] [Imaginary part]** [CT]

Else if current range is enabled (see command 0x97 and 0x98)

[CT] 0B **[ID] [Current Range] [Real part] [Imaginary part]** [CT]

Else if time stamp in ms and current range are enabled (see command 0x97 and 0x98)

[CT] 0F **[ID] [Time stamp] [Current Range] [Real part] [Imaginary part]** [CT]

Else if time stamp in µs and current range are enabled (see command 0x97 and 0x98)

[CT] 10 **[ID] [Time stamp] [Current Range] [Real part] [Imaginary part]** [CT]

[ID]

- ID number of the frequency point
- Length: 2 byte
- Data format: unsigned integer

[Time stamp]

- time stamp of the frequency point
- Length: 4 byte or 5 byte (depending on configured time stamp)
- Data format: unsigned integer
- Unit: ms or µs (depending on configured time stamp)

[Current Range]

- the current range of that frequency point; in case of auto ranging this could change from point to point
- Length 1 byte
- Data format: unsigned integer

[Real part]

- real part of impedance
- Length: 4 byte
- Data format: float

[Imaginary part]

- imaginary part of impedance
- Length: 4 byte
- Data format: float

Data format

Please refer to the Appendix for a description of the data format.

Multichannel Measurements

For multichannel measurements, the complete spectrum of each channel is transmitted in the order defined by the channel stack (see 0xB0 “Set FE Settings” for the channel stack configuration).

DC measurements

In case of a DC measurement (excitation frequency = 0Hz), the real part contains the resistance in Ohm and the Imaginary part equals zero.

Stop measurement

Stops the measurement.

Syntax

- Syntax: **[CT] 01 00 [CT]**

Start measurement

Starts the measurement.

Syntax

- Syntax: **[CT] 03 [OB] [CD] [CT]**

[OB]

Function	code
Stop measurement	0x00
Start EIS measurement	0x01
Start EIT measurement	0x02

[CD]

- number of spectra to be measured
- Length: 2byte
- Data format: integer value
- The setting 0 starts a continuous measurement. Send the command (B8 01 00 B8) to stop the continuous run.
- example EIS: B8 03 01 00 01 B8 - to start a measurement and stop it automatically after one measurement spectra per channel configuration.
- example EIT: B8 03 02 00 01 B8 - to start a measurement and stop it automatically after one measurement frame per channel configuration.

ⓘ If the measurement has been started with a repeat greater than 0 no stop command is required. The data will be transmitted as soon as each frequency point has been measured. The system therefore **does not** wait for the spectrum to be completed before it will send the data.

8.5.13 0xB9 - Set Sync Time

Set the synchronization time in µs. (Time between the measurement of two spectra)

ⓘ This is also the time the SyncOut signal is in its low state between two measurements.
Use save settings command (0x90) to save this parameter persistent.

General syntax

[CT] [LE] **[Sync time]** [CT]

Return

ACK

[Sync time]

- Length: 4 byte
- Data format: integer value
- Unit: µs
- Default: 0µs
- Min: 0µs
- Max 180s = 180E6 µs (0x0ABA9500)

8.5.14 0xBA - Get Sync Time

Reads the currently configured synchronization time.

General syntax

[CT] 00 [CT]

Return

[CT] 04 **[Sync time]** [CT]

ACK

[Sync time]

- Length: 4 byte
- Data format: integer value
- Unit: µs

8.5.15 0xBD - Set Ethernet Configuration

Configure DHCP setting and IP address.

General Syntax

[CT] **[LE] [OB] [CD]** [CT]

Return

ACK

[LE]

- Represents the byte count of the command frame and varies with the amount [data] bytes. [length] is always the amount of [data]-bytes + 1.

[CD]

- Command data whose syntax and content depends on the used [option byte].

[OB]

Function	code
IP address	0x01
DHCP on/of	0x03

IP address

- Set static IPv4 address of the device.
- This address will only be used, when DHCP is disabled.
- The default setting is 0.0.0.0
- Syntax: **[CT] [05] [01] [address] [CT]**
- [address]
  - Desired address.
  - Length: 4 byte
  - Data format: byte-wise unsigned integer
  - Value range of each byte: 0 ... 255

DHCP on/off

- Activate/deactivate DHCP usage.
- To apply this setting, a system reboot is required.
- Syntax: **[CT] [02] [03] [switch] [CT]**
- [switch]
  - Length: 1 byte
  - Values:
    - 0x00 - off
    - 0x01 - on (default)

To save these settings persistently, the "save settings" command must be sent before powering off the device.

8.5.16 0xBE - Get Ethernet Configuration

Read out parameters like DHCP setting, MAC and IP address.

General Syntax

[CT] **[LE] [OB]** [CT]

Return

[CT] **[LE] [OB] [CD]** [CT]
ACK

[LE]

- Represents the byte count of the command frame and varies with the amount [data] bytes. [length] is always the amount of [data]-bytes + 1.

[CD]

- Command data whose syntax and content depends on the used [option byte].

[OB]

Function	get
IP address	0x01
MAC address	0x02
DHCP on/off	0x03

IP address

- Read out currently allocated IPv4 address of the device. This is the actual address the device
- Get syntax: **[CT] [01] [01] [CT]**
- Return syntax: **[CT] [05] [01] [address] [CT]**
- [address]
    - Current address.
    - Length: 4 byte
    - Data format: byte-wise unsigned integer

MAC address

- Read out currently allocated MAC address of the device. This is the actual address the device
- Get syntax: **[CT] [01] [02] [CT]**
- Return syntax: **[CT] [07] [02] [address] [CT]**
- [address]
    - MAC address.
    - Length: 6 byte
    - Data format: byte-wise unsigned integer

DHCP on/off

- Read out the current DHCP setting.
- Get syntax: **[CT] [01] [03] [CT]**
- Return Syntax: **[CT] [02] [03] [switch] [CT]**
- [switch]
    - Length: 1 byte
    - Values:
        - 0x00 - off
        - 0x01 - on

8.5.17 0xCF - TCP connection watchdog

General Syntax

[CT] **05 00 [interval]** [CT]

Return

ACK

Description

The device checks periodically if the connection from the TCP socket to the client is still active. This is done by the device by periodically sending the TCP socket acknowledge command (see chapter "Acknowledge messages"). If the command cannot be send, i.e. because of a lost client connection, the socket server state will be reset to "listening". Thus, a new socket-client connection can be established.

This command provides access to setting the interval for sending the TCP socket acknowledge command.

[interval]

- Set the interval for TCP communication watchdog
- Length: 4 Byte
- Data format: unsigned integer
- Values:
  - Minimum: 1 s
  - Maximum: 600 s
  - Increment: 1 s
  - Default: 60 s

⚠ To save these settings persistently, the "save settings" command must be sent before powering off the device.

8.5.18 0xD0 - Get ARM firmware ID

Syntax

D0 00 D0

Return

D0 06 [developer information] **[revision number] [build number]** D0
ACK

Description

- Reads out version number of ARM firmware.

[developer information]

- This information is for internal development purposes only.
- Length: 2 Byte

[revision number]

- Length: 2 Byte
- Data format: unsigned integer

[build number]

- Length: 2 Byte
- Data format: unsigned integer

8.5.19 0xD1 - Get Device ID

Syntax

[CT] 00 [CT]

Return

[CT] [LE] **[general information]** [developer information] [CT]
ACK

Description

- Read out device information
- Includes all information necessary to identify a Sciospec device.

[general information]
This information can also be found on the serial number label on the case of the device.

Byte	Description	Comments
1	version of the general information part	• Defines the format of the following items. There is only version 1 so far. • data format: unsigned integer
2-3	device identifier	• Unique number, which identifies the device types. • data format: unsigned integer
4-5	serial number	• Unique identifying number for all Sciospec devices. • data format: unsigned integer
6-7	date of delivery	• Byte 6: Year since 2010 (0x00 = 2010, ..., 0xFF = 2265) • Byte 7: Month (0x01 = january, ..., 0x0C = december)

[developer information]

This information is for internal development purposes only.

8.5.20 0xD2 - Get FPGA firmware ID

Syntax

D2 00 D2

Return

D2 09 [developer information] **[revision number] [build number]** D2
ACK

Description

- Reads out version number of FPGA firmware.

[developer information]

- This information is for internal development purposes only.
- Length: 5 Byte

[revision number]

- Length: 2 Byte
- Data format: unsigned integer

[build number]

- Length: 2 Byte
- Data format: unsigned integer

8.5.21 Example of impedance compensation procedure

This example demonstrates the communication required to compensate an already configured setup.

Direction of communication:

PC to ISX-3 / ISX-3mini

ISX-3 / ISX-3mini to PC

↓	#	Command	Communication
	1	Initiating the compensation routine	**B6** **01** **10** **B6**
	2	Acknowledge	**18** **01** **83** **18**
	3	Request to leave the terminals open	**B7** **01** **13** **B7**
	4	compensation acknowledge	**B6** **01** **11** **B6**
	5	Acknowledge	**18** **01** **83** **18**
	compensation measurement on open terminal is performed		
	6	Request to short all terminals	**B7** **01** **14** **B7**
	7	compensation acknowledge	**B6** **01** **11** **B6**
	8	Acknowledge	**18** **01** **83** **18**
	compensation measurement of shorted terminals is performed		
	9	Request to connect a known load impedance to the device	**B7** **01** **15** **B7**
	10	Set known value of the load impedance (e.g. 1000hms)	**B6** **06** **16** **01** **42** **C8** **00** **00** **B6**

	11	Acknowledge	**18** **01** **83** **18**
	12	compensation acknowledge	**B6** **01** **11** **B6**
	13	Acknowledge	**18** **01** **83** **18**
	compensation measurement of load terminals is performed		
	14	compensation done	**B7** **01** **10** **B7**

8.5.22 Example of dc bias compensation procedure

This example demonstrates the communication required to compensation an already configured setup. It is essential the frontend settings (channel, measurement mode and range) are correctly initialized before starting the dc bias compensation routine

Direction of communication:

PC to ISX-3 / ISX-3mini

ISX-3 / ISX-3mini to PC

↓	#	Command	Communication
	1	Setting the dc bias voltage	**B6** **05** **33** **3F** **00** **00** **00** **B6**
	2	Acknowledge	**18** **01** **83** **18**
	3	Init the dc bias compensation	**B6** **01** **30** **B6**
	4	Acknowledge	**18** **01** **83** **18**
	compensation measurement is performed		
	5	compensation DC Bias acknowledge	**B7** **01** **31** **B7**
	6	Getting the dc bias voltage	**B7** **01** **33** **B7**
	7	DC Bias	**B7** **05** **33** **3E** **FF** **FE** **B0** **B7**

	8	Acknowledge	18 01 83 B18

8.6 Code Example (ANSI C)

```c
/*

============================================================================
==
 Name        : ISX-3_C-Code.c
 Author      : Sciospec
 Version     : Revision 1
 Copyright   : Sciospec 2022
 Description: ISX-3 Communication Demo in C, Ansi-style

============================================================================
==
 */


#include <stdio.h>
#include <stdlib.h>

byte connectToDevice(HANDLE* handle);
byte readAck(HANDLE handle);
void readData(HANDLE handle, byte* buffer, int bytesToRead);
void writeDataToDevice(HANDLE handle, byte* data, DWORD dataCount);

int main(void) {

    HANDLE handle;
    if(!connectToDevice(&handle))
        return -1;

    //This will force stdout to be unbuffered.
    setvbuf (stdout, NULL, _IONBF, 0);

    printf("Connection established\n");
    byte *cmd, *readBuffer;



    /* ************************** *
     *                            *
     *     Initialize Setup       *
     *                            *
     * ************************** */
    printf("Initialize Setup.\n");
    int numberOfBytes = 4;
    cmd = (byte*)malloc(sizeof(byte)*numberOfBytes);
        cmd[0] = 0xB6;  cmd[1] = 0x01; cmd[2] = 0x01; cmd[3] = 0xB6;
```

```
43          writeDataToDevice(handle, cmd, numberOfBytes);
44      free(cmd);
45      readAck(handle);
46      printf("\n");
47
48
49      /* *************************** *
50       *                             *
51       *     Initialize Freq.Block   *
52       *                             *
53       * *************************** */
54
55      float startFrequency    = 100;      //100Hz
56      float stopFrequency     = 100e3;    //100kHz
57      float frequencyCount    = 80;
58      float precision         = 1;
59      float amplitude         = 0.1;  //V
60      byte   scale            = 1;    //log
61      printf("Set Setup-config: Frequency-block (%fHz .. %fHz, scale=%i,
    prec=%f, amplitude=%fV)\n", startFrequency, stopFrequency, scale,
    precision, amplitude);
62      numberOfBytes = 0x16 + 3;
63      cmd = (byte*)malloc(sizeof(byte)*numberOfBytes);
64          cmd[0] = 0xB6;  cmd[1] = numberOfBytes-3; cmd[2] = 0x03;
65
66          unsigned long tmp = *(unsigned long*)&startFrequency;
67          byte counter = 3;
68          cmd[counter++] = (tmp>>24)&0xFF;
69          cmd[counter++] = (tmp>>16)&0xFF;
70          cmd[counter++] = (tmp>>8)&0xFF;
71          cmd[counter++] = (tmp)&0xFF;
72
73          tmp = *(unsigned long*)&stopFrequency;
74          cmd[counter++] = (tmp>>24)&0xFF;
75          cmd[counter++] = (tmp>>16)&0xFF;
76          cmd[counter++] = (tmp>>8)&0xFF;
77          cmd[counter++] = (tmp)&0xFF;
78
79          tmp = *(unsigned long*)&frequencyCount;
80          cmd[counter++] = (tmp>>24)&0xFF;
81          cmd[counter++] = (tmp>>16)&0xFF;
82          cmd[counter++] = (tmp>>8)&0xFF;
83          cmd[counter++] = (tmp)&0xFF;
84
85          cmd[counter++] = scale;
86
87          tmp = *(unsigned long*)&precision;
88          cmd[counter++] = (tmp>>24)&0xFF;
89          cmd[counter++] = (tmp>>16)&0xFF;
90          cmd[counter++] = (tmp>>8)&0xFF;
91          cmd[counter++] = (tmp)&0xFF;
92
93          tmp = *(unsigned long*)&litude;
94          cmd[counter++] = (tmp>>24)&0xFF;
```

```
        cmd[counter++] = (tmp>>16)&0xFF;
        cmd[counter++] = (tmp>>8)&0xFF;
        cmd[counter++] = (tmp)&0xFF;

        cmd[numberOfBytes-1] = 0xB6;
        writeDataToDevice(handle, cmd, numberOfBytes);
    free(cmd);

    readAck(handle);
    printf("\n");


    /* *************************** *
     *                             *
     *     Set FrontEnd Settings   *
     *                             *
     * *************************** */
    printf("Clear Channel Stack\n");
    numberOfBytes = 6;
    cmd = (byte*)malloc(sizeof(byte)*numberOfBytes);
    cmd[0] = 0xB0;    cmd[1] = 0x03;    cmd[numberOfBytes-1] = cmd[0];
    cmd[2] = 0xFF;    cmd[3] = 0xFF;    cmd[4] = 0xFF; //Delete channel
stack
    writeDataToDevice(handle, cmd, numberOfBytes);
    readAck(handle);

    printf("\n");    printf("Set Channel Stack: to channel 1\n");
    cmd[3] = 0x02;    //mode = 4Pt measurement
    cmd[4] = 0x01;    //add channel: Channel 1 (BNC),
    cmd[5] = 0x01;    //100Ohm Range,
    writeDataToDevice(handle, cmd, numberOfBytes);
    readAck(handle);
    printf("\n");

    free(cmd);

    /* *************************** *
     *                             *
     *     Start Measurement       *
     *                             *
     * *************************** */
    byte numberOfSpecs = 1;
    printf("Start Measurement\n");
    numberOfBytes = 6;
    cmd = (byte*)malloc(sizeof(byte)*numberOfBytes);
    cmd[0] = 0xB8;  cmd[1] = 0x03;
    cmd[2] = 0x01;
    cmd[3] = 0;
    cmd[4] = 0;
    cmd[numberOfBytes-1] = 0xB8;
    writeDataToDevice(handle, cmd, numberOfBytes);
    free(cmd);
    readAck(handle);
    printf("\n");
```

```
148
149
150     /* ******************** *
151      *                      *
152      *     Receive Specs    *
153      *                      *
154      * ******************** */
155     readBuffer = malloc(14);    //3Byte Framing, 2Byte idNumber, 4Byte RE,
    4Byte Im
156     byte j;
157     byte i;
158     UINT8 ch;
159     UINT16 id;
160     UINT32 tmp32;
161     float re, im;
162     for(j=0; j<numberOfSpecs; j++){
163         printf("Spec#%i:\n", j+1);
164         printf("ch\tid\tre\tim\n");
165         for(i=0; i<frequencyCount;i++){
166
167             readData(handle, readBuffer, 14);
168
169             ch = readBuffer[2];
170             id = (readBuffer[3]<<8) + readBuffer[4];
171             tmp32 = (readBuffer[5]<<24) + (readBuffer[6]<<16) +
    (readBuffer[7]<<8) + (readBuffer[8]);
172             re = *(float*)&tmp32;
173             tmp32 = (readBuffer[9]<<24) + (readBuffer[10]<<16) +
    (readBuffer[11]<<8) + (readBuffer[12]);
174             im = *(float*)&tmp32;
175
176             printf("%i\t%i\t%f\t%f\n", ch, id, re, im);
177         }
178         printf("\n");
179     }
180
181     printf("Stop Measurement");
182     numberOfBytes = 4;
183     cmd = (byte*)malloc(sizeof(byte)*numberOfBytes);
184     cmd[0] = 0xB8;  cmd[1] = 0x01;
185     cmd[2] = 0x00;
186     cmd[numberOfBytes-1] = 0xB8;
187     writeDataToDevice(handle, cmd, numberOfBytes);
188     free(cmd);
189     readAck(handle);
190     printf("\n");
191
192     CloseHandle(handle);//Closing the Serial Port
193
194 //  sprintf();
195
196     return 1;
197 }
198
```

```
199 /*
200  * Connect to COM6 if available
201  * @param handle - the pointer to USB-Conection-Handle
202  * @return True if connection was successfull else false
203  */
204 byte connectToDevice(HANDLE* hSerial){
205     *hSerial = CreateFile(  "\\\\.\\COM6",
206                             GENERIC_READ|GENERIC_WRITE,
207                             0,
208                             NULL,
209                             OPEN_EXISTING,
210                             FILE_ATTRIBUTE_NORMAL,
211                             NULL );
212         if (hSerial == INVALID_HANDLE_VALUE)
213         {
214                 fprintf(stderr, "Error\n");
215                 return 0;
216         }
217 
218         return 1;
219 }
220 
221 void writeDataToDevice(HANDLE handle, byte* cmd, DWORD dataCount){
222 
223     byte i;
224     for(i=0; i<dataCount;i++)
225         printf("%.2X ", cmd[i]);
226     printf("\n");
227 
228     WriteFile(handle, cmd, dataCount, NULL, NULL);
229 
230 
231 }
232 
233 /** Tries to Read ACK Frame
234  *
235  * @return true if ACK is received, else false
236  */
237 byte readAck(HANDLE handle){
238 
239     int i = 0;
240     byte* readBuffer;
241     readBuffer = (byte*)malloc(sizeof(byte)*4);
242 
243     readData(handle, readBuffer, 4);
244 
245     printf("ACK-Frame: ");
246     for(i=0; i<4;i++){
247         printf("%.2X ", readBuffer[i]);
248     }
249     printf("\n");
250 
251     if(readBuffer[2] == 0x83){
252         free(readBuffer);
```

```
253             return TRUE;    //ACK
254         }else{
255             free(readBuffer);
256             return FALSE;   //NOT ACK
257         }
258     }
259
260     void readData(HANDLE handle, byte* buffer, int bytesToRead){
261         int i;
262         for (i = 0; i < bytesToRead; i++) {
263             unsigned char ch = 0;
264             DWORD read = 0;
265
266             /* Wait for characters */
267             while(!read) {
268                 ReadFile(handle, &ch, 1, &read, NULL);
269                 if (read) {
270                     buffer[i] = ch;
271                 }
272             }
273         }
274     }
```

9 Features

9.1 MUX32 / MUX64

9.1.1 Technical Specifications

General Overview

The MUX32 / MUX64 is a multiplexer module which offers 32 channels or 64 channels. Every channel can be any of the functions: Counter, Reference, Working Sense or Work in 2, 3 or 4 point configuration. Measurements can be configured from any to any other channel(s). Measurements are acquired sequentially (multiplexed).

- 32 /64 channel multiplexer module with any to any functional multiplexing for 2, 3 and 4 electrode configurations
- high isolation, ultra-low parasitic reed relay switches with superior performance compared to semiconductor switches
  - >10 TΩ || 0.4 pF isolation at power off or software selected cell disconnect
  - extremely low parasitic capacitance < 0.7 pF offers close to no cell loading
  - max. series resistance 200 mΩ in connected state
  - zero leakage current in on state
  - zero charge injection, no distortion, no noise contributions beyond thermal noise
- 4 IOs (digital IOs with highly flexible digital IO functions through the Sciospec ***InterfacePort*** standard)
- available frontpanel connectors: edge-card connector (MEC1-150-02-F-D-EM2), MDR40 connector (10240-1210PE). The mating connectors for the MDR40 connector are the 10140-3000PE (solder lugs) or the 10140-6000EC (IDC wire mount).

ⓘ All specifications are stated for a operating at temperatures of 0°C to 40°C unless specified otherwise.
Warm-up time must be greater than or equal to 30 minutes after power on to comply with all specifications.

measurement port protection



Be aware that the measurement terminals are very sensitive to electrostatic discharge, over-current and over-voltage. Protection of the terminals of the instrument and strict adherence to the specified maximum ratings has to be ensured by the user. For further information on how to connect specific device under test to the instrument contact Sciospec or an authorized representative directly.

9.1.2 General Specifications

dimensions MUX32	115.6mm x 35.3mm x 156.3mm (width x height x depth) see Figure 4
dimensions MUX64	115.6mm x 48.7mm x 156.3mm (width x height x depth) see Figure 5
weight MUX32 / MUX64	1 kg
operating conditions	0°C to 40°C, <80% relative humidity non condensing, 0...3000m altitude
non-operating conditions	-25°C to 80°C, <80% relative humidity non condensing The temperature gradient should not exceed 1K/min to reach operating conditions.

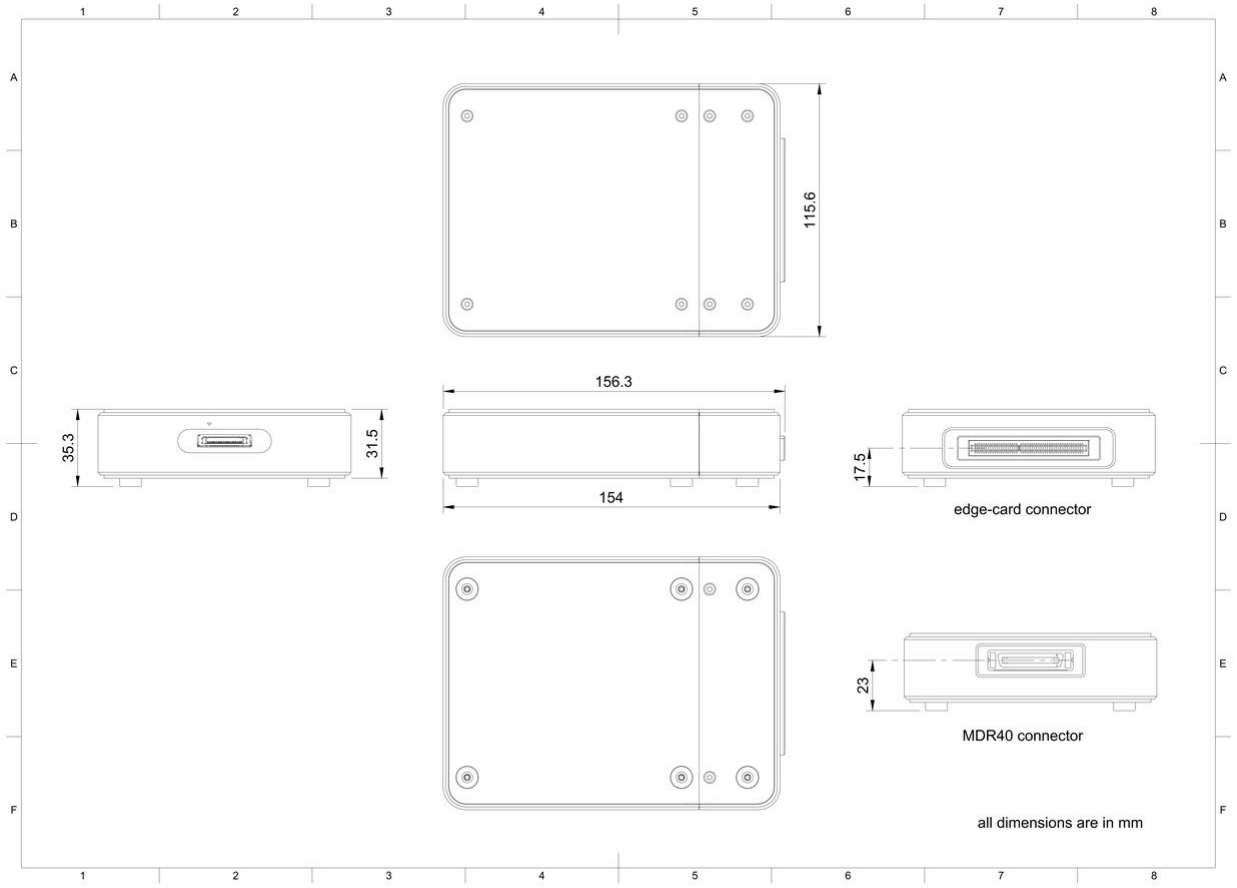


Figure 3: dimensioned drawings of the MUX32

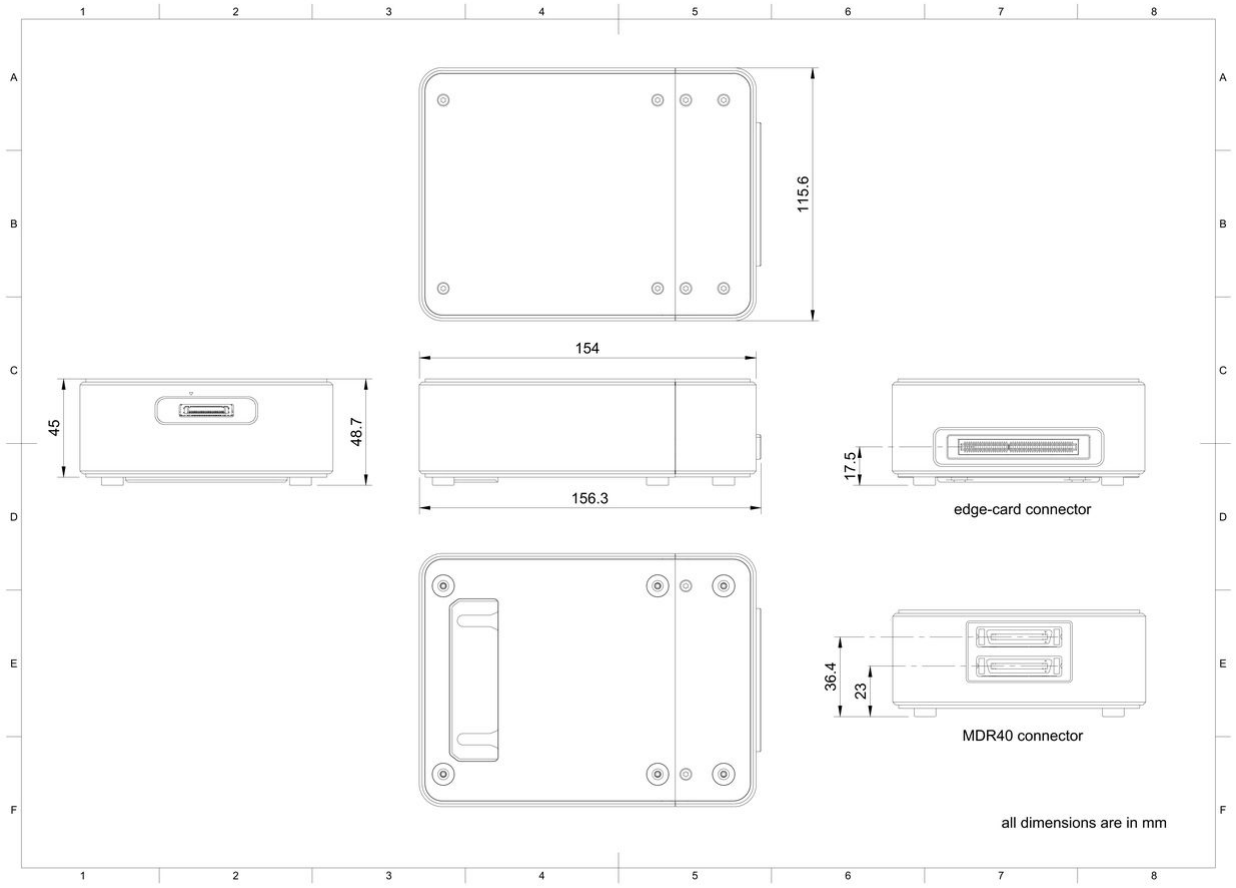


Figure 4: dimensioned drawings of the MUX64

9.1.3 Measurement Software Description

The measurement channel of the *MUX32 / MUX64* is done in the Channel-Settings dialog by selecting **MUX32/64** as the primary channel, see Figure 3. Now the 2/3 or 4 point configuration and channel selection can be configured.

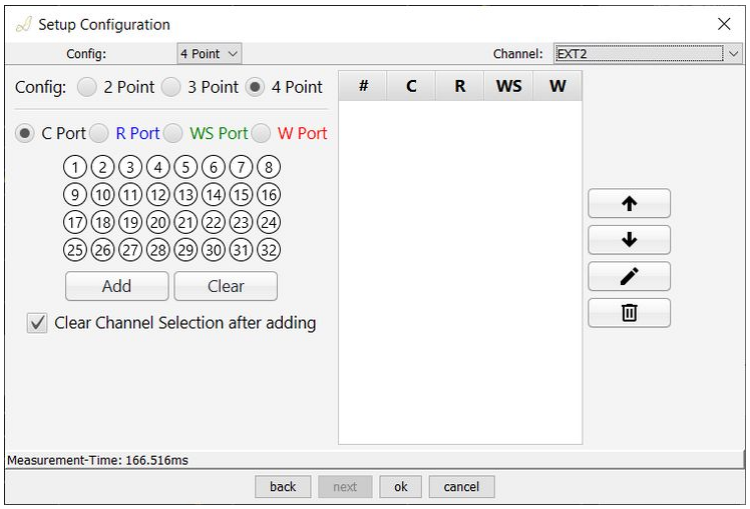
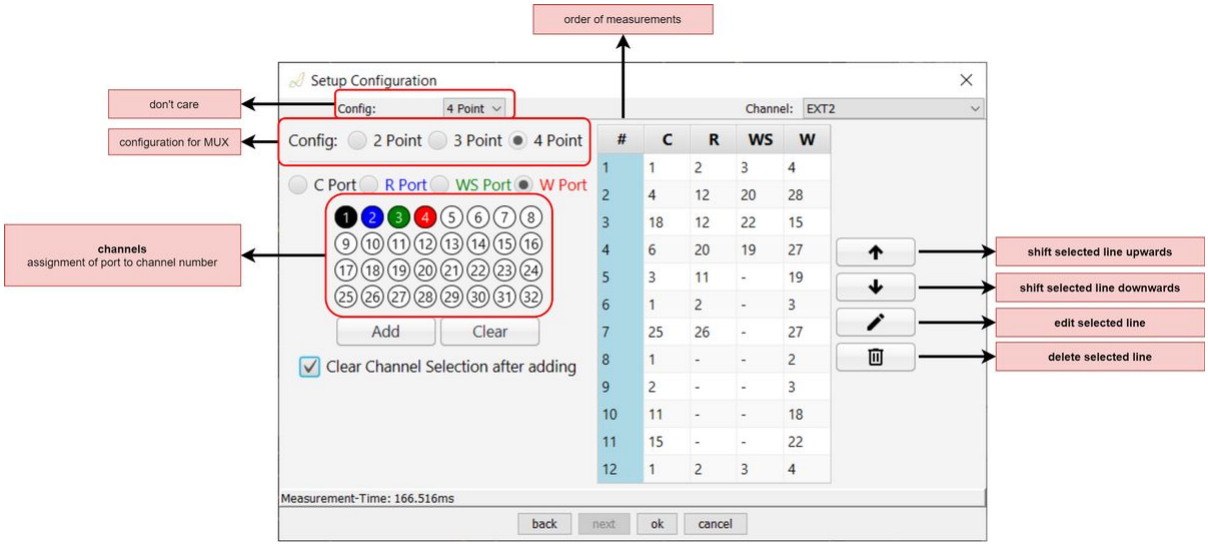


Figure 5: Channel configuration dialog



9.1.4 Sciospec COMInterface

Command list

The leading hex code of each command heading represents the [command code] of the respective function.

- 0xB3 - Get ExtensionPort Channel
  - 0xB2 - Add ExtensionPort Channel
  - 0xB3 - Get ExtensionPort Channel
  - 0xCB - FrontIOs

Command description

0xB2 - Add ExtensionPort Channel

Adding an ExtensionPort channel configuration to the stack.

> The device has a stack length of **128** (for InternalMux). So only 128 ExtensionPort Channel configurations can be stored. Sending the set ExtensionPort Channel command more than 128 times results in a NACK return. The stack will be emptied using the "empty FE settings" command: **B0 03 FF FF FF B0**

General syntax

[CT] 04 [CP] [RP] [WS] [WP] [CT]

Return

ACK

ExtensionPort- channel settings

[CP]	(C-Port): Counter - Port Selection

[RP]	(R-Port): Reference - Port Selection
[WS]	(WS-Port): Working Sense - Port Selection
[WP]	(W-Port): Work - Port Selection

All Ports numbers are send in integer straight binary numbers running from 1 to 32.

Example

- B2 04 **01 02 03 04** B2
  Adding the channel configuration C1, R2, WS3 and W4 for 4 point configuration
- B2 04 **01 02 03 03** B2
  Adding the channel configuration C1, R2 and W3 for 3 point configuration
- B2 04 **01 01 02 02** B2
  Adding the channel configuration C1 and W2 for 2 point configuration

0xB3 - Get ExtensionPort Channel

Read the currently set ExtensionPort configuration stack.

General syntax

[CT] 00 [CT]

Return

[CT] [LE] [channel config 1] [channel config 2] .... [channel config n] [CT] *

ACK

* O*ne data frame is limited to a total of 255 bytes of data. If the returned data exceed this amount of bytes, the returning command will be split into multiple separate frames.*

[LE]

- Length: 1 byte
- depends on the number of configuration currently set

[channel config #]

- # = 1, 2 ... n
- Length: 4 byte (for 2 /3 and 4 point configuration, not used ports are filled with 00)
- **[channel config #] = [CP] [RP] [WS] [WP]**

CP	(C-Port): Counter - Port Selection
RP	(R-Port): Reference - Port Selection
WS	(WS-Port): Working Sense - Port Selection
WP	(W-Port): Work - Port Selection

Example:

B3 0C 01 02 03 04 09 0A 0B 0C 11 00 00 14 B3

- 01 02 03 04 → C = 1, R = 2, Ws = 3, W = 4
- 09 0A 0B 0C → C = 9, R = 10, Ws = 11, W = 12
- 11 00 00 14 → C = 17, W = 20 - 2-point config

0xCB - FrontIOs

Configure and read the state of the available IOs.

General Syntax

[CT] **[LE] [OB] [CD]** [CT]

Return

[CT] [LE] [OB] [CD] [CT] *

ACK

* only for get-commands

[LE]

- Length: 1 byte
- depends on the number of configuration currently set

[OB]

description	set	get
IO direction	0x01	0x81
IO state	0x02	0x82

IO direction

Set/get the direction of an IO port

- Syntax set: **[CT] 03 01 [channel] [direction] [CT]**
  - Return: **ACK**
- Syntax get: **[CT] 02 81 [channel] [CT]**
  - Return: **[CT] 03 81 [channel] [direction] [CT]; ACK**
- Length: 1 byte
- Data format: unsigned integer

[channel]

- channel to set/get IO direction
- Length: 1 byte
- Data format: unsigned integer
- Interpretation:
  - 0x01 - IO 1
  - 0x02 - IO 2
  - 0x03 - IO 3
  - 0x04 - IO 4

[direction]

- direction of requested channel
- Length: 1 byte
- Data format: unsigned integer
- Interpretation:
  - 0x00 - input
  - 0x01 - output
- default: 0x00

IO state

Get / set the input/output state of an IO port

- Syntax set: **[CT] 03 02 [channel] [state] [CT]**
  - Return: **ACK**
- Syntax get: **[CT] 02 81 [channel] [CT]**
  - Return: **[CT] 03 82 [channel] [state] [CT]; ACK**
- Length: 1 byte
- Data format: unsigned integer

[channel]

- channel to set/get IO direction
- Length: 1 byte
- Data format: unsigned integer
- Interpretation:
  - 0x01 - IO 1
  - 0x02 - IO 2
  - 0x03 - IO 3
  - 0x04 - IO 4

[state]

- State of requested channel
- Length: 1 byte
- Data format: unsigned integer
- Interpretation:
  - 0x00 - low
  - 0x01 - high
- Default: 0x00

Example:

- Set IO direction for IO 1 to output
  - Send: CB 03 01 01 01 CB
  - Return: ACK
- Set IO state of IO 1 to low
  - Send: CB 03 02 01 00 CB
  - Return: ACK

- Get Input state of IO 1:
  - Send: CB 02 82 01 CB
  - Return: CB 03 82 01 01 CB
    ACK

9.2 IsoIOport

9.2.1 Functional Description

The *IsoIOport* is Sciospecs interface standard for connecting peripheral equipment while keeping it isolated from the power line with electrical ratings adequate for medical safety requirements. The standard offers

- Pins:
  - 8 isolated output pins
  - 8 isolated input pins
  - 2x isolated I2C on 4 additional pins (1x independent; 1x shared functionality with two of the outputs)
  - isolated 3.3V output rail
- Functionality:
  - Input and output ports.
  - 1x SPI Master on Bank A and B, assignable to any input/output
  - 1x asynchronous serial interface (UART) on Bank A and B, assignable to any input/output
  - 1x I^2C interface on Bank B and C, assignable to dedicated I^2C-ports
  - synchronization input port(s) and synchronization output port(s)

⚠ Due to the directional nature of the isolators some additional restrictions have to be respected when assigning IO functionality and when assigning serial interface line to a particular input or output.
- Input functions such as 'SPI SCLK' or 'UART Rx' may only be asserted to input ports.
- Output functions such as 'SPI MOSI' or 'UART Tx' may only be asserted to output ports.
- The I^2C function may only be assigned to the dedicated I^2C-ports.

The *Sciospec IsoIOport* standard provides two IO banks with each 8 pins, which can be nearly freely configured to serve as a number of different interface types. These particular interface functionalities are described below.

9.2.2 Interfaces

Digital Inputs

- This is the default configuration for all input ports.
- A weak pull up resistor (10kΩ) to VDD is connected to each I/O.
- The logic stack of all I/Os can be read out by command ("get IO state register"). The command provides a bit vector, where every bit represents the actual logic state of one particular I/O, even if it is configured as "digital output".

Digital Outputs

- This is the default configuration for all output ports.
- The logic state of each pin can be set independently, using the "set output register" command.
- Default state of this register ist 0 for every I/O.

SPI Master

- Type: 4-wire (MOSI as well as MISO can be left unused).
- The polarity of the SPI (CPOL) is configurable by command ("set SPI polarity").
- CPHA = 0
- The SCLK-frequency is configurable within the range
  (mSPI_f_SCLK_source ÷ 2) ≥ f_SCLK ≥ mSPI_f_SCLK_min
- Word length = 8 Bit
- Send and receive a maximum of "mSPI_max_byte" Byte in a single communication process framed by #CS = '0'.
- Data, read from MOSI, always will be transmitted to the master interface where the command came from, which initialized the communication process.
- Application dependend parameters:
  - mSPI_f_SCLK_source
  - mSPI_f_SCLK_min
  - mSPI_max_byte

UART

- Parameters:
  - Baud is rate configurable from BaudrateMin to BaudrateMax (Baudrate = BaudrateMax/ BaudrateDivider).
  - Parity configurable (odd, even, no parity)
  - Startbit = 1
  - Stopbit = 1
  - Idle state = high
  - Receive timeout always is 10 word durations.
  - Send and receive a maximum of 128 Bytes in one continuous transmission.
- Fully asynchronous reception
- Configurable output interface for received data (any of the master interfaces)

I^2C

- SCK and SCL line open drain
- Parameters:
  - fSCK = 100kBaud
  - Send and Receive a maximum 15 Bytes

Synchronization Signals

For synchronization of the measurement process with other customer hardware the Sciospec device can be equipped with the Synchronization Signals on the *IsoIOport*. It offers one “Sync Out” and one “Sync In” signals located at the *IsoIOport* of the instrument.

The “Sync Out” port gives information about the current state of the measurement.

- Logic high → Measurement is running
- Logic low → Measurement is paused between two measurement sweeps.

The “Sync In” port allows to control the measurement process

- By default the signal is logic high, which allows for a continuous measurement.
- By connecting this port to ground (logic low) the next measurement will be paused until the signal is released again.
- An already running sweep will not be interrupted by the “Sync In” signal.

⚠ Do not apply any voltage source directly to the synchronization ports!

9.2.3 Technical Specifications

isolated inputs		
high level input voltage	0.7 × Vsupply	
	2.31 V (min)	@ 3.3 V supply
low level input voltage	0.3 × Vsupply	
	0.66 V (max)	@ 3.3 V supply
input current (max)	±10 µA (max)	
absolute max input voltage	Vsupply + 0.5 V	
data rate	0...100 Mbps	
pull-up resistors	10 kΩ (default)	to Vsupply
Isolation ratings	5kVrms, 12.8 kV surge, ±100 kV/µs Typical CMTI reinforced insulation according to 60950-1 (800 Vrms) and 60601-1 (2 MOPP @ 250Vrms)	
ESD protection	±8 kV contact discharge protection per IEC 61000-4-2 ±6 kV human body model (HBM), per ANSI/ESDA/JEDEC JS-001 ±1.5 kV charged device model (CDM) per JESD22-C101	
isolated outputs		

high level output voltage	3V (min)	@ 3.3V supply, Iout ≤ 2 mA
	4.6V (min)	@ 5V supply, Iout ≤ 4 mA
low level output voltage	0.3 V (max)	@ 3.3V supply, Iout ≤ 2 mA
	0.4V (max)	@ 5V supply, Iout ≤ 4 mA
output current capability	±2 mA (min)	@ 3.3V supply
	±4 mA (min)	@ 5V supply
isolation ratings	5kVrms, 12.8 kV surge, ±100 kV/µs Typical CMTI reinforced insulation according to 60950-1 (800 Vrms) and 60601-1 (2 MOPP @ 250Vrms)	
ESD protection	±8 kV contact discharge protection per IEC 61000-4-2 ±6 kV human body model (HBM), per ANSI/ESDA/JEDEC JS-001 ±1.5 kV charged device model (CDM) per JESD22-C101	
isolated bidirectional open drain I^2C ports		
high level input voltage	0.7 x Vsupply (min)	
low level input voltage	0.3 x Vsupply (max)	
low level output voltage	900 mV (max)	
current sink capability	30 mA (max)	
pull-up resistors	10 kΩ (default)	to Vsupply
operating frequency	400 kbps (max)	
Isolation ratings	5kVrms reinforced insulation according to 60950-1 (380 Vrms) and 60601-1 (2 MOPP @ 250Vrms)	
ESD protection	±12 kV contact discharge protection per IEC 61000-4-2 ±15 kV air gap discharge protection per IEC 61000-4-2	
isolated supply		
voltage	3.3 V (typ)	± 180 mV @ 0...10 mA
output current capability	60 mA (max)	
line regulation	2 mV/V (typ)	
load regulation	5% (max)	@ 10...54 mA

output ripple	50 mVpp	20 MHz BW, Cload = 0.1 µF||10 µF, Iout = 54 mA
output noise	130 mVpp	Cload = 0.1 µF||10 µF, Iout = 54 mA
Isolation ratings	5kVrms reinforced insulation according to 60950-1 (415 Vrms), 60601-1 (2 MOPP @ 250Vrms) and 61010-1 (300Vrms mains, 415 V secondary)	
ESD protection	±15 kV air gap discharge protection per IEC 61000-4-2	

9.2.4 Pinout and *InterfacePort* mapping

Used connector type

10240-1210PE, 40 pin .050" Mini D Ribbon (MDR) receptible, SMT right angle, shielded

Pin	Name	*InterfacePort* mapping	Default Function	*Important information*
1	GNDiso			
2	GNDiso			
3	reserved			
4	reserved			
5	reserved			
6	reserved			
7	PowerPlugDetect			High if DC Power Plug is connected Low if DC Power Plug is disconnected Status can also be read by using the Sciospec COMInterface 0xCC Only available for EIT systems
8	I^2C SCL, bank C	Bank C, IO 1	I^2C SCL	Preconfigured *InterfacePort* settings. No configuration necessary and possible!
9	I^2C SDA, bank C	Bank C, IO 2	I^2C SDA	
10	in 8	Bank B, IO 8	input	
11	in 7	Bank B, IO 7	input	
12	in 6	Bank B, IO 6	input	
13	in 5	Bank B, IO 5	input	

Pin	Name	*InterfacePort* mapping	Default Function	*Important information*
14	in 4	Bank A, IO 8	input	
15	in 3	Bank A, IO 7	input	
16	in 2	Bank A, IO 6	input	
17	reserved			
18	reserved			
19	GNDiso			
20	GNDiso			
21	GNDiso			
22	GNDiso			
23	VDDiso			
24	GNDiso			
25	VDDiso			
26	I^2C SCL	Bank B, IO 4	output	Shares functionality with "out 8". This pin may be used as I^2C interface.
27	I^2C SDA	Bank B, IO 3	output	Shares functionality with "out 7". This pin may be used as I^2C interface.
28	n.c.			
29	in 1	Bank A, IO 5	input	
30	out 8	Bank B, IO 4	output	Shares functionality with "I^2C SCL". This pin may **not** be used as I^2C interface.
31	out 7	Bank B, IO 3	output	Shares functionality with "I^2C SDA". This pin may **not** be used as I^2C interface.
32	out 6	Bank B, IO 2	output	
33	out 5	Bank B, IO 1	output	
34	out 4	Bank A, IO 4	output	
35	out 3	Bank A, IO 3	output	

Pin	Name	*InterfacePort* mapping	Default Function	*Important information*
36	out 2	Bank A, IO 2	output	
37	out 1	Bank A, IO 1	output	
38	reserved			
39	GNDiso			
40	GNDiso			

9.2.5 Sciospec COMInterface

9.2.6 0xCA - Configure IsoIOport

General Syntax

[CT] [LE] **[OB] [bank] [command]** [CT]

Return

[CT] [LE] **[OB] [bank] [command]** [CT]*

ACK

** This frame will only be returned (1) on get commands, (2) on a read operation on synchronous interfaces or (3) if data has been received on asynchronous interfaces.*

Due to the directional nature of the isolators some additional restrictions have to be respected when assigning IO functionality and when assigning serial interface line to a particular input or output.

- Input functions such as 'SPI SCLK' or 'UART Rx' may only be asserted to input ports.
- Output functions such as 'SPI MOSI' or 'UART Tx' may only be asserted to output ports.
- The I^2C function may only be assigned to the dedicated I^2C-ports.

[LE]

- Represents the byte count of the command frame and varies with the amount [data] bytes.

[OB]

Function	set	get
IO functionality	0x00	0x80
UART Rx data sink	0x01	0x81

Function	set	get
interface data	0x02	0x82
interface settings	0x03	0x83

[bank]

- Select the target bank of a command. In a returned frame this element indicates the source bank.
- Length: 1 byte
- Data format: unsigned integer
- Values:
  - 0x01 ... Bank A
  - 0x02 ... Bank B
  - 0x03 ... Bank C

IO functionality

- Defines the functionality of a certain Input or Output.
- Syntax set: **[CT] 04 00 [bank] [IO#] [function] [CT]**
- Syntax get: **[CT] 03 80 [bank] [IO#] [CT]**
  - Return get: **[CT] 04 80 [bank] [IO#] [function] [CT]**
- [IO#]:
  - Defines the target Input or Output of the command.
  - Values:
    - 0x01 ... 0x08 for IO 1 to 8
- [function]:
  - Defines the functionality that will be assigned to the Input or Output.
  - Values:
    - 0x00 - Input
    - 0x01 - Output
    - 0x02 - I^2C SCL (only for Bank B, IO 4 and IO 3)
    - 0x03 - I^2C SDA (only for Bank B, IO 4 and IO 3)
    - 0x04 - SPI Master SCLK
    - 0x05 - SPI Master CS#
    - 0x06 - SPI Master MOSI
    - 0x07 - SPI Master MISO
    - 0x08 - UART Tx
    - 0x09 - UART Rx
    - 0x0A - Sync Out
    - 0x0B - Sync In

UART Rx data sink

- Defines output interface for received data of the asynchronous peripheral interfaces.
- Syntax set: **[CT] 04 01 [bank] [interface] [sink] [CT]**
- Syntax get: **[CT] 03 81 [bank] [interface] [CT]**
  - Return get: **[CT] 04 81 [bank] [interface] [sink] [CT]**
- [interface]:
  - Names the source interface of data which a certain sink should be assigned to.

    - Values:
        - 0x04 - UART Rx
- [sink]:
    - Values:
        - 0x00 - High speed USB
        - 0x01 - Full speed USB (default)

interface data

- Commit/Receive data to a certain peripheral interface.
- Syntax set: **[CT] [LE] 02 [bank] [interface] [data] [CT]**
- Syntax get: **[CT] 03 82 [bank] [interface] [CT]** (only defined for [interface] = "Input register" and "Output register")
- Return: **[CT] [LE] 82 [bank] [interface] [data] [CT]**
- [interface]:
    - Defines the target interface of the command. The content of [data] will be committed to this interface.
    - Values:
        - 0x00 - IO state register
        - 0x01 - Output register
        - 0x02 - I^2C
        - 0x03 - SPI Master
        - 0x04 - UART
- [data]:
    - Carries the data for/from the target interface. Length and structure of this element depends on the interface and the amount of transmitted data.
    - IO state register
        - Used only within "get" command. Returns NACK when used with "set" command.
        - [data] contains one byte where every bit represents the actual logic state of one certain Input or Output. 1st bit = IO1, 2nd bit = IO2 and so on.
        - Bit order: most significant bit first
        - [data] is part of the return frame which will be sent on the "get IO state register" command.
    - Output register
        - [data] contains one byte where every bit represents the logic state which will be driven to the output pin. 1st bit = IO1, 2nd bit = IO2 and so on.
        - Bit order: most significant bit first
        - [data] is both part of the "set output register" command and the return on the "get output register" command.
    - I^2C
        - Used only within "set" command. Returns NACK when used with "get" command.
        - Write process: [data] = **[I^2C address + R/W bit] [txDataCount] [txData]**
            - [I^2C address + R/W bit]
                - I^2C address:
                    - Address of the I^2C device
                    - length: 7 Bit
                    - data format: unsigned integer
                - R/W bit

- Indicates the communication direction.
- must be '0' (= Write)

- [txDataCount]
  - Number of bytes to be sent.
  - length: 1 Byte
  - data format: unsigned integer
  - value margin: 0 ... 15
- [txData]
  - Actual data to be sent.
  - length: Must match [txDataCount].

- Read process: [data] = **[I²C address + R/W bit] [txDataCount] [txData] [rxDataCount]**
  - [I²C address + R/W bit]
    - I²C address:
      - Address of the I²C device
      - length: 7 Bit
      - data format: unsigned integer
    - R/W bit
      - Indicates the communication direction.
      - '0' = Write
      - '1' = Read
  - [txDataCount]
    - Number of bytes to be sent before read operation is executed.
    - length: 1 Byte
    - data format: unsigned integer
    - value margin: 0 ... 15
    - ℹ If [txDataCount] is ≠ 0, then a sequential read process is executed. First [txData] is sent to the device. The stop condition of this write process is omitted. Then [rxDataCount] is read from the device.
  - [txData]
    - Actual data to be sent.
    - length: Must match [txDataCount].
  - [rxDataCount]
    - Number of bytes to be received.
    - length: 1 Byte
    - data format: unsigned integer
    - value margin: 0 ... 15
- Return: [data] contains only the data read from the I²C slave. A return frame will only be sent after a successful read process.

- SPI Master
  - Used only within "set" command. Returns NACK when used with "get" command.
  - ⚠ Read process: Besides the data to be written to the MOSI line the [data] part of the set command contains an additional byte before this write data, which carries the information how many bytes shall be read after finishing the transmission of the data to be written. Data format of this byte is unsigned integer. (See example below)
  - In a return, which will be sent on a set command where the number of bytes to read ≠ 0, [data] only contains the read data bytes.

        - Constraint: [data] can contain a maximum of 200 bytes.
    - UART
        - Used only within "set" command. Returns NACK when used with "get" command.
        - Only contains transmit or received data without any additional information. A frame with received data will be returned everytime the device received data on the UART Rx line. As the interface is of asynchronous nature, this can occur anytime and without being initialized by user command.
        - Constraint: [data] can contain a maximum of 200 bytes

interface settings

- Set miscellaneous settings such as SPI SCLK-rate.
- Syntax set: **[CT] [LE] 03 [bank] [interface] [setting] [value] [CT]**
- Syntax get: **[CT] [LE] 83 [bank] [interface] [setting] [CT]**
- Syntax return: **[CT] [LE] 83 [bank] [interface] [setting] [value] [CT]**
- [interface]:
    - Defines the target interface of the command. The setting will be applied to this interface.
    - Values:
        - 0x03 - SPI Master
        - 0x04 - UART
- [setting]:
    - Defines the setting to be applied.
    - Possible values depend on the [interface]:
        - "SPI Master"
            - 0x00 - SPI SCLK-rate
            - 0x01 - SPI polarity
        - "UART"
            - 0x00 - UART baud rate
            - 0x01 - UART parity
- [value]:
    - Actual set value.
    - Length and possible values of this element depend on the chosen [setting]:
        - "SPI SCLK-rate"
            - Defines the SCLK frequency (f_SCLK).
            - 32 bit, unsigned integer
            - f_SCLK = mSPI_f_SCLK_source ÷ [value]
            - mSPI_f_SCLK_source = 115MHz
            - f_SCLK-range: (mSPI_f_SCLK_source ÷ 2) ≥ f_SCLK ≥ 1kHz
            - Default: 1MHz
        - "SPI polarity"
            - Defines the SPI polarity, thus the idle state of the SCLK signal.
            - values:
                - 0x00 - CPOL = 0 (idle low) (default)
                - 0x01 - CPOL = 1 (idle high)
        - "UART baud rate"
            - Defines the baud rate of the UART interface.
            - 32 bit, unsigned integer

- UartBaudRate = UartBaudrate_source ÷ [value]
- UartBaudRate_source = 115MHz
- UartBaudRate-range: 3 MBaud ≥ UartBaudRate ≥ 9600 Baud
- Default: 9600Baud

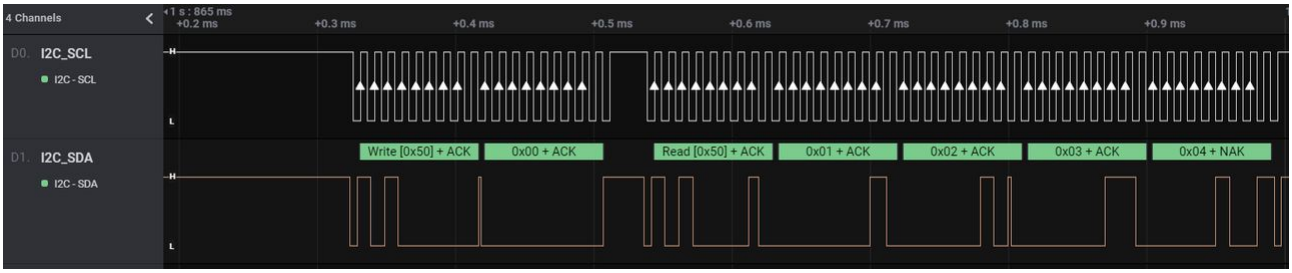
▪ "UART parity"

- Defines the use and polarity of the parity bit.
- values:
  - 0x00 - no parity (default)
  - 0x01 - even parity
  - 0x02 - odd parity

Examples

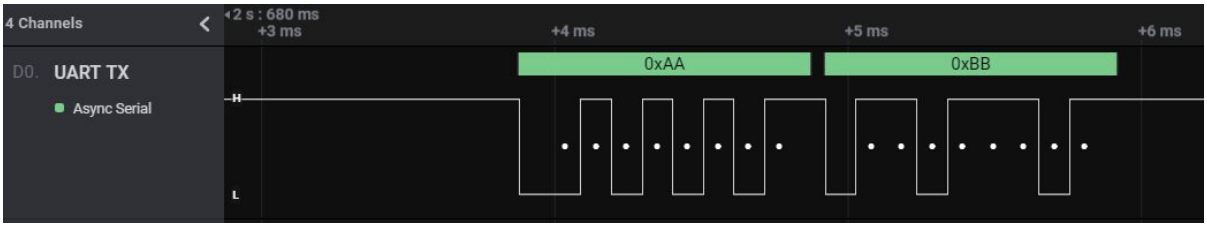
I^2C - Read / Write process

- Set command interface data: **CA 07 02 03 02 A1 01 00 04 CA** → write 00 and read 4 byte of I^2C Bank 3
- Set command IO functionality (set Bank B, IO 3 to I^2C SDA): **CA 04 00 02 03 03 CA**
- Set command IO functionality (set Bank B, IO 4 to I^2C SCL): **CA 04 00 02 04 02 CA**
- Set command interface data: **CA 07 02 02 02 A1 01 00 04 CA** → write 00 and read 4 byte of I^2C Bank 2
  - Return: **CA 07 82 02 02 01 02 03 04 CA**
- Set command interface data: **CA 0A 02 02 02 A0 05 00 01 02 03 04 CA** → write 5 byte to I^2C Bank



UART

- Set command: **CA 04 01 01 04 00 CA** → set UART Rx data sink to HS USB
- Set command: **CA 04 00 01 01 08 CA** → set IO 1 Bank A to UART Tx
- Set command: **CA 04 00 01 06 09 CA** → set IO 6 Bank A to UART Rx
- Set command: **CA 05 02 01 04 AA BB CA** → send AA BB on UART Bank1
  - Return: **CA 05 82 01 04 AA BB CA**



SPI

- Get command: **CA 04 83 01 03 00 CA** Read SPI SCLK-rate.
  - Return: **CA 08 83 01 03 00 00 00 00 73 CA** → 1MHz (because f_SCLK = mSPI_f_SCLK_source ÷ [value], mSPI_f_SCLK_source = 115MHz)
- Get command: **CA 04 83 01 03 01 CA** Read SPI polarity.

- Return: **CA 05 83 01 03 01 00 CA** → CPOL = 0 (idle low)

- Set command: **CA 04 00 01 01 05 CA** → set IO 1 Bank A to SPI Master CS#
- Set command: **CA 04 00 01 02 04 CA** → set IO 2 Bank A to SPI Master SCLK
- Set command: **CA 04 00 01 03 06 CA** → set IO 3 Bank A to SPI Master MOSI
- Set command: **CA 04 00 01 05 07 CA** → set IO 5 Bank A to SPI Master MISO
- Set command: **CA 07 02 01 03 02 03 AA BB CA** → send AA BB to SPI Master Bank A and reads 3 Bytes



9.3 Battery Operation

The Sciospec ISX-3 and ISX-3mini can be equipped with an integrated battery pack for measurements up to 2-3 h. The Sciospec Medical Research ISX-3 is by default equipped with the integrated battery pack.

9.3.1 Technical Specifications

Battery Pack

- Lithium Ion battery pack
- no. of cells: 3 cells; 1S3P
- capacity nom. 10 Ah
- 36Wh according to UN38.3 TSR
- measurement time ~2-3h (determined under certain circumstances, not generally valid!)
- ambient temperature range

0°C to +45°C	charge
-20°C to +60°C	discharge
-20°C to +30°C	storage max. 1 year
-20°C to +45°C	storage max. 3 month

9.3.2 Functional Description

The Sciospec ISX-3 / ISX-3mini can be configurated to measure on battery operation or mains operation. The configuration can be done in the Sciospec software (see Measurement Software Description) or using the Sciospec COMInterface (see Sciospec COMInterface). If the device is measuring on battery operation a minimum capacity can be configured, where the device changes to mains operation but the measurement continuous. The minimum capacity can be changed between 5 % and 100 %. By default 10 % is configured.

ⓘ Please charge the device at 10% remaining capacity. Please do not fully discharge the battery.

9.3.3 Sciospec COMInterface

0xC2 - Set BatMod

General Syntax

[CT] [LE] **[OB] [CD]** [CT]

Return

ACK

[OB]

Function	set
Measure on Battery	0x02
Measure on Battery min capacity	0x03

Measure on Battery

Set the Battery Mode to measure on battery.

Syntax

- Syntax set: **[CT] 02 02 [CD] [CT]**

[CD]

- 0x00: Disable
- 0x01: Enable

Measure on Battery min capacity

Set the Battery Mode to change to external power supply when the battery min capacity is reached. Measurements will be continued.

Syntax

- Syntax set: **[CT] 02 03 [CD] [CT]**

[CD]

- minimum state of charge in percent (≤ 100)

0xC3 - Get BatMod

General Syntax

[CT] [LE] **[OB] [CD]** [CT]

Return

[CT] [LE] **[OB] [status] [remaining capacity]** [CT]

ACK

[OB]

Function	get
Status / remaining capacity	0x01
Measure on Battery	0x02
Measure on Battery min capacity	0x03

Status / remaining capacity

Gets Status of Battery Mode and remaining capacity of battery.

Syntax

- Syntax get: **[CT] 01 01 [CT]**
  - Return: **[CT] [LE] 01 [status] [remaining capacity] [CT]**

[status]

1 Byte bit coded

MsBit - Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	LsBit - Bit 0
d. c.	d. c.	d. c.	d. c.	d. c.	d. c.	powers source	charging

power source:

- 1 = battery
- 0 = external

[remaining capacity]

- 1 byte hexadecimal remaining capacity in percent (≤ 100 percent)

Measure on Battery

Gets the status of "Measure on Battery".

Syntax

- Syntax get: **[CT] 01 02 [CT]**
  - Return: **[CT] [LE] 02 [Measure on Battery] [CT]**

[Measure on Battery]

0x00: Disable
0x01: Enable

Measure on Battery min capacity

Get the minimum battery capacity at where the system automaticaly switch to external power.

Syntax

- Syntax get: **[CT] 01 03 [CT]**
  - Return: **[CT] [LE] 03 [Measure on Battery min capacity] [CT]**

[Measure on Battery min capacity]

minimum state of charge in percent (≤ 100)

9.3.4 Measurement Software Description

If the Sciospec ISX-3 or ISX-3mini contains a battery pack, the main window of the software will show the battery status at the right top, see *Fig: Main Window battery operation.* By hovering over the battery status symbol with the mouse pointer, the capacity left in % will be displayed. The possible battery statuses are shown in *Tab: Battery statuses*.



Fig: Main Window battery operation

symbol	status	remaining capacity		symbol	status	remaining capacity
	battery	100%-75%			charging	100%-75%
	battery	75%-50%			charging	75%-50%

symbol	status	remaining capacity		symbol	status	remaining capacity
	battery	50%-25%			charging	50%-25%
	battery	25%-0%			charging	25%-0%

Tab: Battery statuses

Battery Operation Configurations

Under [Extra] → [Battery Mode] a Battery Mode Configuration Window can be opened.

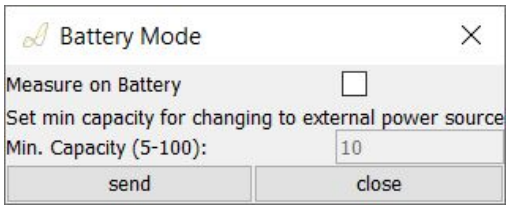


Fig: Battery Mode Configuration Window

Following configurations are available:

- Measure on Battery
- Set minimum capacity for changing to external power supply, **measurements will be continued**
  - selectable between 5-100%

10 Accessories

10.1 BNC and MCX Measurement Cable Sets

10.1.1 Kelvin Clips

- PN: SA000275
- for clamp/clip connection
- use with ISX-3, LCR-1
- cable length: 1 m



10.1.2 BNC - Banana

- PN: EA001555
- use with ISX-3, LCR-1
- length: 1 m

10.1.3 BNC - BNC

- PN: EA000749
- use with ISX-3, LCR-1
- length: 1 m

10.1.4 BNC - MCX

- PN: EA002696
- use with ISX-3, ISX-3mini, LCR-1
- length: 0.8 m

10.1.5 MCX - Banana

- PN: EA001726
- use with ISX-3mini
- length: 1 m

10.2 Multiplexer Wires

10.2.1 MUX Cable Set

- PN: SA000248
- RG178

- MDR - banana plug (+ 32 croco clips)
- length: 1 m



10.2.2 Medical trunk cable

- LEMO - 4 x 4mm Banana Plug
- length: 1 m



10.3 Grounding Wires

10.3.1 Grounding Cable 2.5 mm^2

- PN: SA000442
- green/yellow 2.5 mm^2 14AWG 1100 mm.
- side A: 2 mm banana plug (solder method)
- side B: M5 eye clamp

10.3.2 Grounding Cable M3 0.75 mm^2

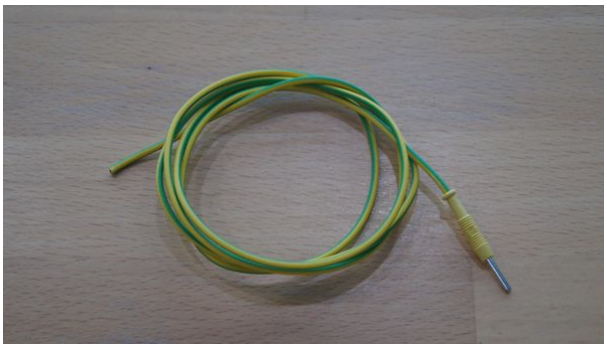
- PN: SA000441
- green/yellow 0.75 mm^2 20AWG 1500 mm.
- side A: 2 mm banana plug (solder method)
- side B: M3 eye clamp





10.3.3 Grounding Cable Blunt Cut 0.75 mm^2

- PN: SA000416
- green/yellow 0.75 mm^2 20AWG 1000 mm.
- side A: 2 mm banana plug (solder method)
- side B: Blunt Cut



10.4 Test Fixtures

10.4.1 Component Test Fixture 10 MHz

- PN: EA002764
- for easy connection of wired components
- for measurements up to 10 MHz
- short plate included
- size: 105 mm x 25 mm x 68 mm (width x height x depth)
- weight: ca. 150 g

10.4.2 Component Test Fixture 30 MHz

- PN: EA002793
- for easy connection of wired components
- for measurements up to 30 MHz
- short plate included
- size: 98 mm x 25 mm x 65 mm (width x height x depth)
- weight: ca. 250 g





10.4.3 Impedance Test Fixture

- PN: SA000460
- for measuring SMD and THT components soldered on test boards
- includes a set of 12 test boards
  - 3 resistors (82Ohm, 10kOhm, 1MOhm)
  - 3 inductors (1µH, 100µH, 10mH)
  - 3 capacitors (1pF, 10nF, 100µF)
  - 2 compensation boards (open, short)
  - 1 RC circuit (1kOhm + 1kOhm || 10nF)
- size: 118 mm x 29 mm x 80 mm (width x height x depth)
- weight: ca. 250 g
- includes case



10.4.4 SMD Test Fixture

- PN: EA002733
- for connection of SMD components
- SMD package types from 0201 to 2920 (imperial)
- size: 123 mm x 64 mm x 88 mm (width x height x depth)
- weight: ca. 200 g



10.4.5 Dielectric Test Fixture

- PN: EA002763
- for connection of foils and thin films with thickness up to 10 mm

10.4.6 Liquid Test Fixture

- PN: EA002792
- for measuring liquids and solutions
- size: 115 mm x 129.5 mm x 159 mm (width x height x depth)

- size: 115 mm x 155 mm x 126 mm (width x height x depth)
- weight: ca. 850 g



- weight: ca. 1700 g



10.5 Compensation Kit

- PN: SA000278
- consists of 6 high-precision wired resistors (1 Ohm, 10 Ohm, 100 Ohm, 1 kOhm, 10 kOhm, and 100 kOhm)
- for connection to different fixtures, e.g., component test fixture 10 MHz or 30 MHz or Kelvin clip leads
- including case for safe keeping
- size: 215 mm x 52 mm x 133 mm (width x height x depth)
- weight: ca. 350 g



11 Appendix

11.1 Data formats

All data sent to the device or sent from the device is either “Integer-Straight-Binary” or 32 Bit Float coded (according to IEEE 745). Either way the data is in Big-Endian format.

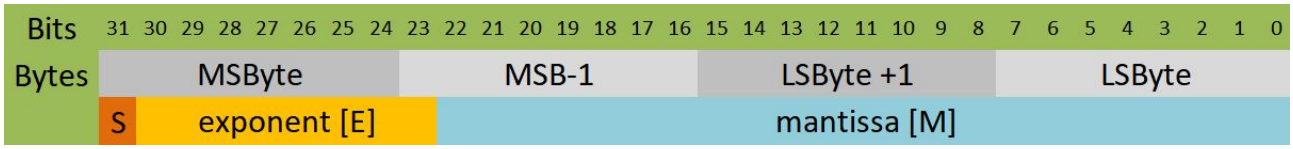
Big-Endian means that if a value consists of more than one byte the byte order is according to their value:

Example 4 byte Big-Endian Value: [MSByte] [MSByte-1] [LSByte-1] [LSByte]

The bit order is within one byte MSBit-first:

[MSBit] [MSBit-1] [MSBit-2] [MSBit-3] [LSBit+3][LSBit+2] [LSBit+1] [LSBit]

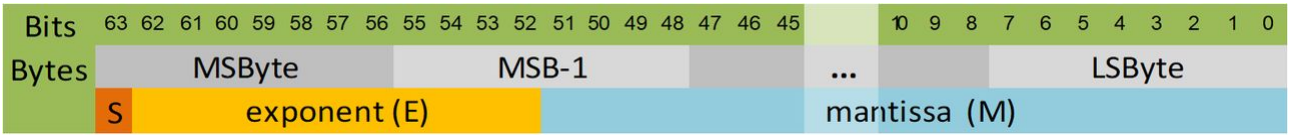
11.1.1 Float data format - single precision



The float value is calculated as follows:

$$float = (-1)^S * (1 + M * 2^{-23}) * 2^{E-127}$$

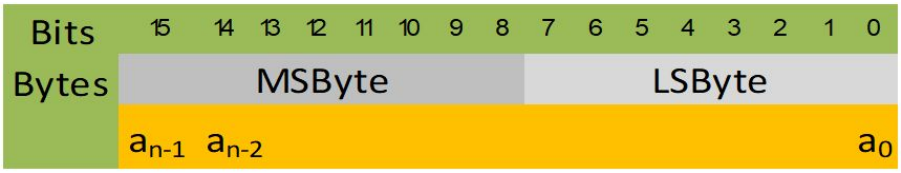
11.1.2 Float data format - double precision



The double precision value is calculated as follows:

$$float = (-1)^S * (1 + M * 2^{-52}) * 2^{E-1023}$$

11.1.3 Integer straight binary data format



The numerical value is calculated as follows:

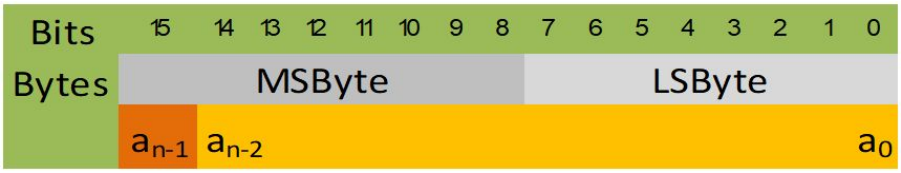
$$x_{ISB} = 2^{n-1} * a_{n-1} + 2^{n-2} * a_{n-2} + \cdots + 2^1 * a_1 + 2^0 * a_0$$

with

- n - bit width (in this example n=16)
- a_i - bit numbering i (a_{n-1}...most significant bit; a_0...least significant bit)

11.1.4 Two's complement data format

Example: 16bit two's complement



The numerical value is calculated as follows:

$$x_{twos} = -2^{n-1} * a_{n-1} + 2^{n-2} * a_{n-2} + \cdots + 2^1 * a_1 + 2^0 * a_0$$

with

- n - bit width (in this example n=16)
- a_i - bit numbering i (a_{n-1}...most significant bit; a_0...least significant bit)

11.2 Declaration of Conformity

scio spec scientific instruments	**Declaration of Conformity** **According to EN ISO/IEC 17050-1:2010**	CE
Manufacturer's Name	Sciospec Scientific Instruments GmbH	
Manufacturer's Address	Leipziger Str. 43b, 04828 Bennewitz Germany	
Declares under sole responsibility that the product as originally delivered		
Product Name	Impedance Analyzer	
Model Number	Medical Research / Isolated ISX-3	
Product Options	This declaration covers all options, cables and accessories of the above products(s).	
Serial Number	Covers all products • 01-003D-XXXX-XXXX (Medical Research / Isolated ISX-3)	

scio spec scientific instruments	**Declaration of Conformity** **According to EN ISO/IEC 17050-1:2010**	CE
complies with the essential requirements of the following applicable European Directives, and carries the CE marking accordingly		
	Low Voltage Directive (LVD)	2014/35/EU
	Electromagnetic compatibility Directive (EMC)	2014/30/EU
	Restriction of Hazardous Substances Directive (RoHS2)	2011/65/EU
	Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)	1907/2006/EU
	Persistent Organic Pollutants Regulation (POP)	2019/1021/EU
and confirms with the following product standards		
	EMC	DIN EN 61326-1:2022-11
	Safety	DIN EN 61010-1:2020-03
	RoHS	DIN EN IEC 63000:2019-05
	REACH: This product complies with the requirements of the REACH Regulation (EC) No. 1907/2006 (including SVHC communication obligations; PFAS assessed as applicable). Based on our current assessment and the information available to us, this product does not contain any Substances of Very High Concern (SVHC) in concentrations above 0.1%, as per the Candidate List of the European Chemicals Agency (ECHA) - current as of date of this declaration's issue.	
The product is to be used for the intended purpose. This product were tested in a typical configuration with Sciospec test systems. When used under conditions other than the specified conditions appropriate protective measures by the user must be taken.		
This DoC applies to above-listed products placed on the EU market after		
Date	April 2nd, 2026	
Place	Bennewitz, Germany	
Name	Martin Bulst	
Signature		

12 Changelog

Revision 6

- Initial release of this document

Revision 7

- Updated accuracy contour plot
- Added product picture

Revision 9

- Updated chapter software installation
- Updated setup configuration in software chapter
- Added intended frequency range for AC coupling interface

Revision 10

- Added option of MUX32 or MUX64
- Updated Declaration of Conformity: EMC to DIN EN 61326-1:2022-11

Revision 11

- renamed calibration to compensation
- Updated 0x97 - Set Options
  - Added new option byte for activating time stamp in µs
  - Changed description for "activation time stamp" to "activating time stamp in ms"
- Updated description of measurement data output (0xB8 - Start Measure)
- Added Autoranging in Set/Get FE Settings command
- Added current range output to Set/Get Options command and added current range output to measurement data description
- Added frequency range to Get Option command
- Updated software connection dialog
- Updated frequency ranges
- Added Autoranging feature

Revision 13

- Updated documentation of Auto-ranging feature

Revision 14

- Added Accessories

Revision 16

- General
  - improved measurement quality
  - added 0.09 V voltage range
- Measurement software description
  - added 0.09 V voltage range
- Technical specifications
  - added 0.09 V voltage range
- COMInterface:

- 0xB0 cmd, option byte "range setting": changed to "current range setting"
- 0xB0 cmd: added option byte "voltage range settings
- 0xB1 cmd, option byte "range setting": changed to "current range setting"
- 0xB1 cmd: added option byte "voltage range settings
- 0xB6 cmd: corrected available option bytes
- 0xB7 cmd, option byte get frequency list: frame splitting

Revision 18

- Technical specifications
  - Frequency resolution specified in more detail
- Functional description
  - Constraints for phase synchronization
- COMInterface
  - Description saving to / loading from slot

Revision 20

- Updated Declaration of Conformity

Revision 22

- Updated Measurement Software Description
- Updated General Specifications

Revision 26

- Added chapter Compensation in the Software
- Removed outdated MUX software description
- Improved internal timing
  (might require a new acquisition of previously created compensation data for high absolut precision)
- updated documentation of Sciospec ComInterface command: 0xB0 - Set FE Settings and 0xB1 - Get FE Settings